

Metabolism

Edwin Maiko

What is Metabolism?

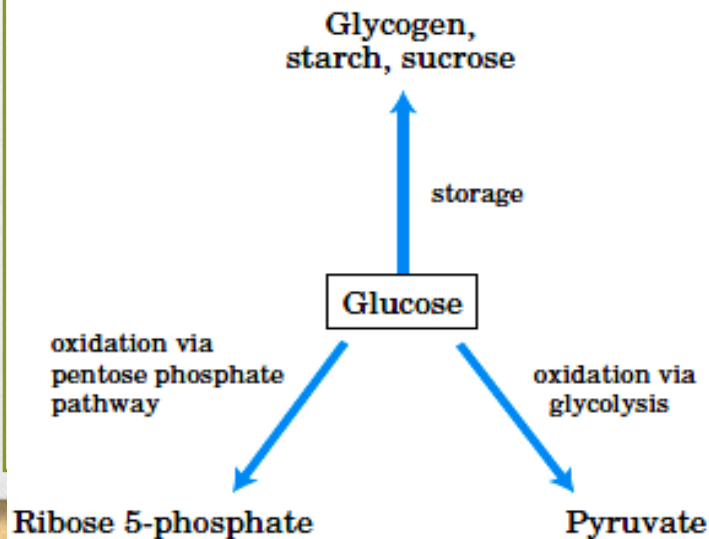
Def:

- The entire chemical reaction network of the body that are catalysed by enzymes.
- The intermediates, substrates or products of these enzyme catalysed reactions are referred to as **Metabolites**.
- The sequence of enzymatic reactions that lead to specific products are referred to as **Metabolic Pathways**. They are of three types:
 - i. **Catabolic pathways (Catabolism):** Breakdown of complex molecules (Carbohydrates, Proteins & Lipids) to small molecules. Energy is generated.
 - ii. **Anabolic pathways (Anabolism):** Synthesis of complex molecules from small molecules. Energy is consumed.
 - iii. **Amphibolic pathways:** Act as links between catabolic pathways and anabolic pathways.

Note: These pathways do not occur in isolation.

Carbohydrate Metabolism

- Glucose occupies a central position in the metabolism of plants, animals and many microorganisms.
 - a. It is relatively rich in potential energy, thus a good fuel. (Complete oxidation to CO_2 and H_2O proceeds with a ΔG value of $-2,840 \text{ KJ/mol}$).
 - b. It is a versatile precursor, capable of supplying a huge array of metabolic intermediates for biosynthetic reactions. e.g. **E. coli** can obtain carbon skeleton for Amino acids, nucleotides, coenzymes, fatty acids or other metabolic intermediates it needs for growth.
- In animals and vascular plants, glucose has three major fates:



How is glucose synthesized?

- ❖ Photosynthetic organisms make glucose by fixing CO_2 to trioses, then convert the trioses to glucose.
- ❖ Non-photosynthetic cells make glucose from simpler three and four-carbon compounds by the process of **gluconeogenesis**

Content:

- Glycolysis.
- Fructose metabolism
- Galactose metabolism
- Gluconeogenesis.
- Pentose Phosphate Pathway.
- Tricarboxylic acid cycle (TCA cycle).
- Electron Transport Chain (Oxidative phosphorylation).

Glycolysis

- ✓ In **glycolysis** a molecule of glucose *is* degraded in a series of enzyme-catalyzed reactions to yield two molecules of the three-carbon compound (pyruvate).
- ✓ The sequential reactions of glycolysis, conserve the free energy released from glucose in form of ATP and NADH.
- ✓ Glycolysis is the central pathway of glucose catabolism in most cells some of which solely depend on it as a source of metabolic energy .e.g.
 - i. Mammalian tissues and cell types (erythrocytes, renal medulla, brain and sperm cell .
 - ii. Plants tissues that are modified to store starch (potato tubers)
 - iii. Most anaerobic microorganisms.
- ✓ The breakdown of the six-carbon (glucose) into two molecules of the three-carbon pyruvate occurs in ten steps, which are divided in to two phases:
 - i. Preparatory phase
 - ii. Pay off phase.

Note: Glycolytic Enzymes are present in the cytosol of cells.

Glycolysis cont....

Preparatory

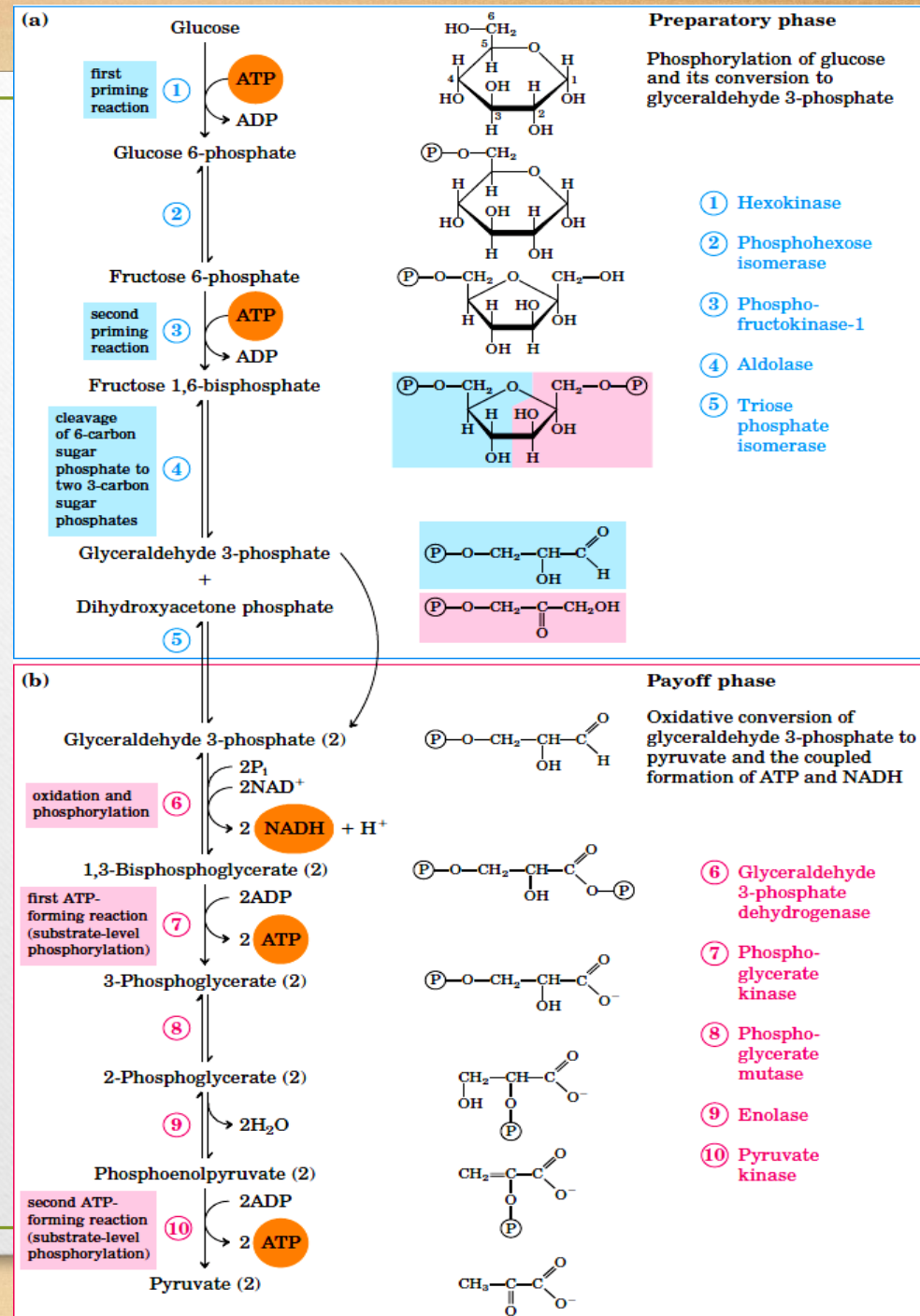
➤ **phase** first five reactions of glycolysis which constitute the preparatory phase the energy of ATP is invested, raising the free-energy content of the intermediates, and the carbon chain of all hexoses are converted into a common product; two molecules of glyceraldehyde 3- phosphate (G3P).

Payoff phase

➤ In the payoff phase of each molecule of G3P is oxidized and phosphorylated by inorganic phosphate (not by ATP) to form 1,3-bisphosphoglycerate.

➤ Energy is then released as the two molecules of 1,3-bisphosphoglycerate are converted to two molecules of pyruvate.

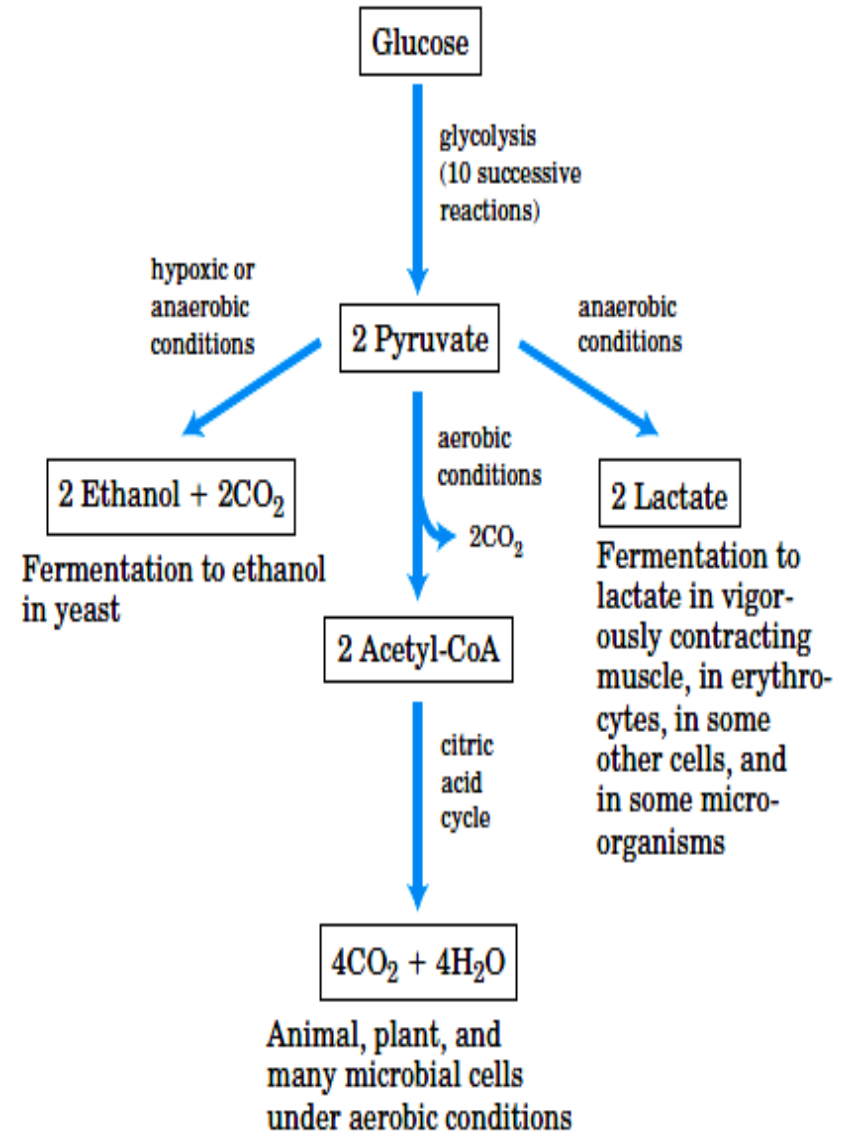
➤ The net yield is two molecules of ATP and two molecules of NADH per molecule of glucose.



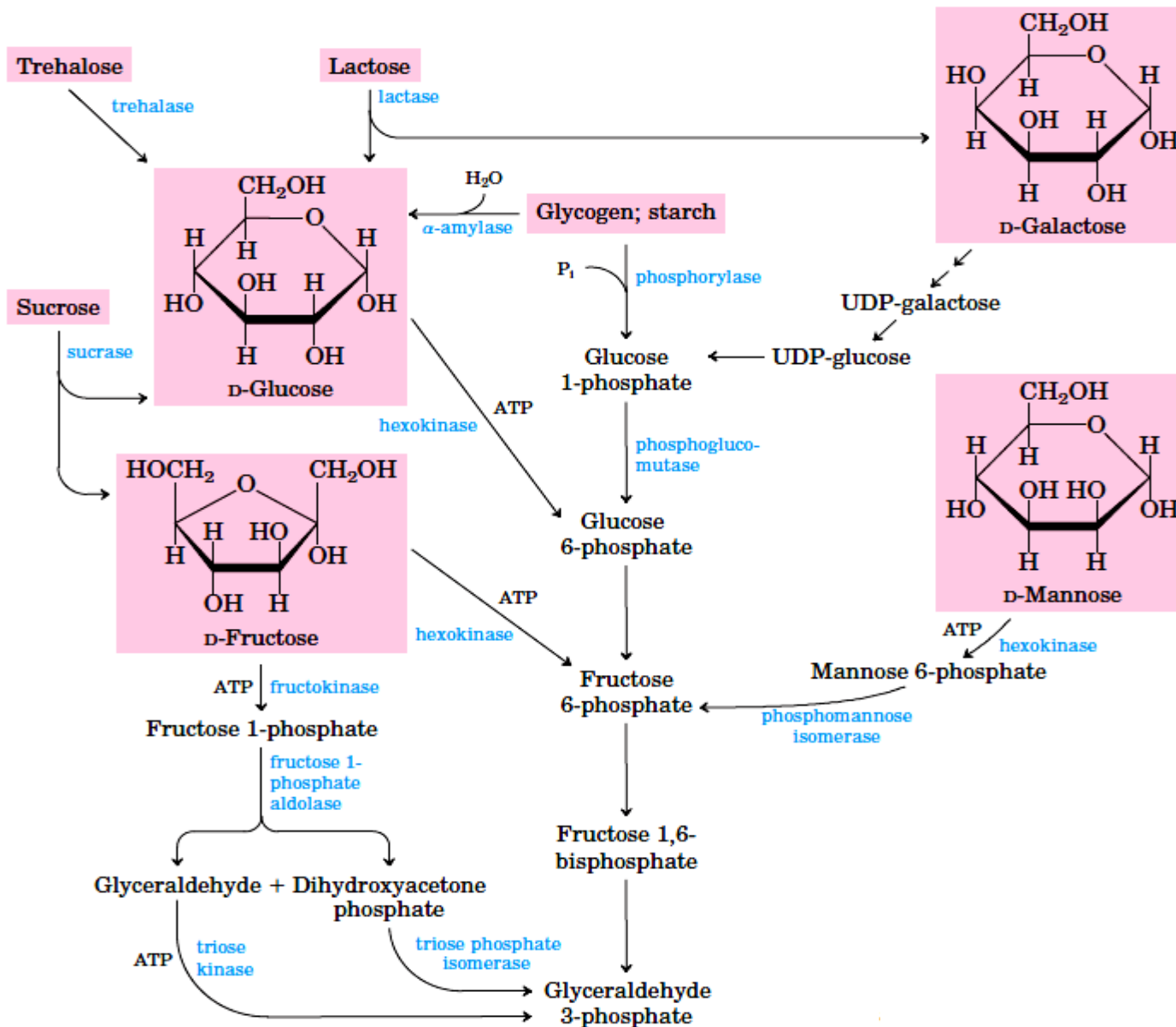
Fates of Pyruvate

The pyruvate formed in glycolysis is further metabolized in three major catabolic routes.

- I. Complete oxidation to CO_2 and water in the citric acid cycle and electron transport chain in the mitochondria. The energy from electron-transfer reactions drive ATP synthesis (e.g. aerobic organisms or tissues).
- II. Pyruvate reduction to lactate via lactic acid fermentation for regeneration of NAD^+ required as an electron acceptor for further oxidation of glucose to pyruvate (e.g. contracting muscle, erythrocytes, retina and anaerobic microorganisms)
- III. Decarboxylation of pyruvate to ethanol in ethanol (alcohol) fermentation under hypoxic or anaerobic conditions (e.g. protists and microorganisms)



Feeder pathways for



Regulation of Glycolysis

- metabolic pathways are regulated by altering activities of few enzymes of that pathway.
- **Hexokinase, phosphofructo kinase and pyruvate kinase** are the regulatory enzymes of glycolysis.
- Their activities are under allosteric and hormonal control.

Allosteric regulation of glycolysis

Phosphofructokinase-1:

- is the major regulatory enzymes of glycolysis.
- It is an allosteric enzyme and catalyzes rate limiting reaction of glycolysis.
- It is inhibited by **ATP** and **citrate**, while **AMP** and **fructose-6-phosphate** are activators of this enzyme.

Pyruvate kinase:

- is the second regulatory enzyme.
- It is inhibited by ATP.

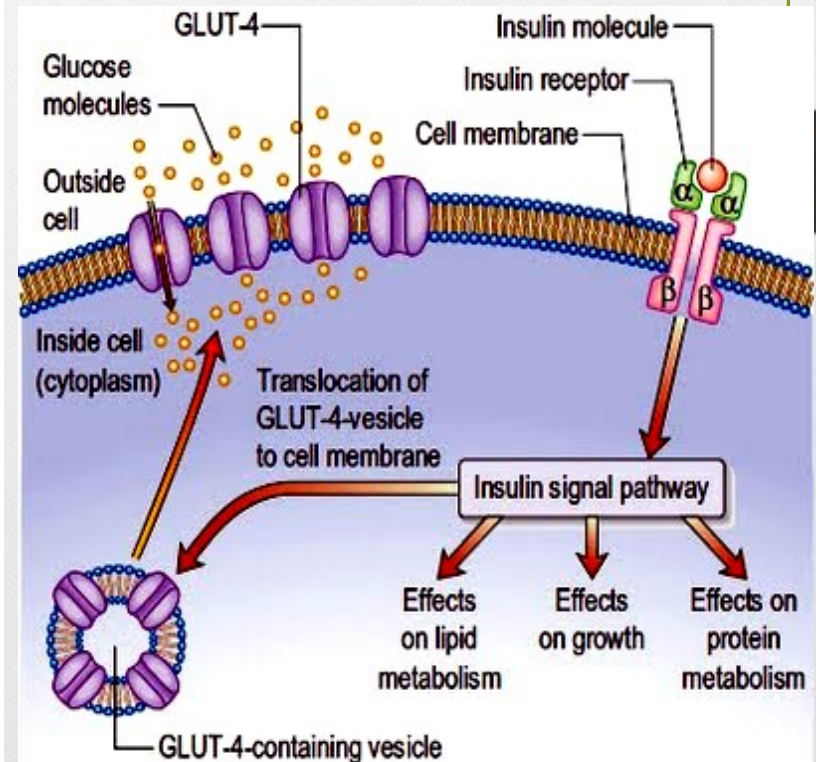
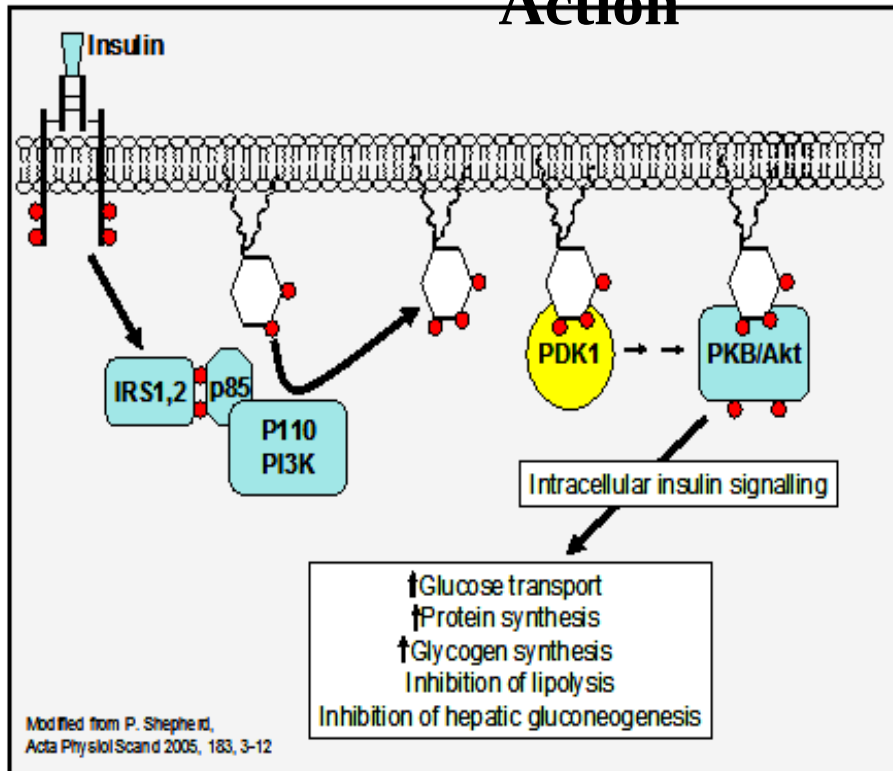
Hexokinase:

- Is inhibited by Glucose-6-phosphate.
- ✓ So, when ATP (energy) concentration is high glycolysis is inhibited.
- ✓ and decrease in ATP level increases rate of glycolysis.

Hormonal regulation of glycolysis

- Insulin increases rate of glycolysis by increasing concentration of **glucokinase**, **phosphofructokinase-1** and **pyruvate kinase**.

Mechanism of Insulin Action



FRUCTOSE METABOLISM

- **Fructolysis** is the term used to refer to the metabolism of fructose from dietary sources.
- Though the metabolism of glucose through glycolysis uses many of the same enzymes and intermediate structures as those in fructolysis, the two sugars have very different metabolic fates in human metabolism.
- Unlike glucose, which is directly metabolized widely in the body, fructose is almost entirely metabolized in the **liver** in humans, where it is directed toward replenishment of liver glycogen and triglyceride synthesis.
- Under 1% of ingested fructose is directly converted to plasma triglyceride.
- Approx. 29% - 54% of fructose is converted in liver to glucose, and about a quarter of fructose is converted to lactate. 15% - 18% is converted to glycogen.
- Glucose and lactate are then used normally as energy to fuel cells

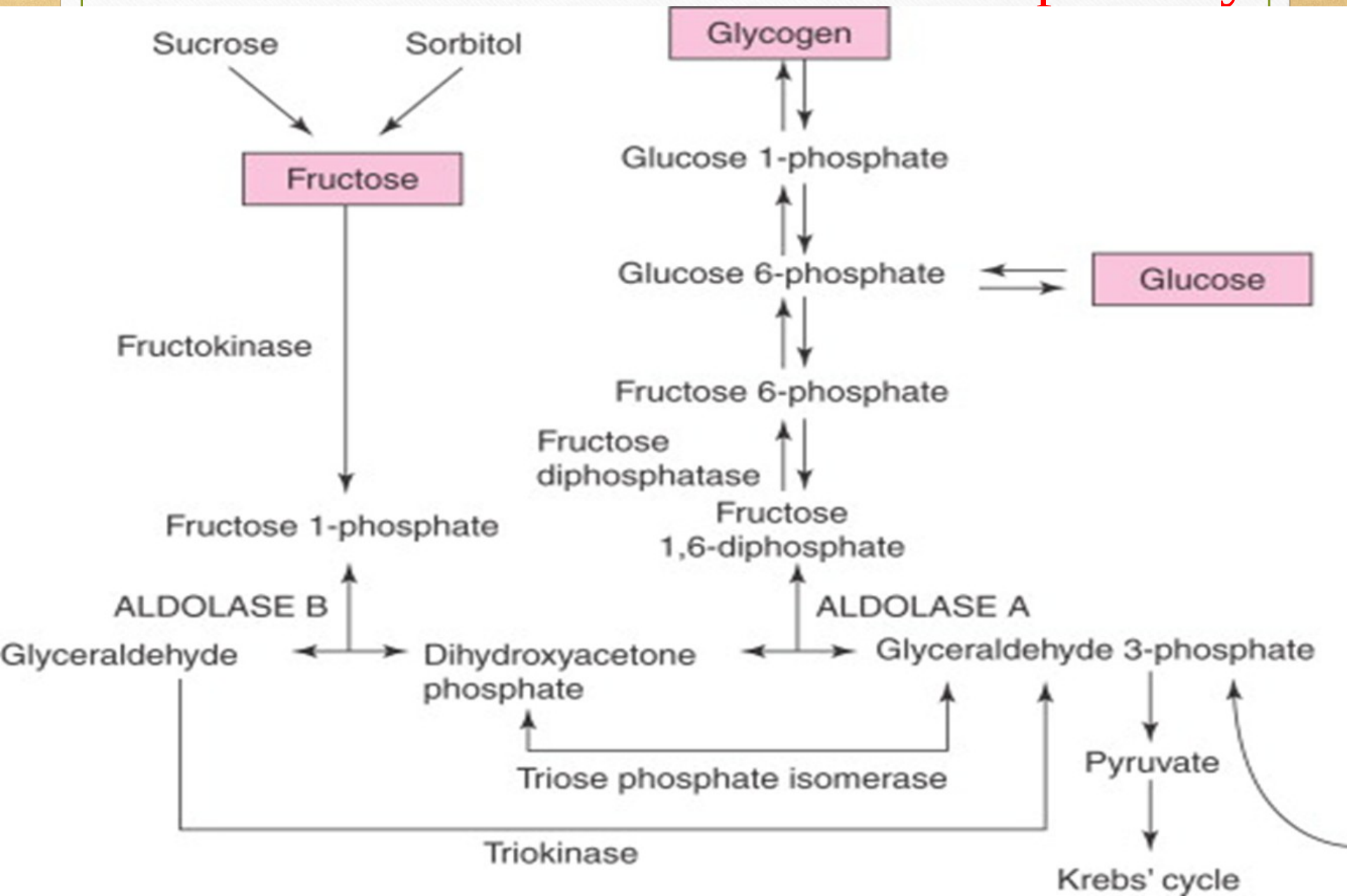
Reactions of Fructose Metabolism.

- Fructose is phosphorylated to fructose-1-phosphate by fructokinase in the liver. This enzyme will not phosphorylate glucose and unlike glucokinase its activity is not affected by fasting or by insulin and that is why fructose disappears from the blood of diabetic patients at a normal rate.
- Fructose-1-phosphate is cleaved by liver aldolase (aldolase B) to dihydroxyacetone phosphate (DHAP) and D-glyceraldehyde. Aldolase-B also functions in glycolysis in the liver attacking fructose-1,6-bisphosphate.
- D-glyceraldehyde is phosphorylated by glyceraldehyde kinase to D-glyceraldehyde-3-phosphate. These two triose phosphates (DHAP and glyceraldehyde-3-phosphate) then may enter the glycolytic pathway, gluconeogenesis or glycogenesis according to the metabolic status of the tissue.
- In extra hepatic tissues, especially the adipose and muscle tissues, fructose can be phosphorylated by hexokinase to fructose-6-phosphate. This however is a slow reaction and occurs only in the presence of high concentration of fructose because when fructose is present with glucose, its phosphorylation is inhibited by glucose.

Continuation.

- about a quarter of fructose is converted to lactate. 15% - 18% is converted to glycogen.
- Glucose and lactate are then used normally as energy to fuel cells all over the body.
- Fructose is a dietary monosaccharide present naturally in fruits and vegetables, either as free fructose or as part of the disaccharide sucrose, and as its polymer inulin.
- It is also present in the form of refined sugars including granulated sugars (white crystalline table sugar, brown sugar, confectioner's sugar, and turbinado sugar), refined crystalline fructose, as high fructose corn syrups as well as in honey.

Illustration of the fructose metabolism pathway

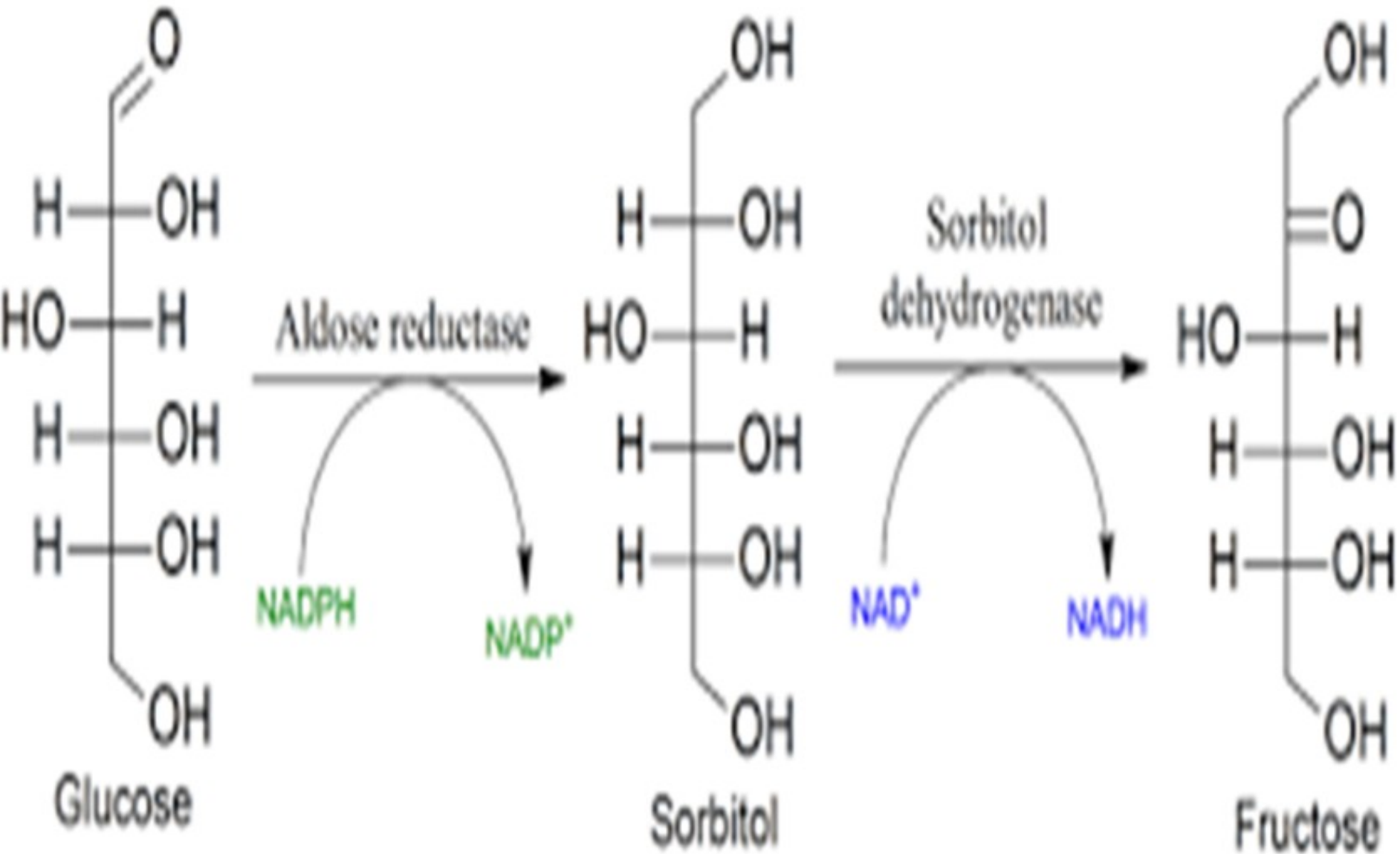


Sorbitol or Polyol Pathway for Formation of Fructose

Sorbitol

- Sorbitol less commonly known as glucitol, is a sugar alcohol with a sweet taste which the human body metabolizes slowly.
- It can be obtained by reduction of glucose, which changes the converted aldehyde group ($-CHO$) to a primary alcohol group ($-C(OH)H_2$).
- Most sorbitol is made from potato starch, but it is also found in nature, for example in apples, pears, peaches, and prunes.
- It is converted to fructose by sorbitol dehydrogenase.

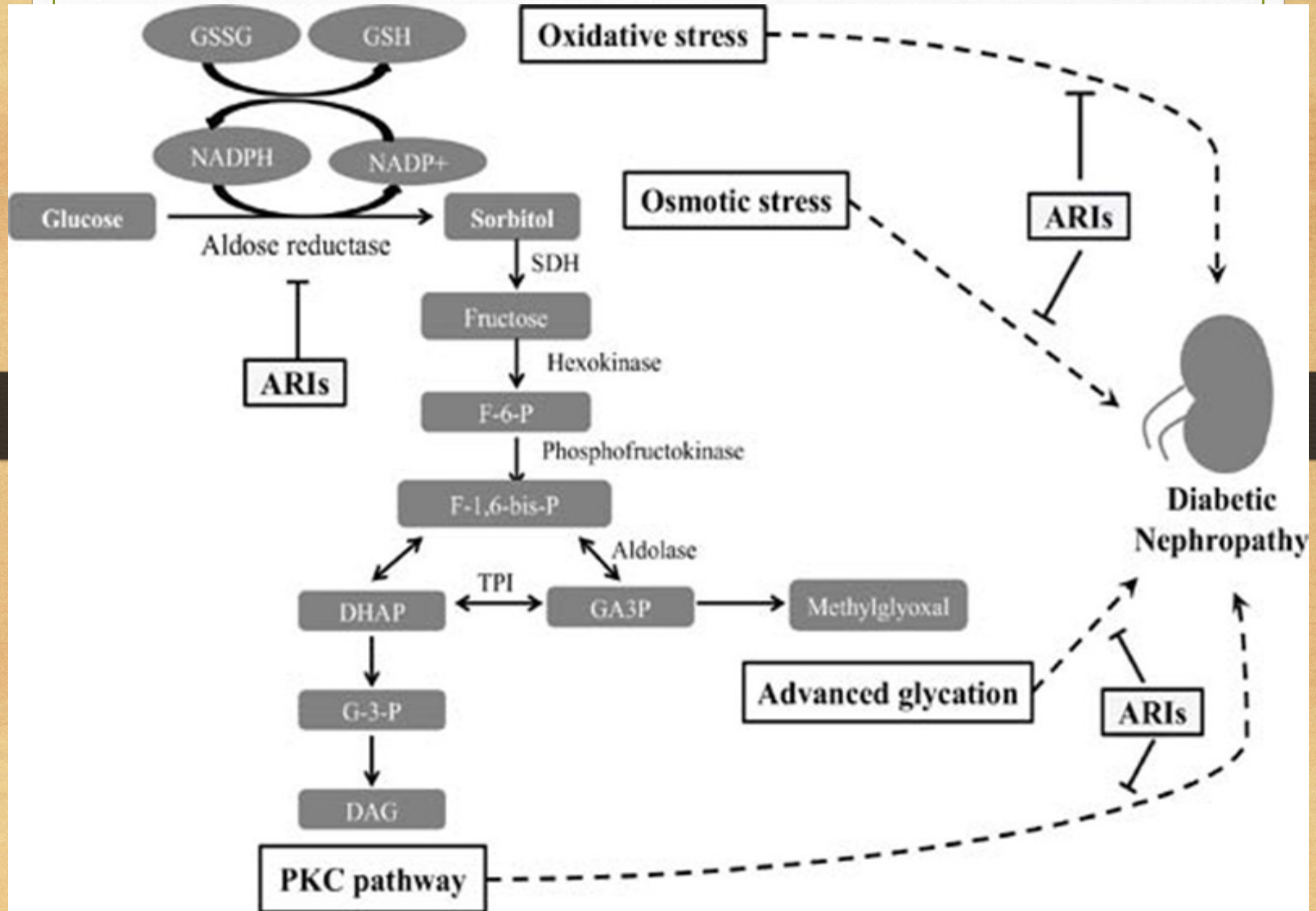
Continuation



Continuation.

- The sorbitol (polyol) pathway is for formation of fructose from glucose.
- Aldose reductase reduces glucose to sorbitol (glucitol).
- It's found in many tissues such as lens, retina, peripheral nerve cells, kidney, placenta, red blood cells, ovaries and seminal vesicles. But it is not present in liver.
- A second enzyme, sorbitol dehydrogenase that can oxidize the sorbitol to fructose is present in liver, ovaries and sperm cell.
- In liver, ovaries and sperm cells, the same enzyme, sorbitol dehydrogenase oxidizes the sorbitol to fructose.
- The pathway from sorbitol to fructose in the liver provides a mechanism by which dietary sorbitol is converted into fructose which can enter glycolysis.

Continuation



Inborn errors in fructose metabolism

- 1. Essential fructosuria
- The absence of fructokinase results in the inability to phosphorylate fructose to fructose-1-phosphate within the cell.
- As a result, fructose is neither trapped within the cell nor directed toward its metabolism.
- Free fructose concentrations in the liver increase and fructose is free to leave the cell and enter plasma.
- This results in an increase in plasma concentration of fructose, eventually exceeding the kidneys' threshold for fructose reabsorption resulting in the appearance of fructose in the urine.
- Essential fructosuria is a benign asymptomatic condition.

2. Hereditary fructose intolerance

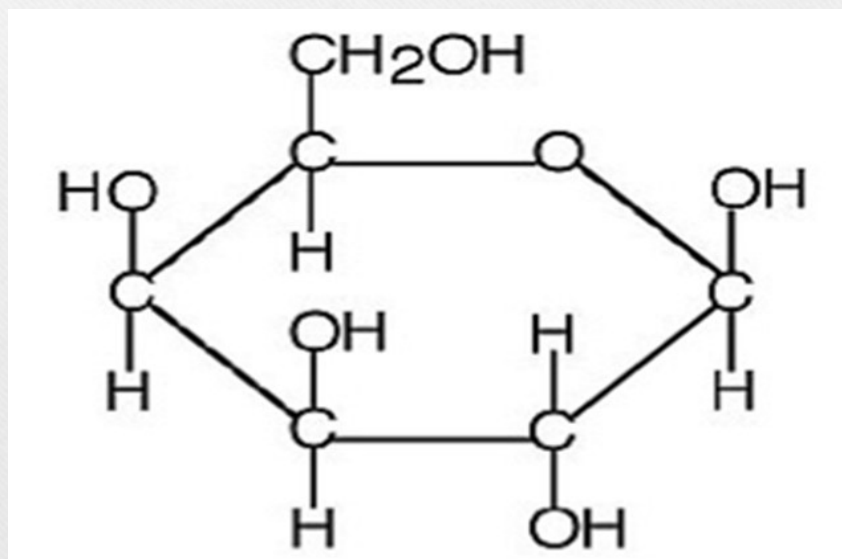
- The absence of fructose-1-phosphate aldolase (aldolase B) results in the accumulation of fructose 1 phosphate in hepatocytes, kidney and small intestines.
- An accumulation of fructose-1-phosphate following fructose ingestion inhibits glycogenolysis (breakdown of glycogen) and gluconeogenesis, resulting in severe hypoglycemia.
- It is symptomatic resulting in severe hypoglycemia, abdominal pain, vomiting, hemorrhage, jaundice, hepatomegaly, and hyperuricemia eventually leading to liver and/or kidney failure and death.

- Intravenous (i.v) infusion of fructose has been shown to lower phosphorylation potential in liver cells by trapping Pi(inorganic phosphate) as fructose 1-phosphate.
- The fructokinase reaction occurs quite rapidly in hepatocytes trapping fructose in cells by phosphorylation. On the other hand, the splitting of fructose-1-phosphate to DHAP and glyceraldehyde by Aldolase B is relatively slow.
- Therefore, fructose-1-phosphate accumulates with the corresponding reduction of intracellular Pi available for phosphorylation reactions in the cell.
- This is why fructose is contraindicated for total parenteral nutrition (TPN) solutions and is never given intravenously as a source of carbohydrate.
- It has been suggested that excessive dietary intake of fructose may also result in reduced phosphorylation potential. However, this is still a contentious issue. Dietary fructose is not well absorbed and increased dietary intake often results in malabsorption.

GALACTOSE METABOLISM

- Galactose, sometimes abbreviated Gal, is a monosaccharide and the C4 epimer of glucose, that is, they differ only for the position of the -OH group on C4. It has the molecular formula $C_6H_{12}O_6$ and molar mass 180.156 g/mol. It has a sweetness equal to 33% of sucrose.

Structure of galactose



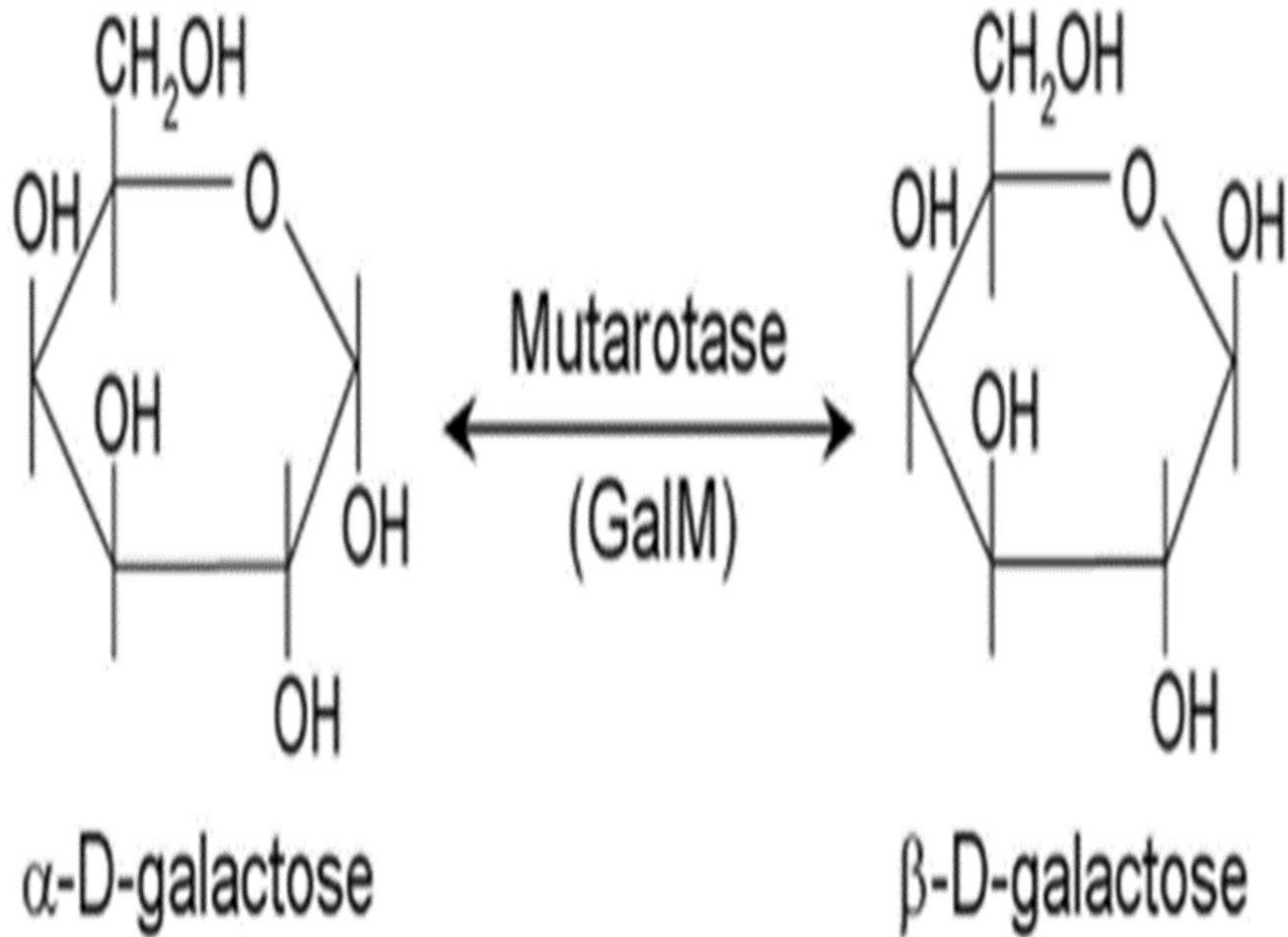
Food sources of galactose

- In human nutrition the most part comes from the hydrolysis of the disaccharide lactose (Galactose + Glucose), the milk sugar, including that of the human milk.
- As mother's milk is the only source of energy and carbohydrates for newborn, galactose has a crucial role in human nutrition.
- The monosaccharide is also bound to caseins, and therefore it is found in all dairy products.
- Findings from studies conducted since the 50s of last century have shown that galactose is present not only in milk and dairy products, but also in plant products such as legumes, grains, nuts, tubers and vegetables. It seems that in these foods it is often engaged in bonds resistant to the attack of human digestive enzymes, and therefore not metabolizable. However, different fruits and vegetables contain it also in free form, in variable amounts;
- Less than 0.1 mg/100 g of edible portion: artichokes, mushrooms, olives, and peanuts; more than 10 mg/100 g of edible portion: bell peppers, date, papaya, watermelon, tomato;
- up to 35.4 mg/100 g of edible portion in persimmon.
- These values are very low, but to take into account in case of galactosemia

Process of Galactose Metabolism.

- Galactose, which is metabolized from the milk sugar, lactose (a disaccharide of glucose and galactose), enters glycolysis by its conversion to glucose-1-phosphate (G-1-P). This occurs through a series of steps that is referred to as the Leloir pathway, named after Luis Federico Leloir who determined the overall process of galactose utilization.
- Galactose can exist in two different stereoisomeric forms; α -D-galactose and β -D-galactose. The α -D-form is that which is metabolized in the Leloir pathway.
- Conversion of the β -form of galactose to the α -form requires the enzyme galactose mutarotase encoded by the GALM gene (also known as aldose 1-epimerase)

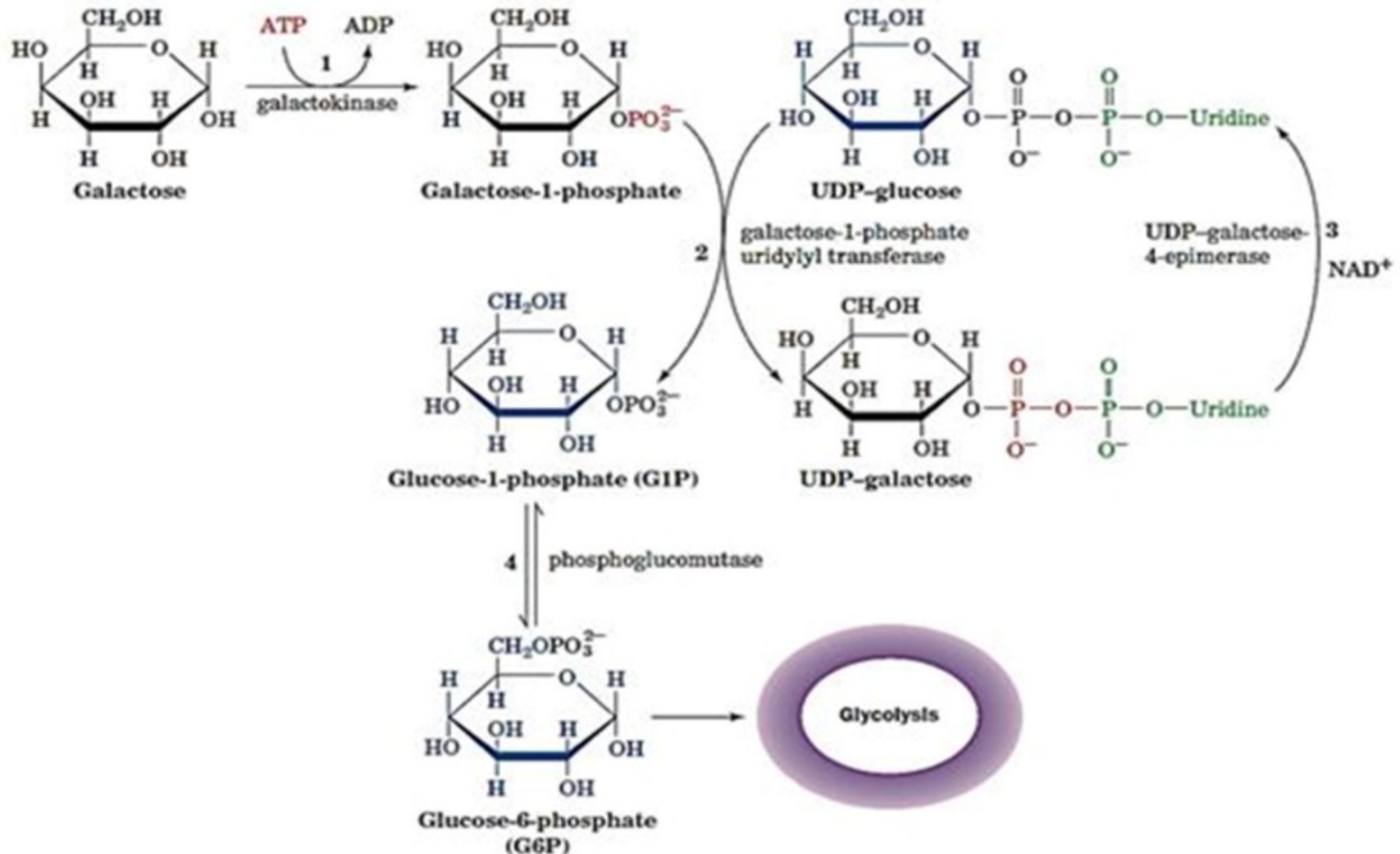
Continuation..



Continuation...

- The first reaction of the Leloir pathway is the phosphorylation of α -D-galactose by galactokinase to yield galactose-1-phosphate.
- Epimerization of galactose-1-phosphate to glucose-1-phosphate (G-1-P) requires the transfer of UDP from UDP-glucose catalyzed by galactose-1-phosphate uridylyltransferase (GALT). The galactose-1-phosphate uridylyltransferase catalyzed reaction generates UDP-galactose and glucose-1-phosphate (G-1-P).
- The UDP-galactose is epimerized to UDP-glucose by UDP-galactose-4 epimerase (GALE). The UDP portion is exchanged for phosphate generating glucose-1-phosphate which then is converted to glucose-6-phosphate (G-6-P) by phosphoglucosemutase. The glucose-6-phosphate then joins glycolytic pathway where it is metabolised further to pyruvate.

Illustration of Leloir pathway (Galactose pathway)

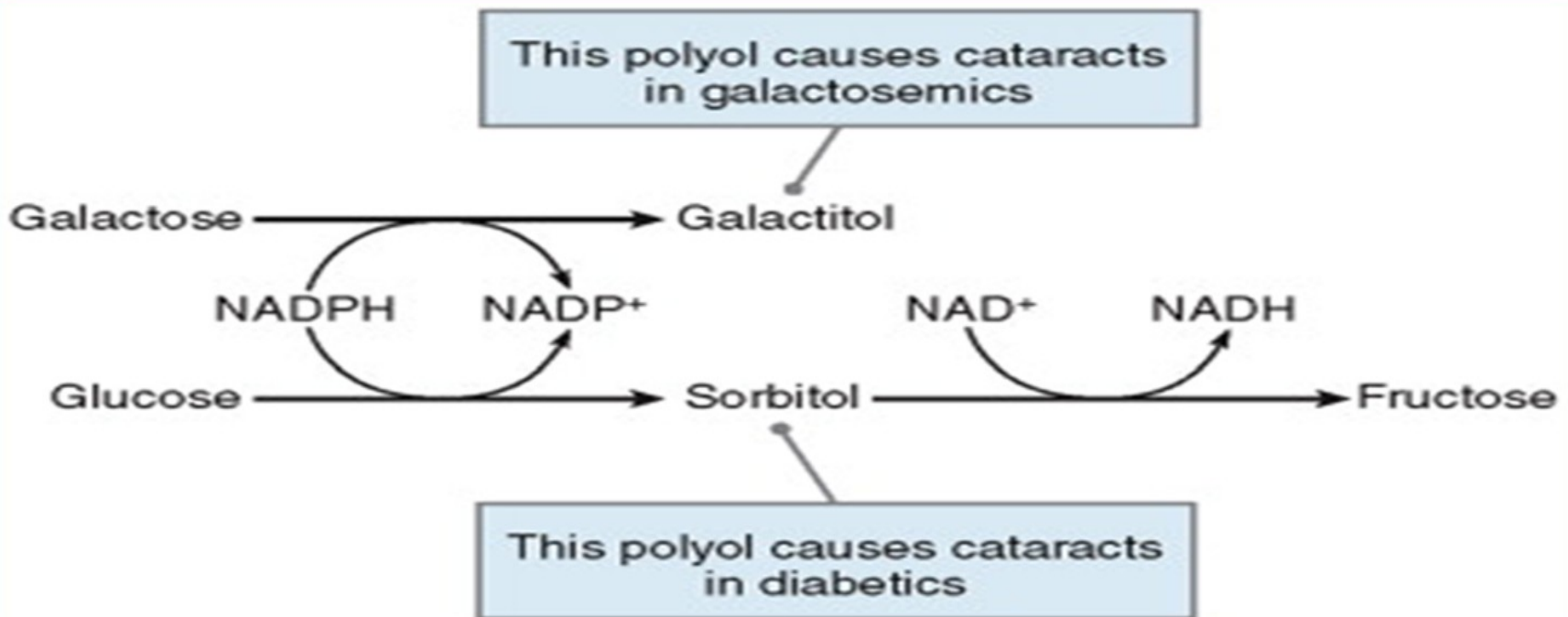


Metabolic fate of UDP-Galactose

- UDP-Gal is an important precursor in the synthesis of glycolipids, such as gangliosides and galactocerebrosides, sphingolipids, mucopolysaccharides and membrane glycoproteins.
- In the adult mammary gland, under the influence of prolactin, UDP-Gal can be joined to glucose to give milk sugar(Lactose).
- Regulated Reactions-No Regulation
- There is no known regulated step in the conversion of galactose to glucose intermediates. The fate of glucose from dietary galactose, toward either glycolysis or glycogenesis, is determined by pathways regulating glucose metabolism in the liver.
- Unique Characteristics-UDP-Glucose Intermediate
- Although the galactose pathway forms UDP-glucose, there is no net synthesis of this intermediate, since it is recycled. The net output of this pathway is in the conversion of a mole of galactose 1-phosphate to glucose 1-phosphate.

Interface with Other Pathways-Polyol Pathway

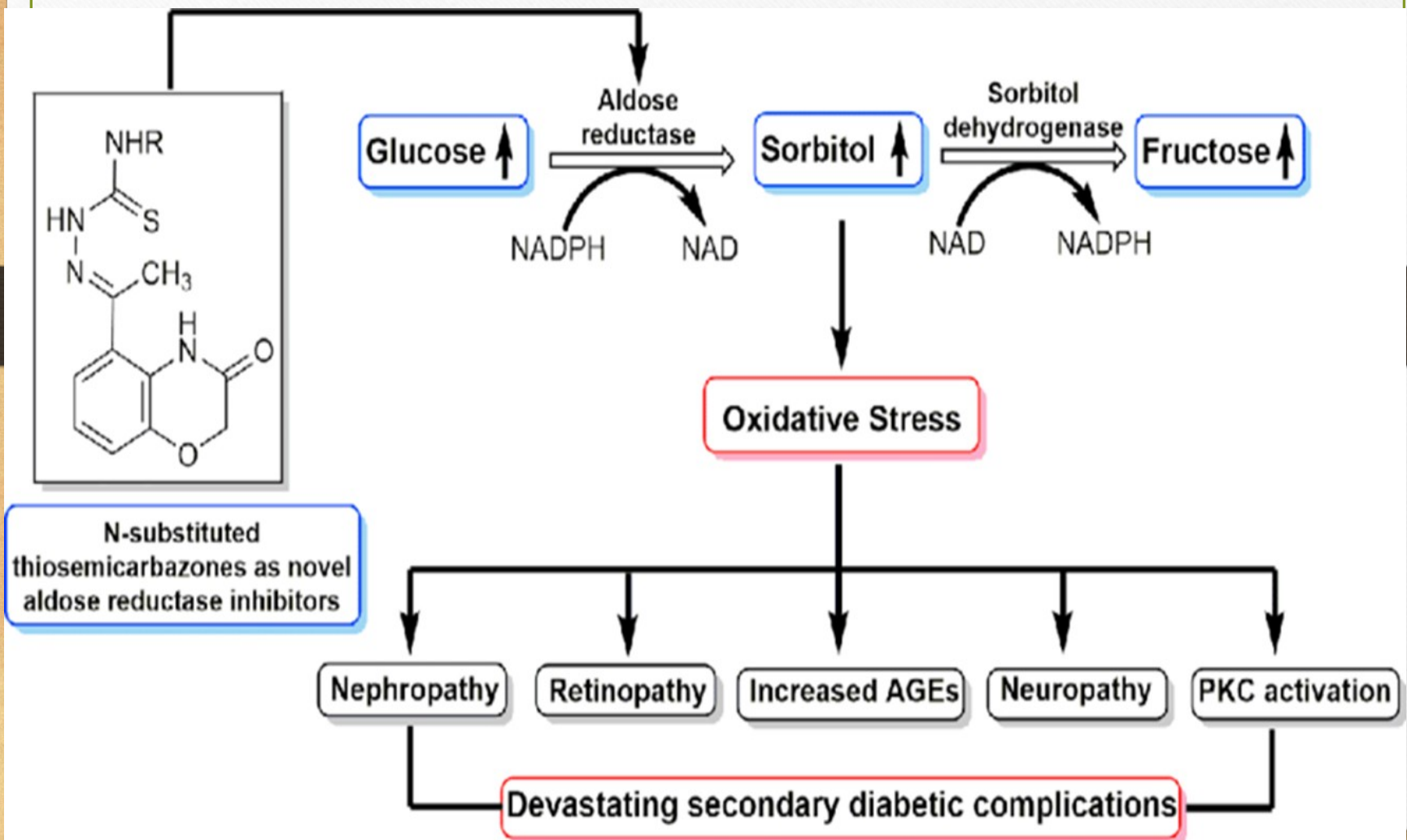
- Free galactose can be reduced by aldose reductase in the polyol pathway to galactitol (Figure below). The polyol pathway is widely distributed in the body including in the lens, where it contributes to the formation of cataracts in both galactosemia and diabetes.



Continuation..(sorbitol implicating diabetes)

- The pathway is implicated in diabetic complications, especially in microvascular damage to the retina, kidney and nerves.
- Sorbitol cannot cross cell membranes, and, when it accumulates, it produces osmotic stresses on cells by drawing water into the insulin-independent tissues.
- Cells use glucose for energy. This normally occurs by phosphorylation via the enzyme hexokinase.
- However, if large amounts of glucose are present (as in diabetes mellitus), hexokinase becomes saturated and the excess glucose enters the polyol pathway when aldose reductase reduces it to sorbitol

Continuation....



Gluconeogenesis

❖ In mammals, some tissues depend almost completely on glucose for their metabolic energy. e.g. Brain and the nervous system, erythrocytes, testes, renal medulla and embryonic tissues.

Note:

- i. Supply of glucose stored from stores (Glycogen) is not always sufficient (i.e. brain requires 120g of glucose/day More than half the glucose stored as glycogen in muscle and liver.
- ii. Between meals and prolonged fasts or vigorous exercise, glycogen is depleted.

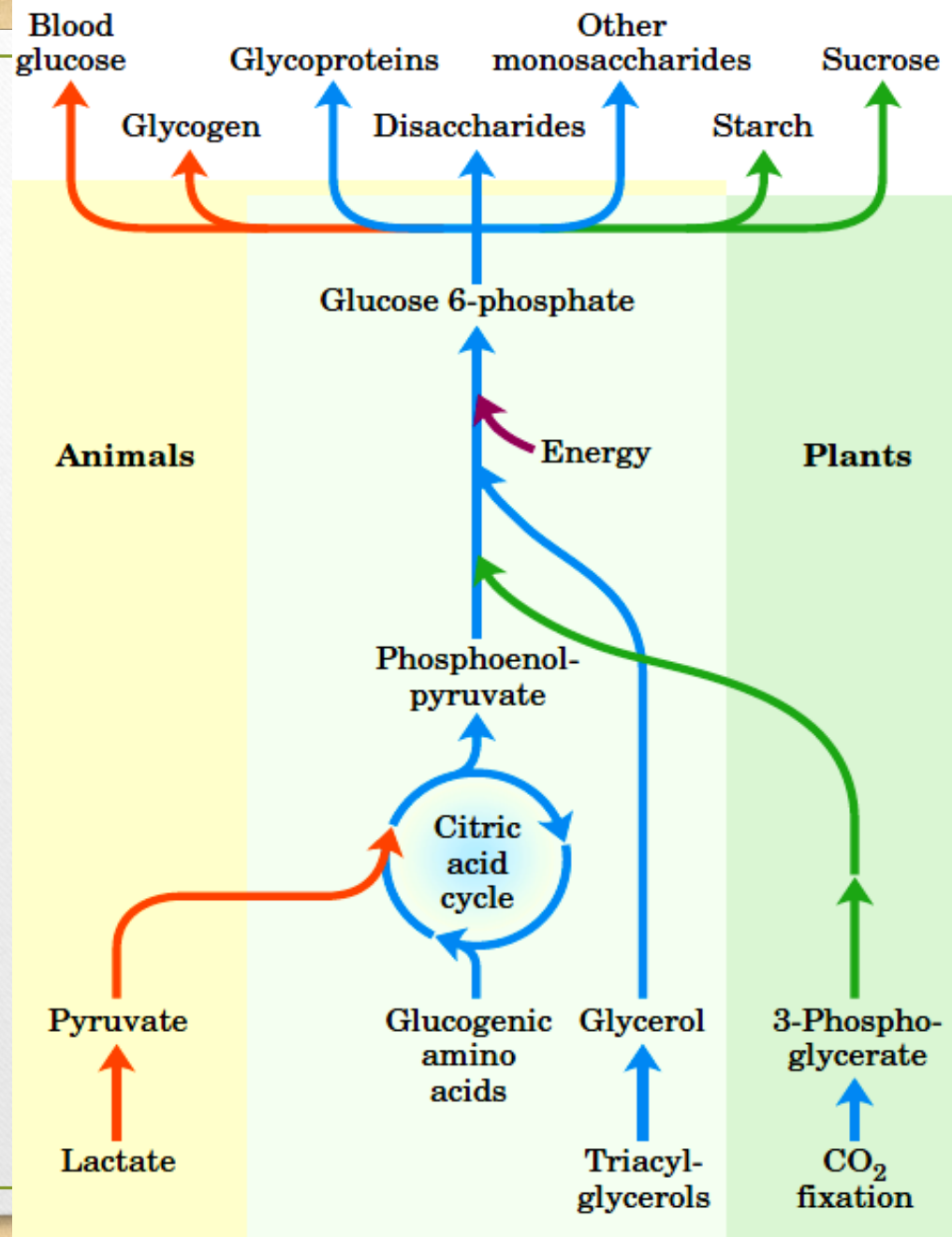
❖ To maintain blood glucose, glucose is synthesised from non-carbohydrate precursors through **Gluconeogenesis**.

❖ **Gluconeogenic pathway** convert pyruvate and related 3 and 4 carbon cpds to glucose.

- ✓ In animals important precursors of glucose include: **lactate, pyruvate, glycerol and some amino acids**.
- ✓ In microorganisms gluconeogenesis starts from simple organic cpds of 2 or 3 carbons such as: **acetate, lactate and propionate**.

❖ In Mammals it takes place mainly in the liver and to a lesser extent in renal cortex.

An overview of Gluconeogenesis in Plants and Animals



Reactions of the gluconeogenic

pathway

❖ Glycolysis and gluconeogenesis share several steps; seven of the ten enzymatic reactions of gluconeogenesis are the reverse of glycolysis

❖ Three Glycolysis reactions have such a large negative ΔG that they are essentially **irreversible**.

- ◆ **Hexokinase** (or Glucokinase)

- ◆ **Phosphofructokinase**

- ◆ **Pyruvate Kinase.**

❖ In Gluconeogenesis the three irreversible steps are **bypassed** by a separate set of enzymes that are irreversible in the direction of glucose synthesis.

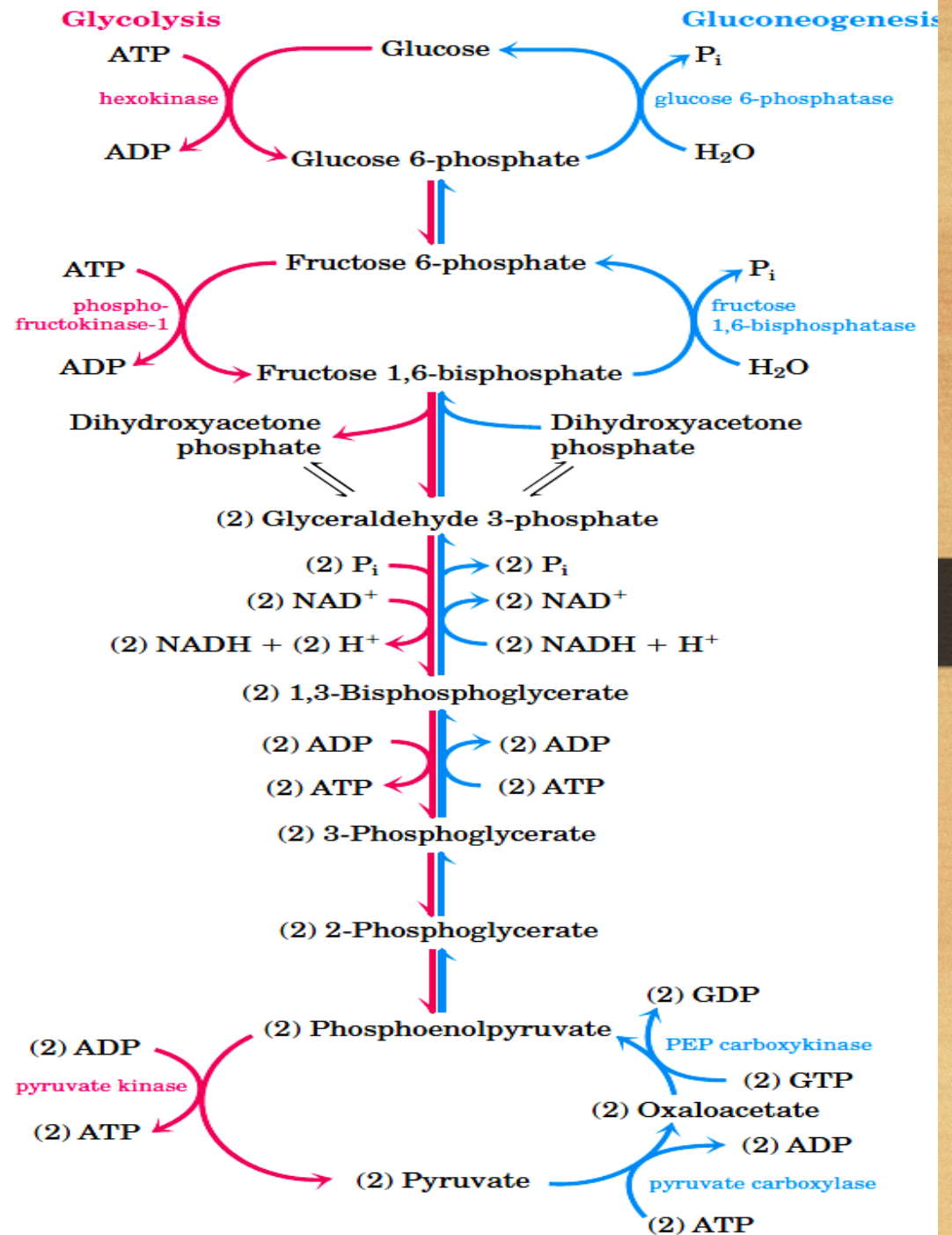
- Conversion of pyruvate to phosphoenol pyruvate.

- Conversion of Fructose 1,6- Bisphosphate to Fructose 6-phosphate.

- Conversion of glucose 6-phosphate to glucose.

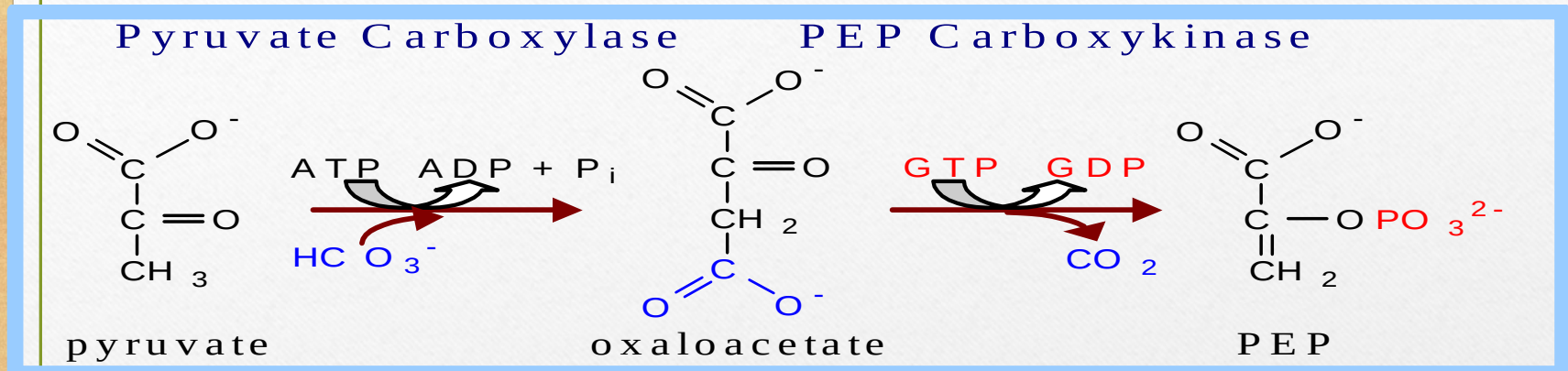
❖ In animals, both pathways occur largely in the cytosol, necessitating their reciprocal and coordinated regulation.

Gluconeogenic pathway



First bypass: Conversion of Pyruvate to phosphoenol pyruvate.

- ❖ This step requires enzymes in both the Cytosol and Mitochondria



- ❖ Pyruvate is transported to the mitochondria where **pyruvate carboxylase** converts pyruvate to **oxaloacetate**.



Note: The reaction requires Biotin as a carrier of activated HCO_3^-

- ❖ Oxaloacetate formed is transported back to the cytosol where it is converted **PEP** by **phosphoenolpyruvate carboxykinase**

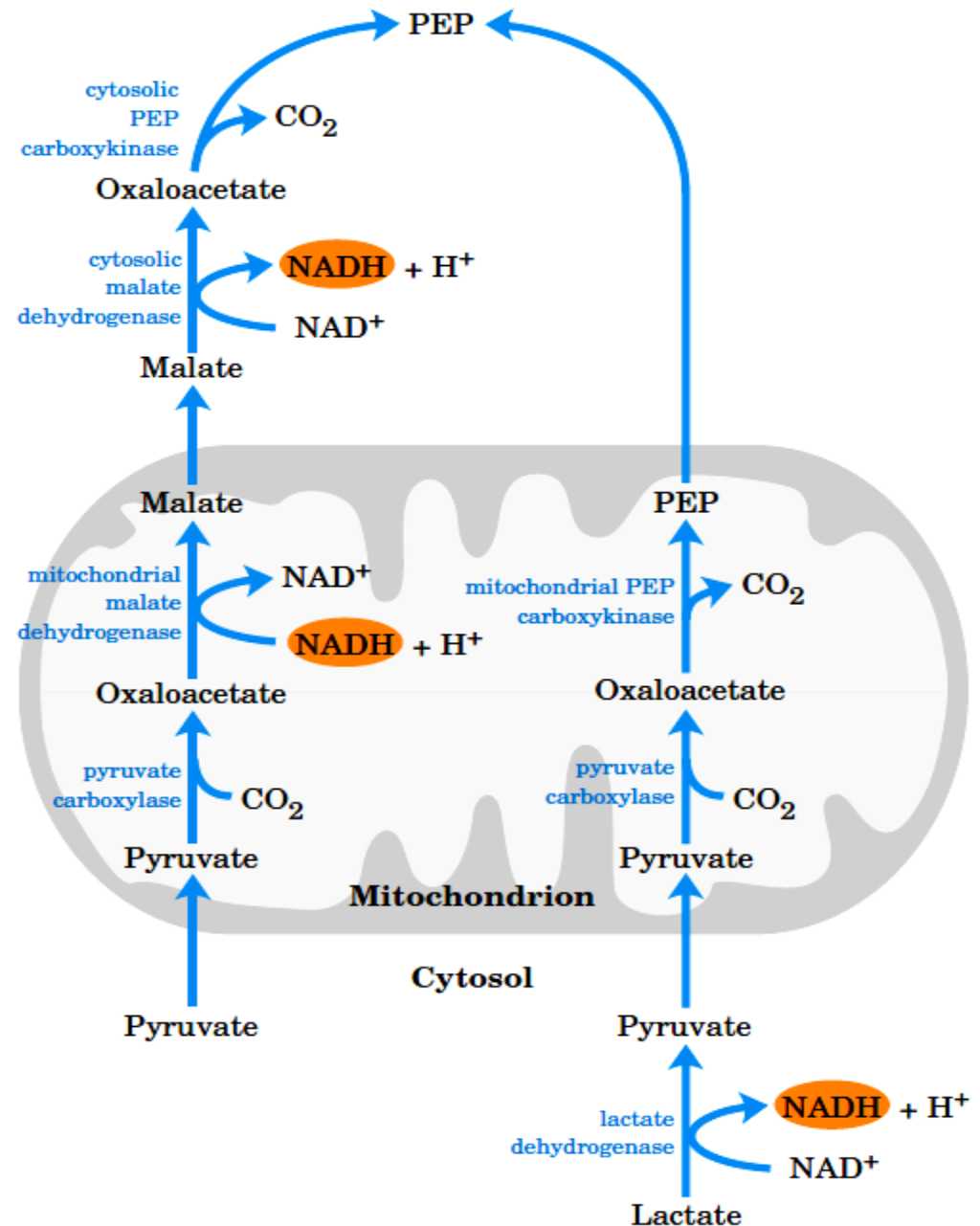


Alternative paths from pyruvate to PEP

- ❖ Pyruvate is transported to the mitochondria where it is converted to Oxaloacetate.
- ❖ Oxaloacetate formed is reduced to malate by mitochondrial malate DH at the expense of NADH.
- ❖ Malate is transported back to the cytosol through a specific transporter where it is reoxidised to oxaloacetate generating cytosolic NADH.
- ❖ Oxaloacetate is then converted to PEP by a **cytosolic PEP carboxykinase**.

OR

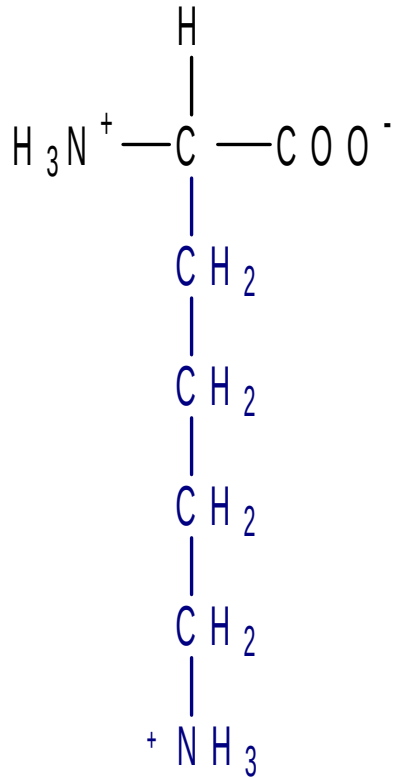
- ❖ Oxaloacetate, can directly be converted to PEP by a mitochondrial **isozyme of PEP carboxykinase**.



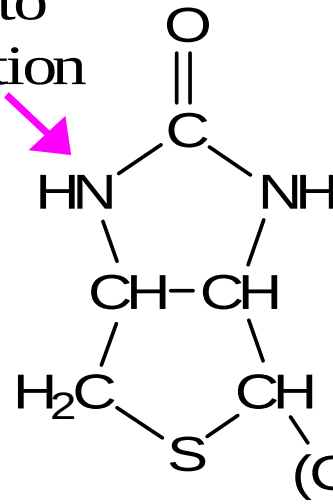
Pyruvate Carboxylase

uses **biotin** as
prosthetic group.

lysine

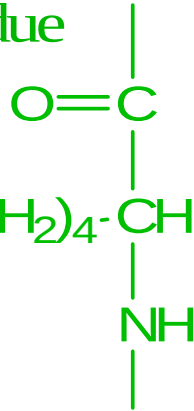


N subject to
carboxylation



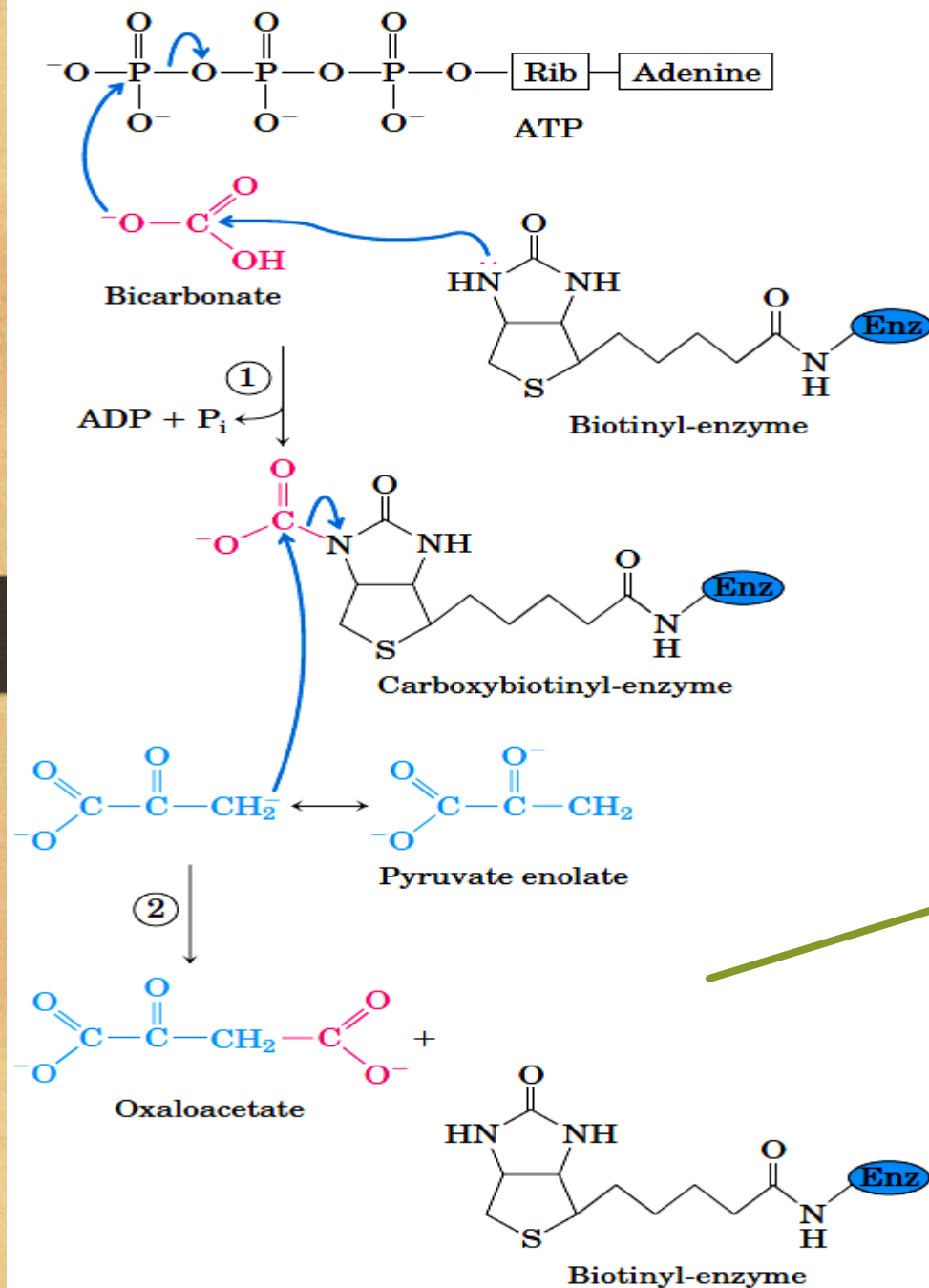
biotin

lysine
residue



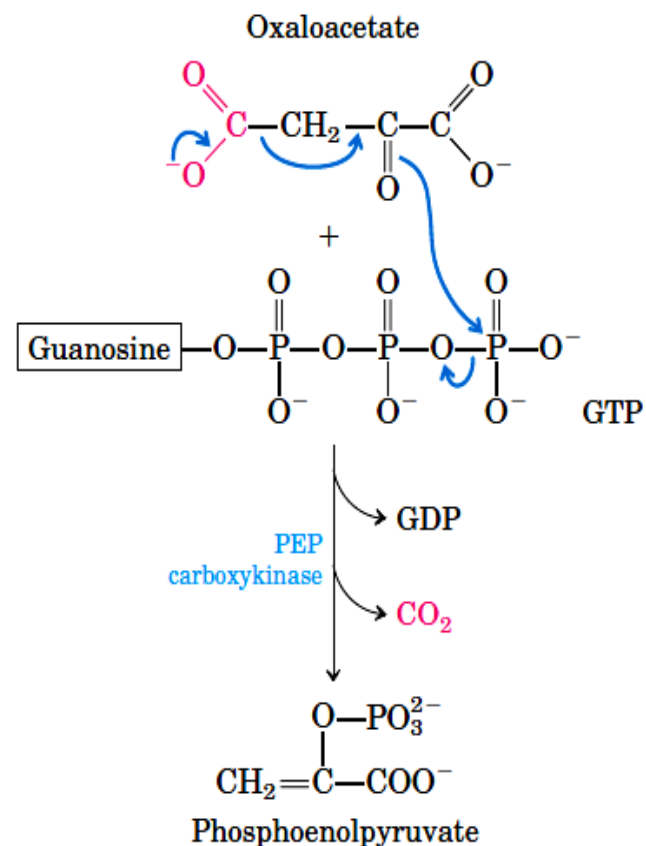
Biotin has a 5-C side chain whose terminal carboxyl is in amide linkage to the ϵ -amino group of an enzyme **lysine**.

The biotin & lysine side chains form a **long swinging arm** that allows the biotin ring to swing back & forth between 2 active sites.



Role of Biotin in the pyruvate carboxylase reaction

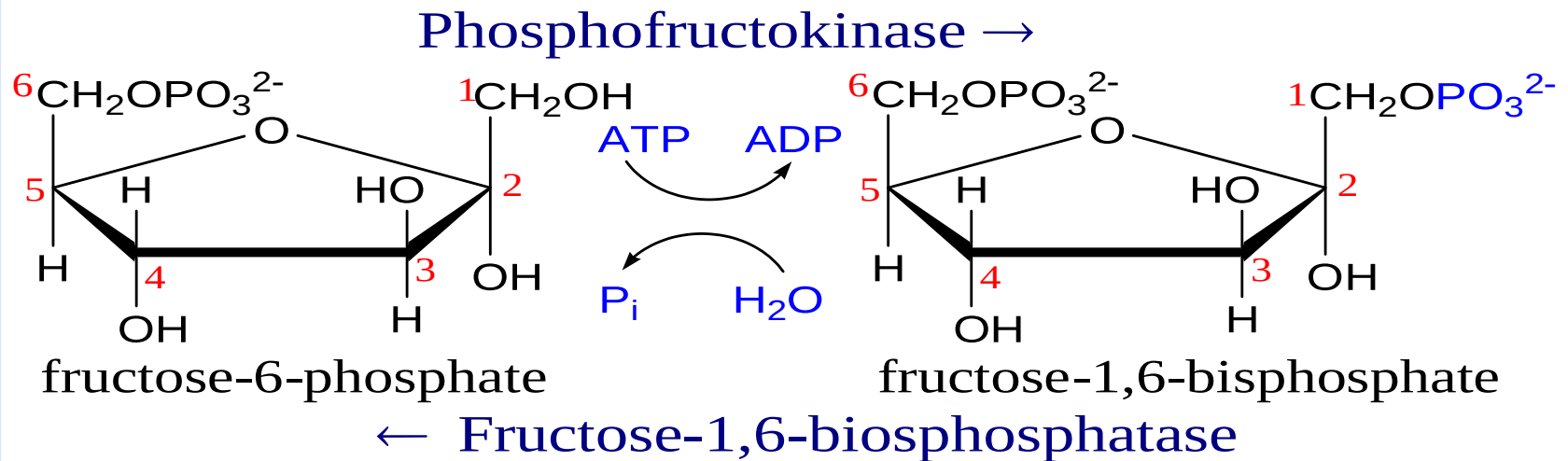
The reaction occurs in two steps at different active sites of pyruvate carboxylase enzyme.



Second bypass: Conversion of F 1,6- Bisphosphate to F 6-

❖ Phosphate

This is catalysed by a different enzyme, Mg^{2+} - dependent fructose 1,6-bisphosphatase (FBPase-1) which promotes the irreversible hydrolysis of C-1 phosphate.



Phosphofructokinase (Glycolysis) catalyzes:

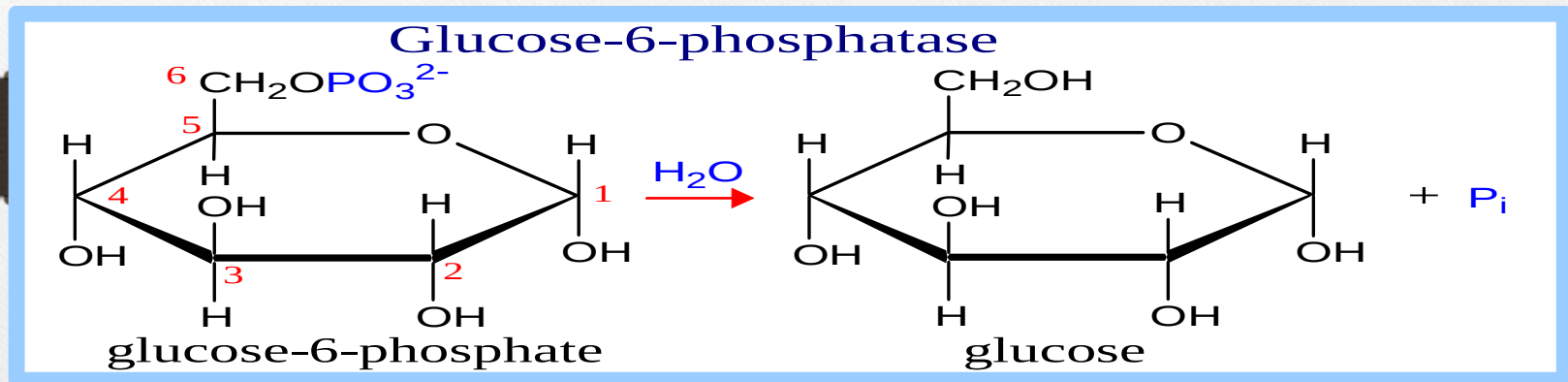


Fructose-1,6-bisphosphatase (Gluconeogenesis) catalyzes:



Third bypass: Conversion of Glc 6- phosphate to

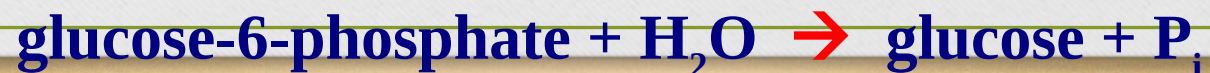
- ❖ glucose. The reaction is catalysed by glucose 6-phosphatase, without synthesis of ATP, it is a simple hydrolysis of a phosphate ester
- ❖ This Mg^{2+} activated enzyme is found on the luminal side of the ER of **Hepatocytes** and **Renal cells**.
- ❖ **Note:** Muscle and brain tissues do not have this enzyme thus cannot carry out gluconeogenesis.



Hexokinase or Glucokinase (Glycolysis) catalyzes:



Glucose-6-Phosphatase (Gluconeogenesis) catalyzes:



The **source of pyruvate and oxaloacetate** for gluconeogenesis during fasting or carbohydrate starvation is mainly **amino acid catabolism**.

Some amino acids are catabolized to pyruvate, oxaloacetate, or precursors of these.

Muscle proteins may break down to supply amino acids. These are transported to liver where they are deaminated and converted to gluconeogenesis inputs.

Glycerol, derived from hydrolysis of triacylglycerols in fat cells, is also a significant input to gluconeogenesis.

Regulation of Gluconeogenesis

Allosteric

1. Pyruvate Carboxylase
(pyruvate $\rightarrow$ oxaloacetate) is
allosterically **activated** by:
acetyl CoA.

[Oxaloacetate] tends to be
limiting for Krebs cycle.

**2. Fructose 1,6
biphosphatase:**

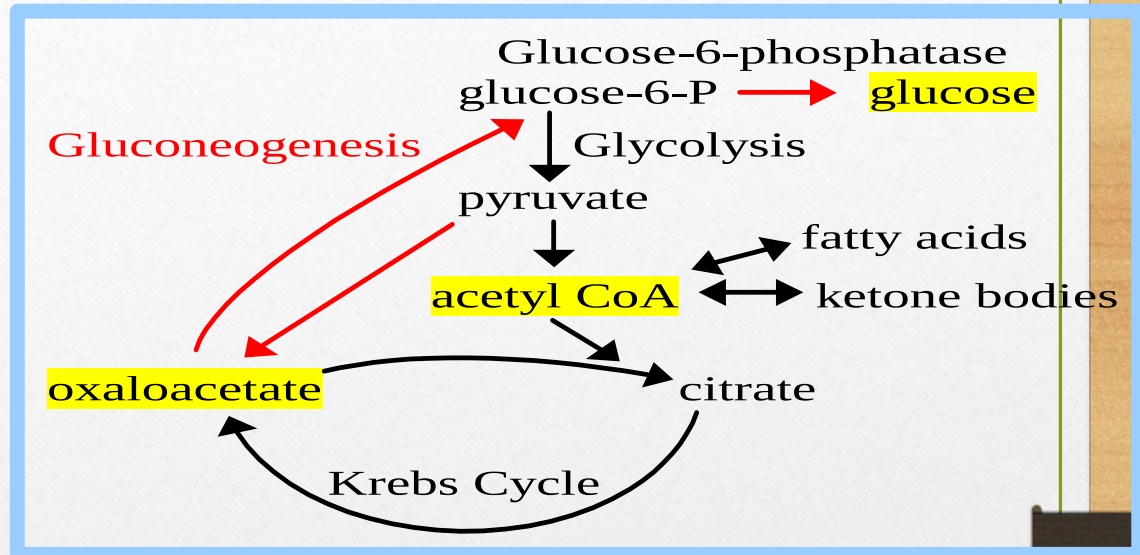
AMP is its allosteric inhibitor.

- ❖ When gluconeogenesis is active in liver, oxaloacetate is diverted to form glucose.
- ❖ Oxaloacetate depletion hinders acetyl CoA entry into Krebs Cycle.
- ❖ The increase in [acetyl CoA] activates Pyruvate Carboxylase to make oxaloacetate.

Hormonal regulation

➤ Insulin decreases the synthesis of key enzymes of gluconeogenesis, thus inhibit gluconeogenesis.

Note: Glycolysis and gluconeogenesis are regulated reciprocally



Note: Avidin, a protein in egg whites, tightly binds biotin.

Excess consumption of raw eggs can cause nutritional deficiency of biotin.

Glycolysis & Gluconeogenesis: both spontaneous.

If both pathways were simultaneously active in a cell, it would constitute a "**futile cycle**" that would waste energy.

Glycolysis:



Gluconeogenesis:



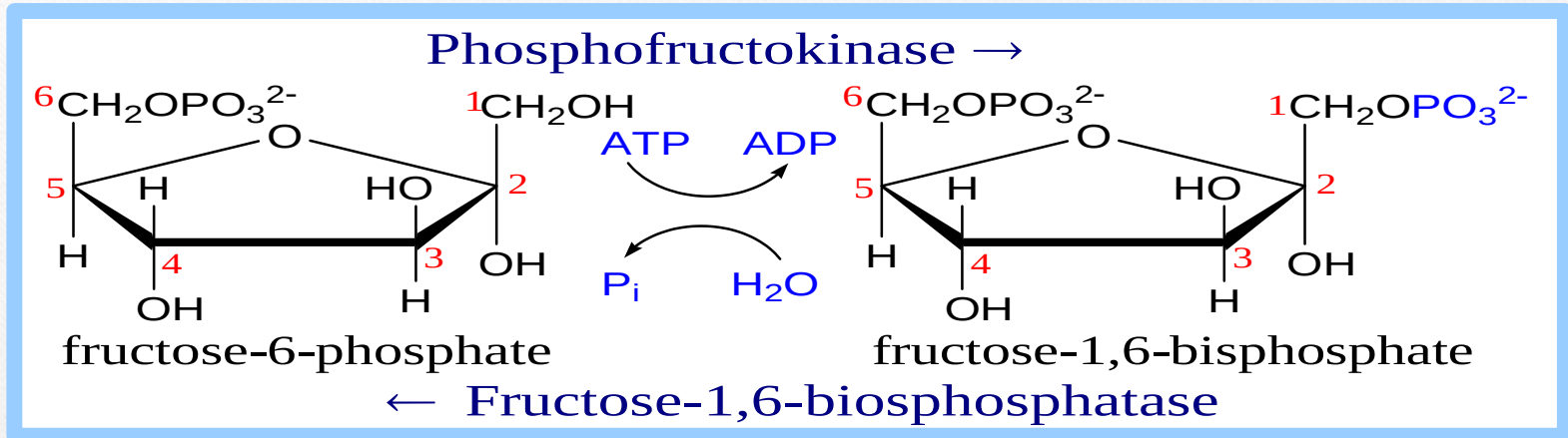
Questions:

1. **Glycolysis** yields **how many ~P** ? **2**
2. **Gluconeogenesis** expends **how many ~P** ? **6**
3. A **futile cycle** of both pathways would waste **how many ~P per cycle** ? **4**

To prevent the waste of a futile cycle, Glycolysis & Gluconeogenesis are **reciprocally regulated**.

Local Control: Includes reciprocal allosteric regulation by **adenine nucleotides**.

- ◆ **Phosphofructokinase** (Glycolysis) is inhibited by ATP and stimulated by AMP.
- ◆ **Fructose-1,6-bisphosphatase** (Gluconeogenesis) is inhibited by AMP.



The **opposite effects of adenine nucleotides** on

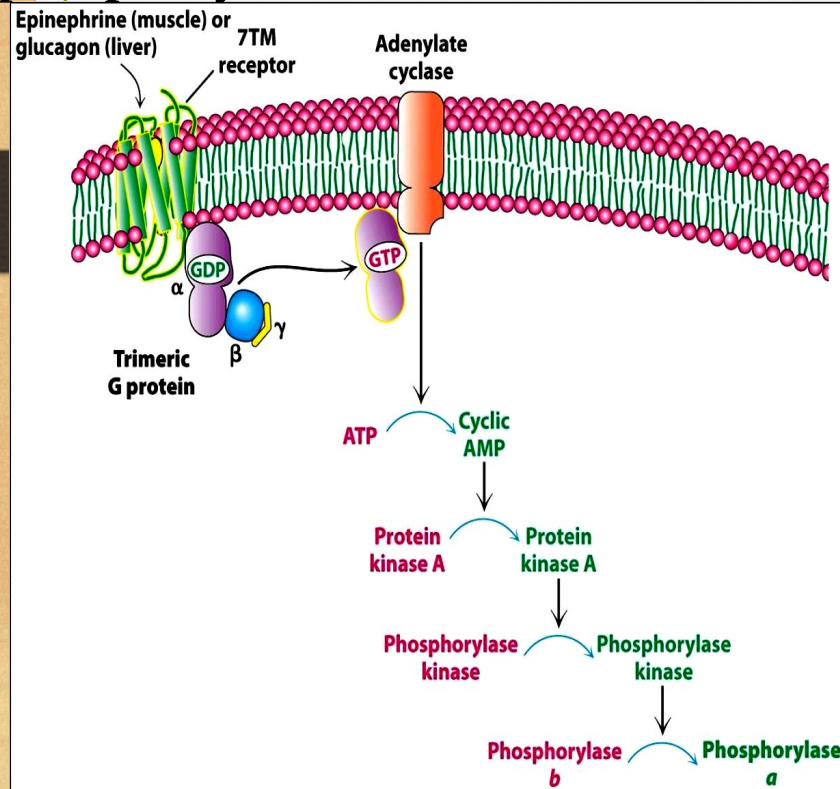
- ◆ **Phosphofructokinase** (Glycolysis)
- ◆ **Fructose-1,6-bisphosphatase** (Gluconeogenesis)

insures that:

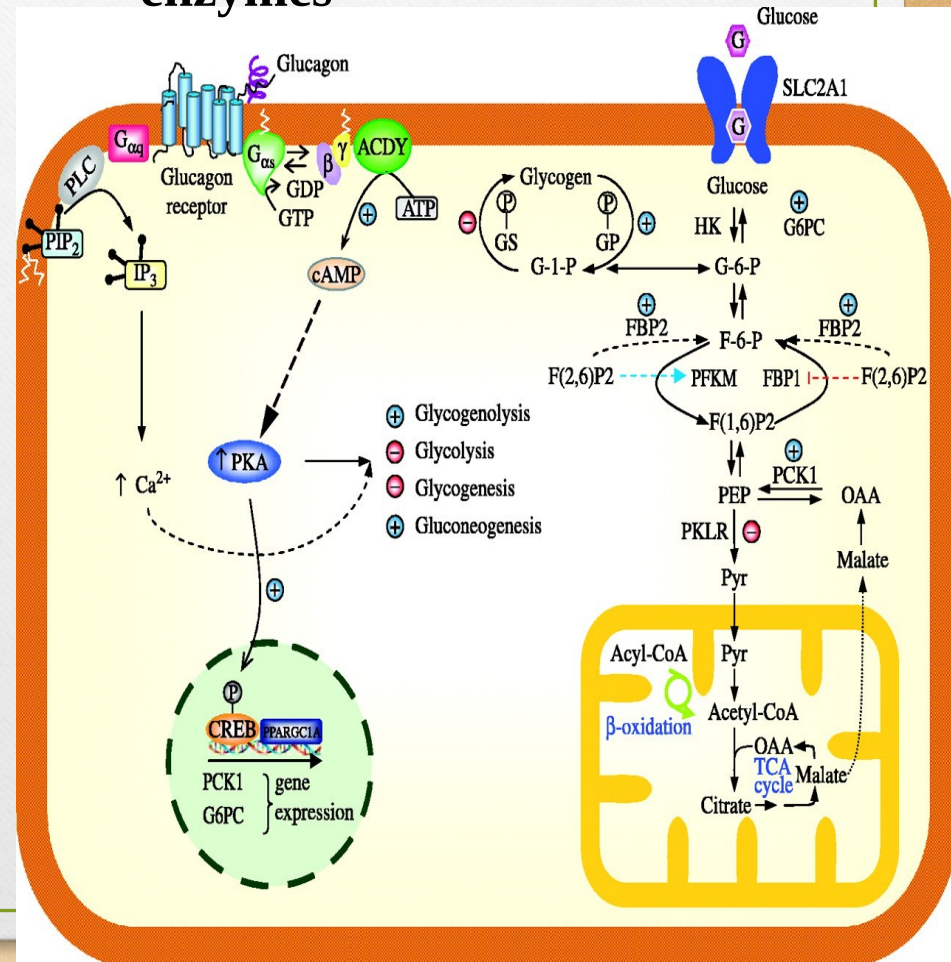
- When cellular ATP is high (AMP would then be low), glucose is not degraded to make ATP.
- When ATP is low (AMP would then be high), the cell does not expend energy in synthesizing glucose.

Global Control: includes reciprocal effects of a **cyclic AMP cascade**, triggered by the hormone **glucagon** when blood glucose is low.

Activation of Glycogen phosphorylase



Activation of other enzymes



Enzymes relevant to these pathways that are **phosphorylated** by Protein Kinase A include:

- ◆ **Pyruvate Kinase**, a glycolysis enzyme that is **inhibited** when phosphorylated.
- ◆ **CREB** (cAMP response element binding protein) which activates, through other factors, transcription of the gene for **PEP Carboxykinase**, leading to **increased gluconeogenesis**.
- ◆ A **bi-functional enzyme (PFK-2)** that makes and degrades an allosteric regulator, **fructose-2,6-bisphosphate** (A potent activator of **PFK-1** and an inhibitor of **Fructose 1,6 bisphosphatase**).

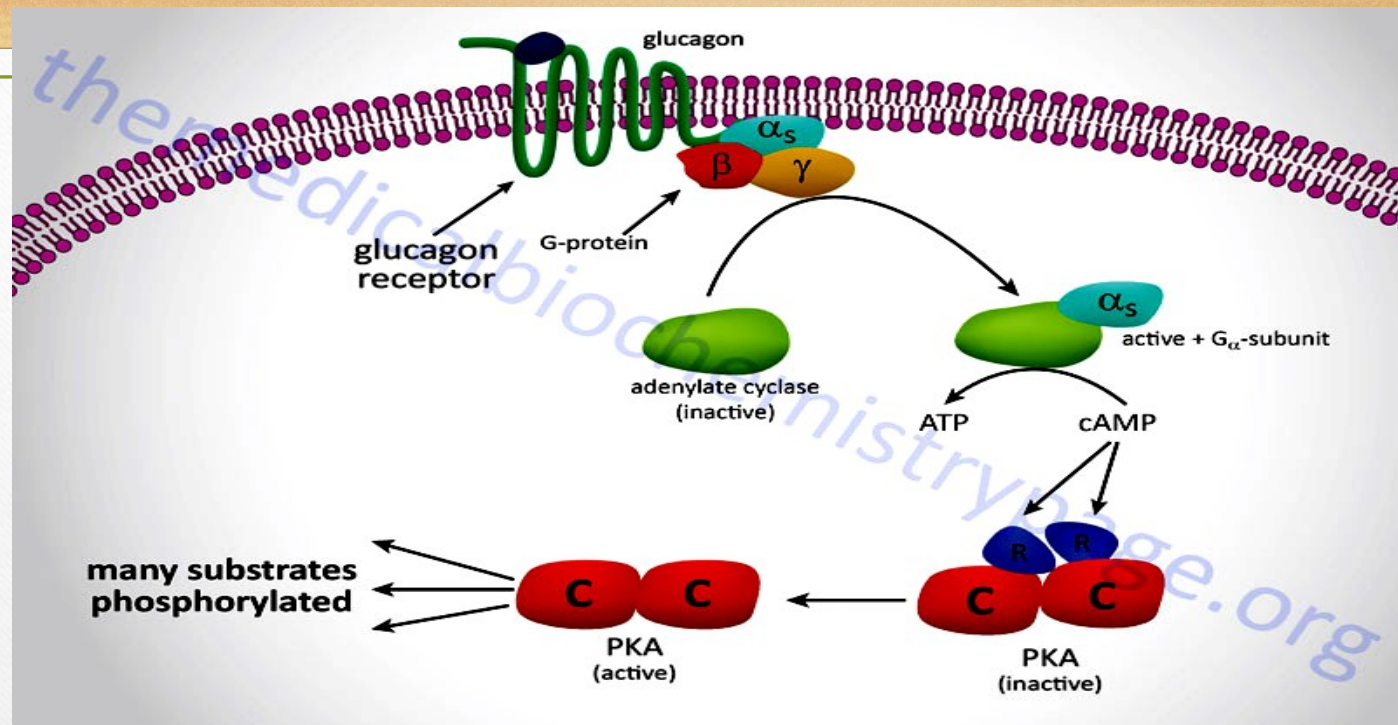
Reciprocal regulation by fructose-2,6-bisphosphate:

- ◆ **Fructose-2,6-bisphosphate stimulates Glycolysis.**

Fructose-2,6-bisphosphate allosterically **activates** the Glycolysis enzyme **Phosphofructokinase 1**.

Fructose-2,6-bisphosphate also **activates transcription** of the gene for **Glucokinase**, the liver variant of Hexokinase that phosphorylates glucose to glucose-6-phosphate, the input to Glycolysis.

- ◆ **Fructose-2,6-bisphosphate allosterically inhibits the gluconeogenesis enzyme Fructose-1,6-bisphosphatase.**



Phosphorylation of enzymes & regulatory proteins in liver by Protein Kinase A (cAMP Dependent Protein Kinase) results in

- ◆ **inhibition of glycolysis**
- ◆ **stimulation of gluconeogenesis,**

making glucose available for release to the blood.

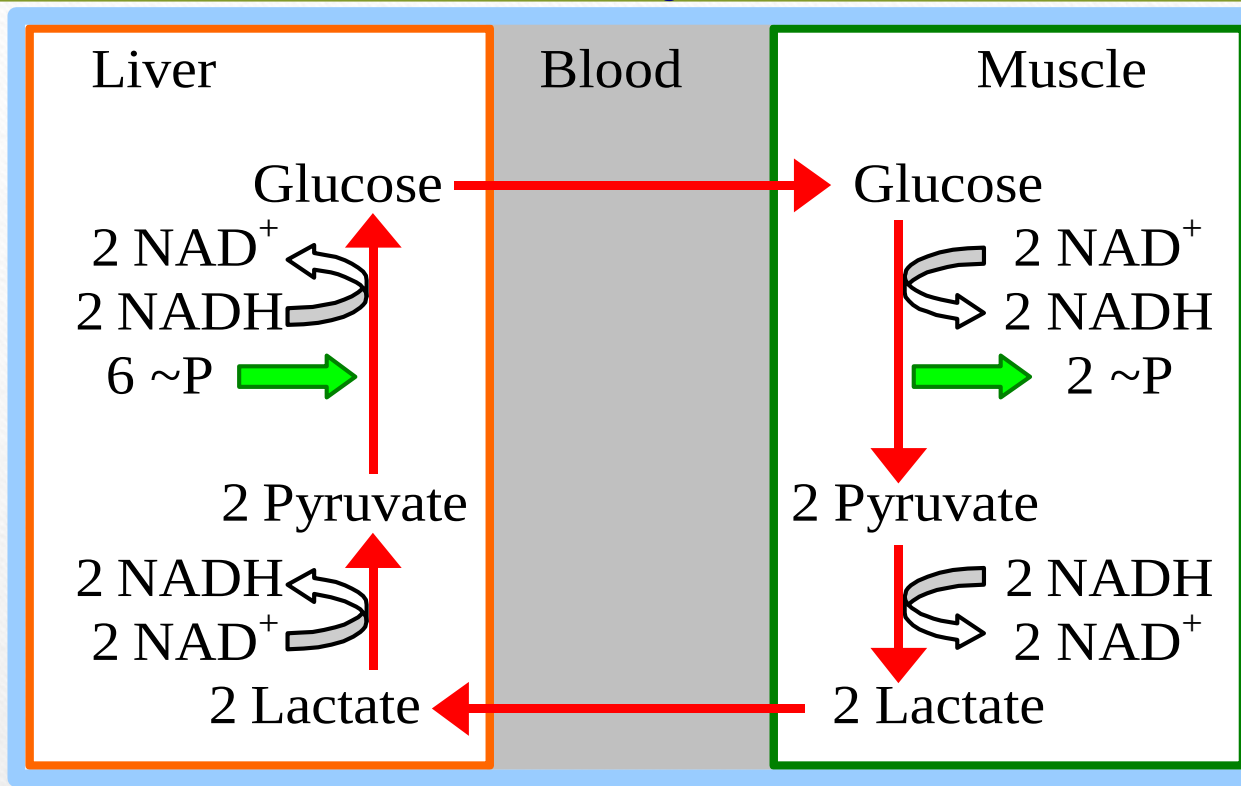
The Cori Cycle

The **Cori Cycle** operates during **exercise**.

For a brief burst of **ATP** utilization, muscle cells utilize **~P** stored as **phosphocreatine**.

Once phosphocreatine is exhausted, ATP is provided mainly by **Glycolysis**, with the input coming from **glycogen** breakdown and from **glucose uptake** from the blood.

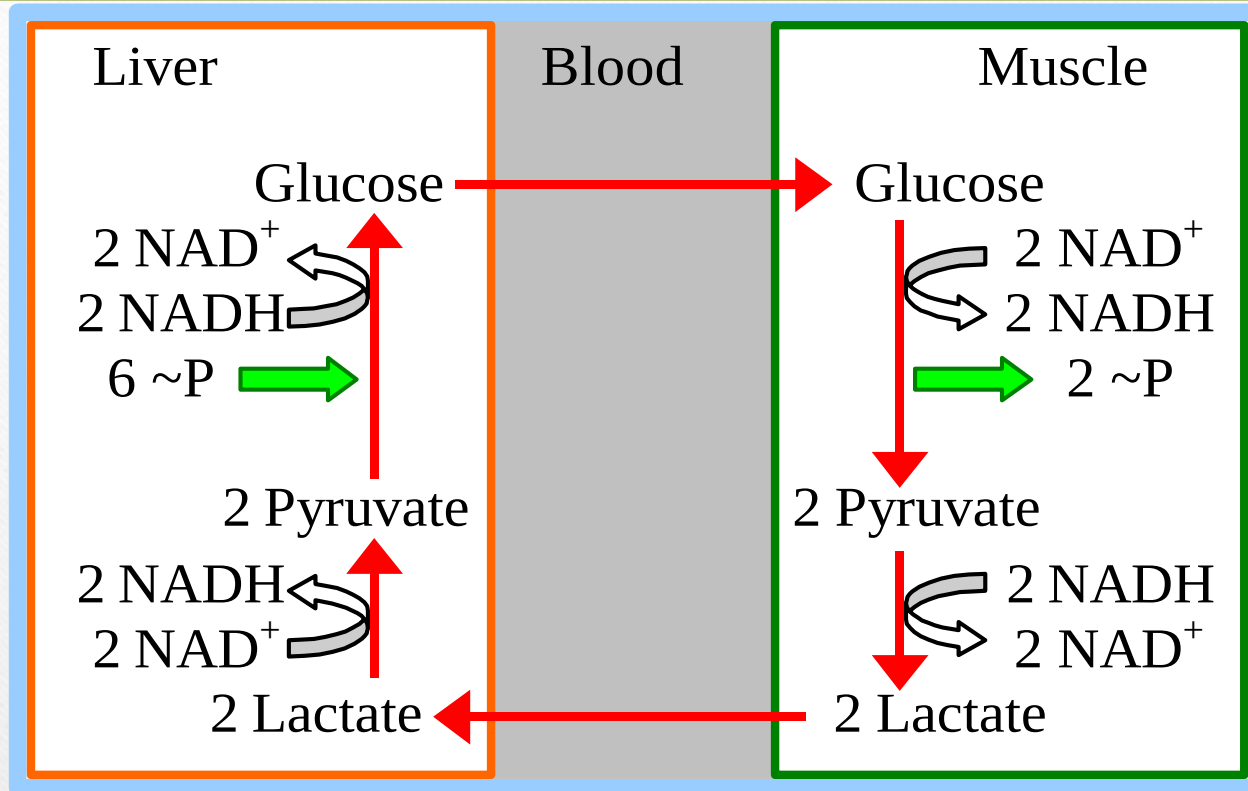
Cori Cycle



Lactate produced from pyruvate passes via the blood to the **liver**, where it may be converted to **glucose**.

The glucose may travel back to the **muscle** to fuel **Glycolysis**.

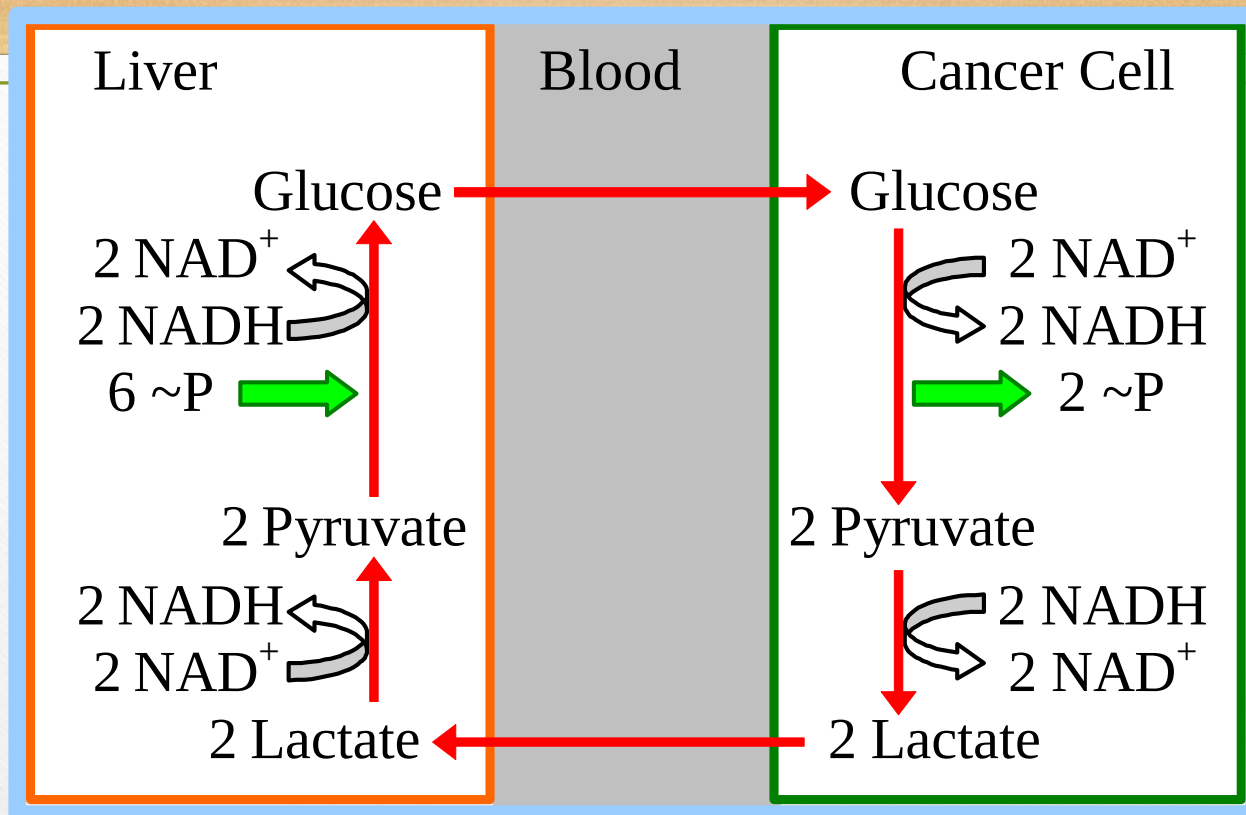
Cori Cycle



- ❖ The Cori cycle **costs 6 ~P** in liver for every **2 ~P** made available in muscle. The **net cost is 4 ~P**.
- ❖ Although costly in ~P bonds, the Cori Cycle allows the organism to accommodate to large fluctuations in energy needs of skeletal muscle between rest and exercise.

The equivalent of the **Cori Cycle** also operates during **cancer**.

If blood vessel development does not keep pace with growth of a solid tumor, **decreased O_2 concentration** within the tumor leads to activation of signal processes that result in a shift to **anaerobic metabolism**.



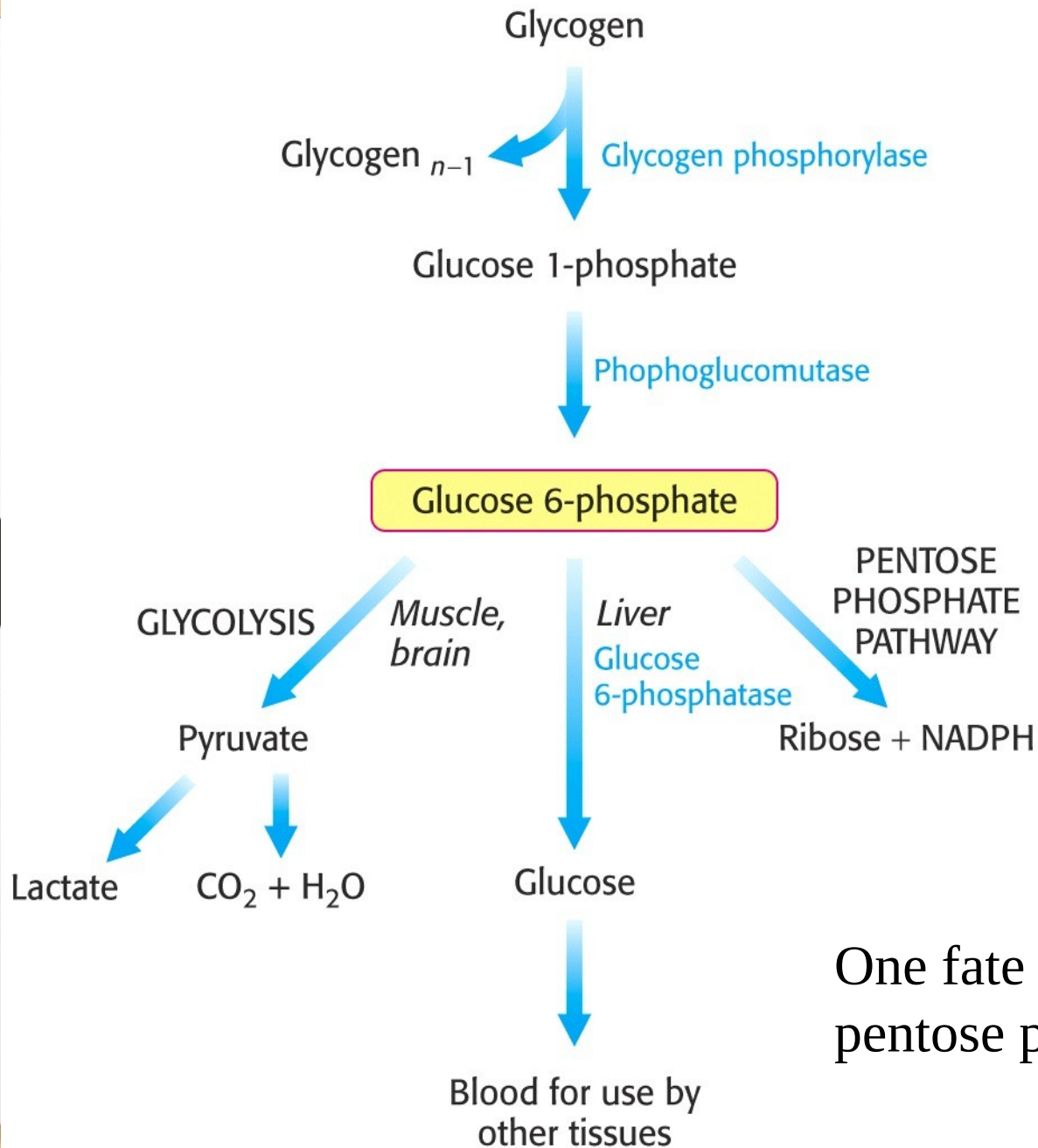
Energy dissipation by the **Cori Cycle**, which expends **6 ~P** in liver for every **2 ~P** produced via Glycolysis for utilization within the tumor, is thought to contribute to the **weight loss** that typically occurs in late-stage cancer even when food intake remains normal.

Pentose Phosphate pathway

Also known as:

- Pentose shunt
- Hexose monophosphate shunt
- Phosphogluconate pathway

❖ It occurs in the cytosol.



One fate of G6P is the pentose pathway.

What does the pentose phosphate pathway achieve?

- The pathway yields reducing potential in the form of NADPH to be used in anabolic reactions requiring electrons.
- The pathway yields ribose 5-phosphate.
 - Nucleotide biosynthesis leading to:
 - DNA
 - RNA
 - Various cofactors (CoA, FAD, NAD⁺/NADP⁺).

TABLE 20.2 **Pathways requiring NADPH**

Synthesis

Fatty acid biosynthesis

Cholesterol biosynthesis

Neurotransmitter biosynthesis

Nucleotide biosynthesis

Detoxification

Reduction of oxidized glutathione

Cytochrome P450 monooxygenases

Tissues with active pentose phosphate pathway

- ❖ Pentose phosphate pathway is especially prominent in tissues actively synthesizing fatty acids and steroids.
- ✓ These tissues use NADPH to reduce the double bonds and carbonyl groups of intermediates in the synthetic process.
- ❖ PPP is also prominent in growing and regenerating tissues and in tumors.

TABLE 20.4 Tissues with active pentose phosphate pathways

Tissue	Function
Adrenal gland	Steroid synthesis
Liver	Fatty acid and cholesterol synthesis
Testes	Steroid synthesis
Adipose tissue	Fatty acid synthesis
Ovary	Steroid synthesis
Mammary gland	Fatty acid synthesis
Red blood cells	Maintenance of reduced glutathione

Reactions of the Pentose

Phosphate Pathway

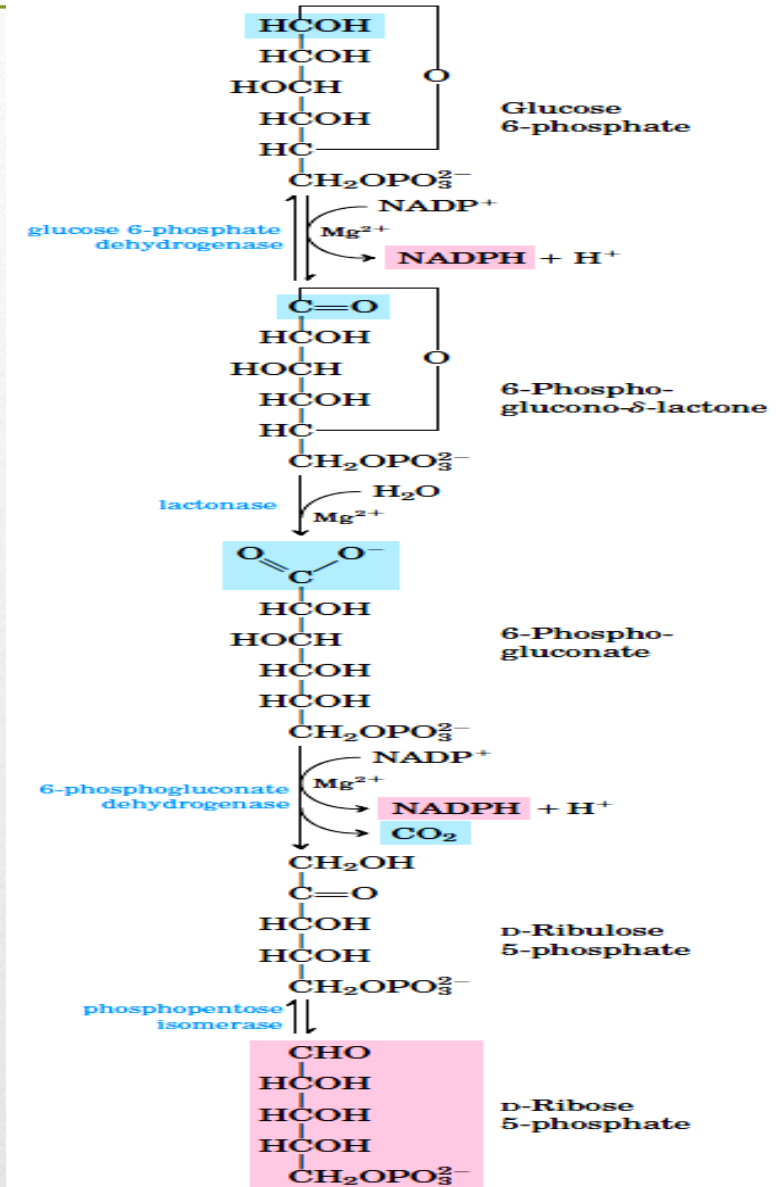
The pentose pathway can be divided into two phases:

- ✓ Oxidative phase.
- ✓ Non-oxidative phase.

Oxidative reactions of the pentose phosphate pathway

Step

- 1s Dehydrogenation of glucose 6-phosphate by glucose 6-phosphate DH
2. The **lactone** is hydrolysed to the free acid; **6-phosphogluconate** by a specific **lactonase**
3. Dehydrogenation and decarboxylation of **6-phosphogluconate** by **6-phosphogluconate dehydrogenase** to **ribulose 5-phosphate**.
4. **Phosphopentose isomerase** converts **ribulose 5-phosphate** to its aldolase isomer **ribose 5-phosphate**.



Non-Oxidative reactions of the pentose phosphate pathway

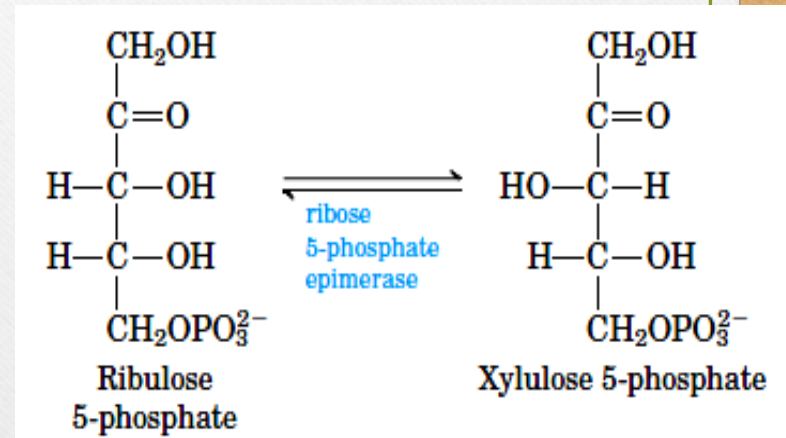
- ❖ Occurs in tissues that require primarily NADPH.
- ❖ The pentose phosphates produced in the oxidative phase of the pathway are recycled into glucose 6-phosphate.

Step

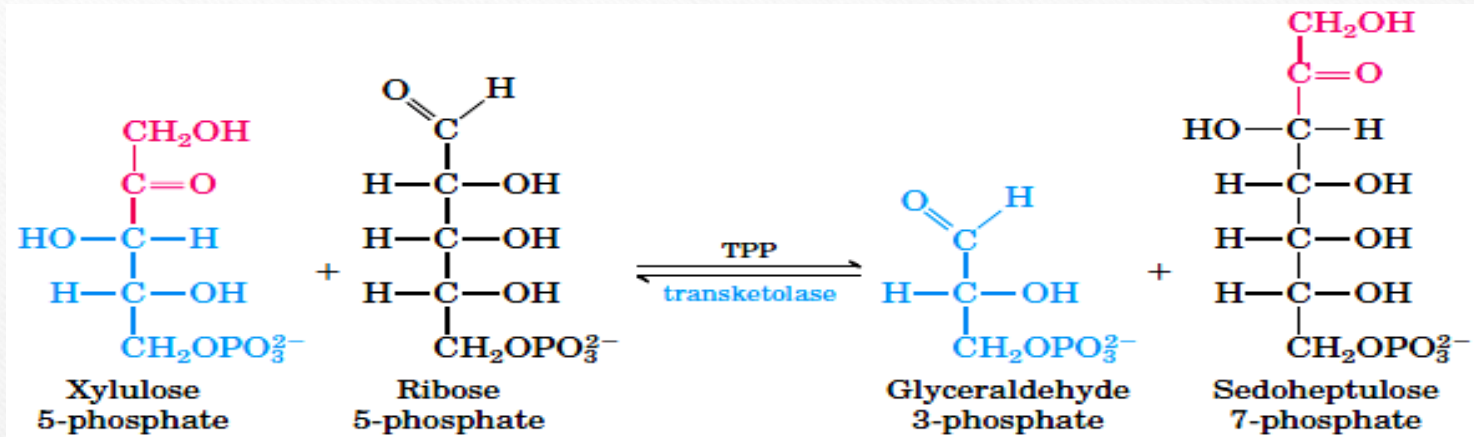
1^s. In this non-oxidative phase, **ribulose 5-phosphate** is first epimerized to **xylulose 5-phosphate**.

In a series of rearrangements of the carbon skeletons **six five-carbon sugar phosphates** are converted to **five six-carbon sugar phosphates**, completing the cycle, allowing continued production of NADPH.

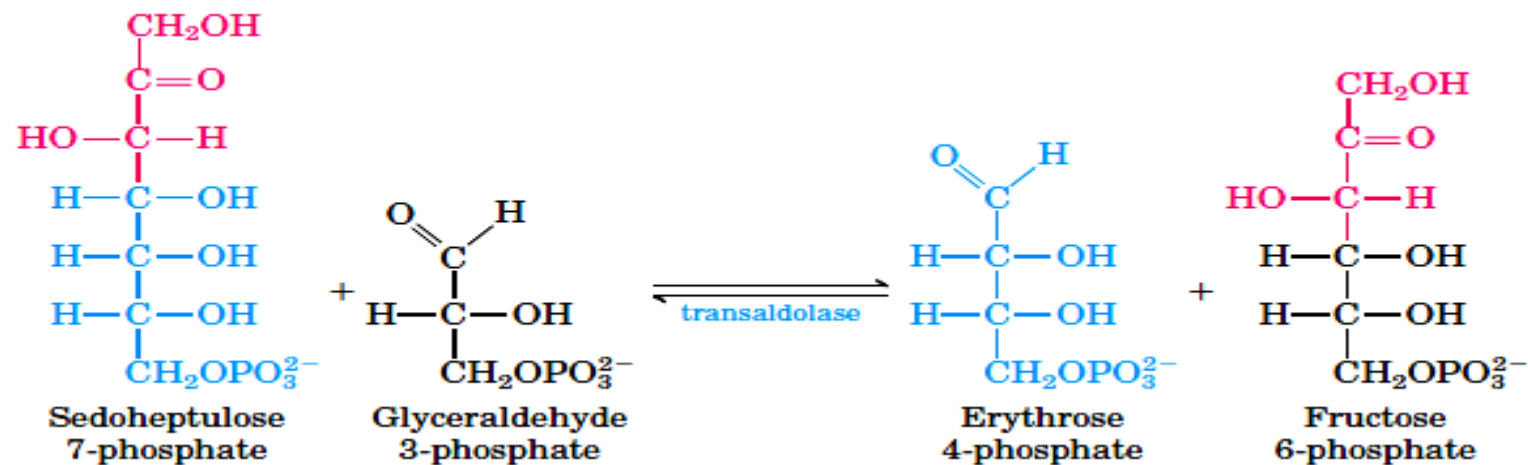
Two enzymes unique to the pentose phosphate pathway act in these interconversions of sugars: **transketolase** and **transaldolase**.



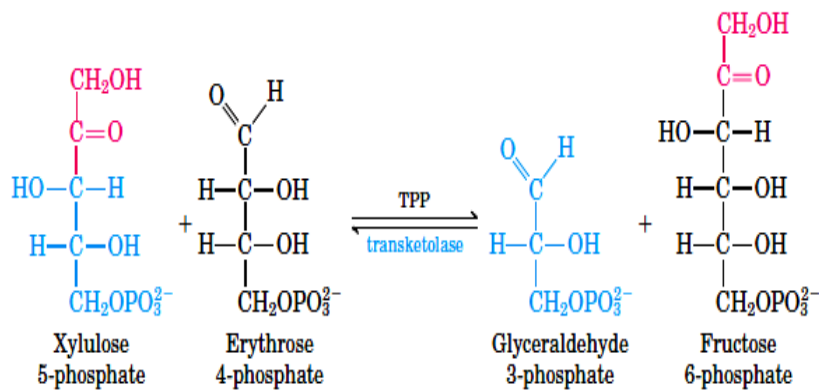
2. **Transketolase** transfers C-1 and C-2 of xylulose 5-phosphate to ribose 5-phosphate, forming the seven-carbon product **sedoheptulose 7-phosphate** and **glyceraldehyde 3-phosphate**.



3. **Transaldolase** transfers a three-carbon fragment from sedoheptulose 7-phosphate to glyceraldehyde 3-phosphate, forming **fructose 6-phosphate** and (the tetrose) **erythrose 4-phosphate**.

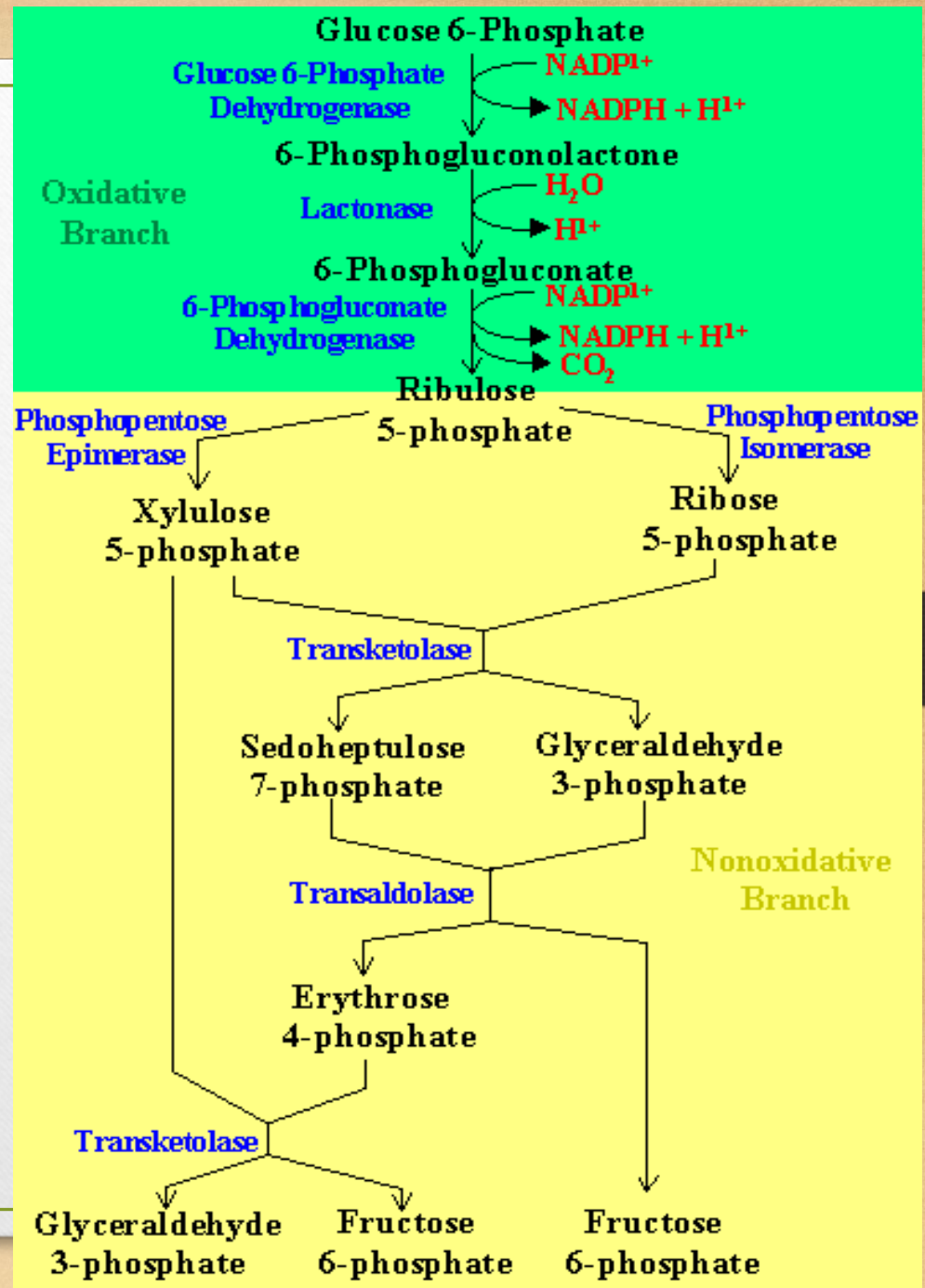


4. **transketolase** acts again, forming **fructose 6-phosphate** and **glyceraldehyde 3-phosphate** from **erythrose 4-phosphate** and **xylulose 5-phosphate**.

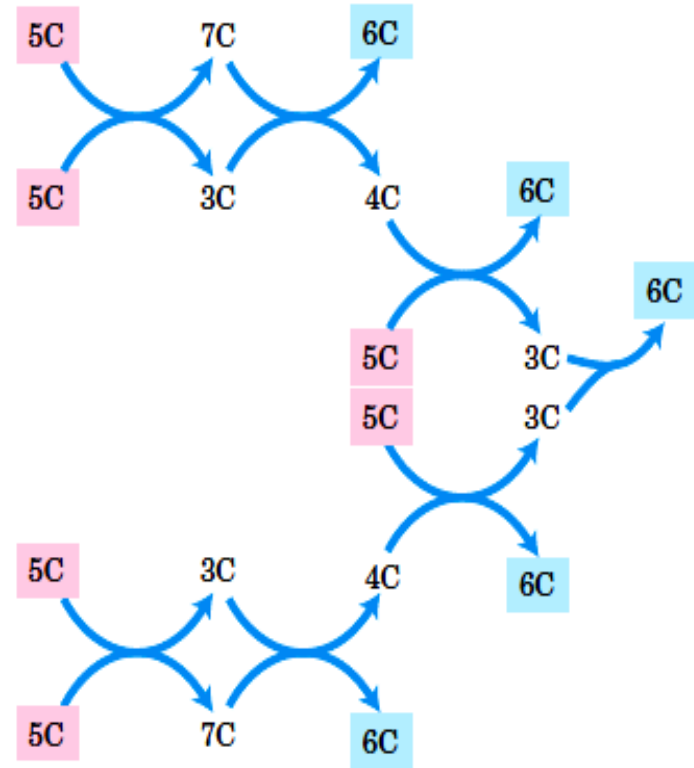
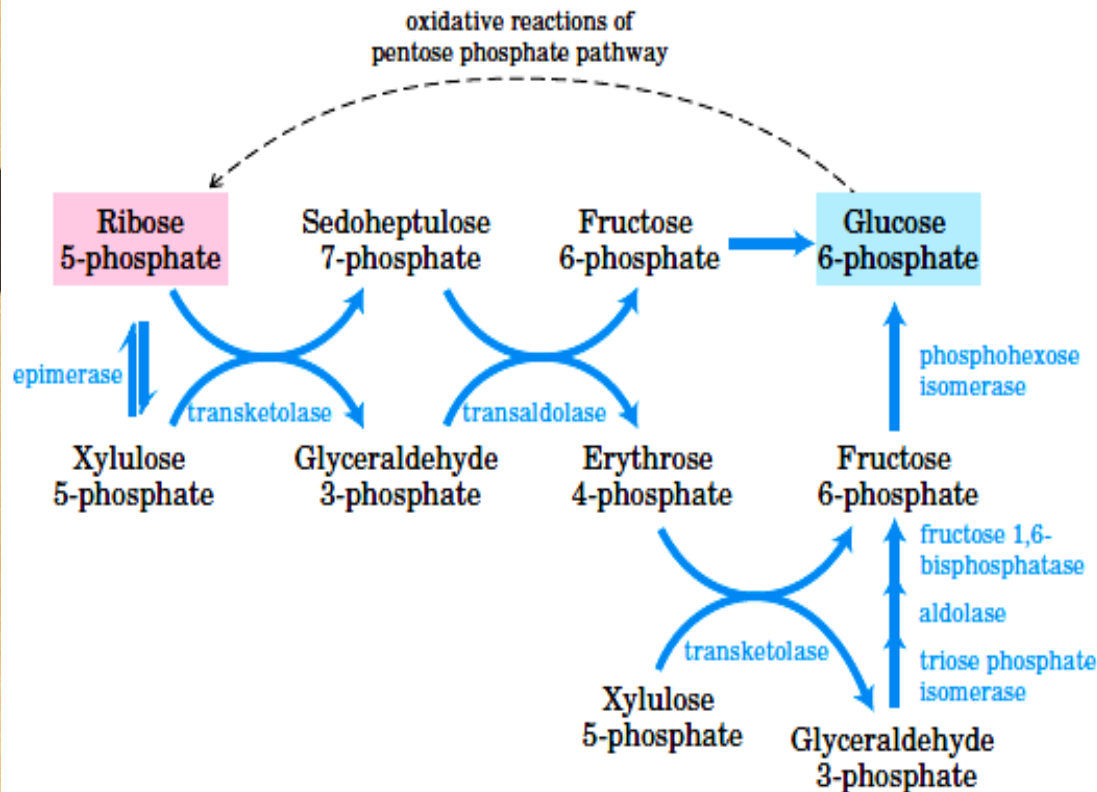


Note:

- ❖ The first two steps in the **oxidative pentose phosphate pathway** are essentially **irreversible** in the cell.
- ❖ The reactions of the **non-oxidative** part are however readily **reversible** and thus provide a means of converting **hexose phosphates** to **pentose phosphates**.



- ❖ Two molecules of **glyceraldehyde 3-phosphate** formed by two iterations of these reactions can be converted to a molecule of **fructose 1,6-bisphosphate** as in gluconeogenesis.
- ❖ In a complete cycle: **six pentose phosphates** are converted to **five hexose phosphates**.



Regulation of the Pentose Pathway

Oxidative Phase

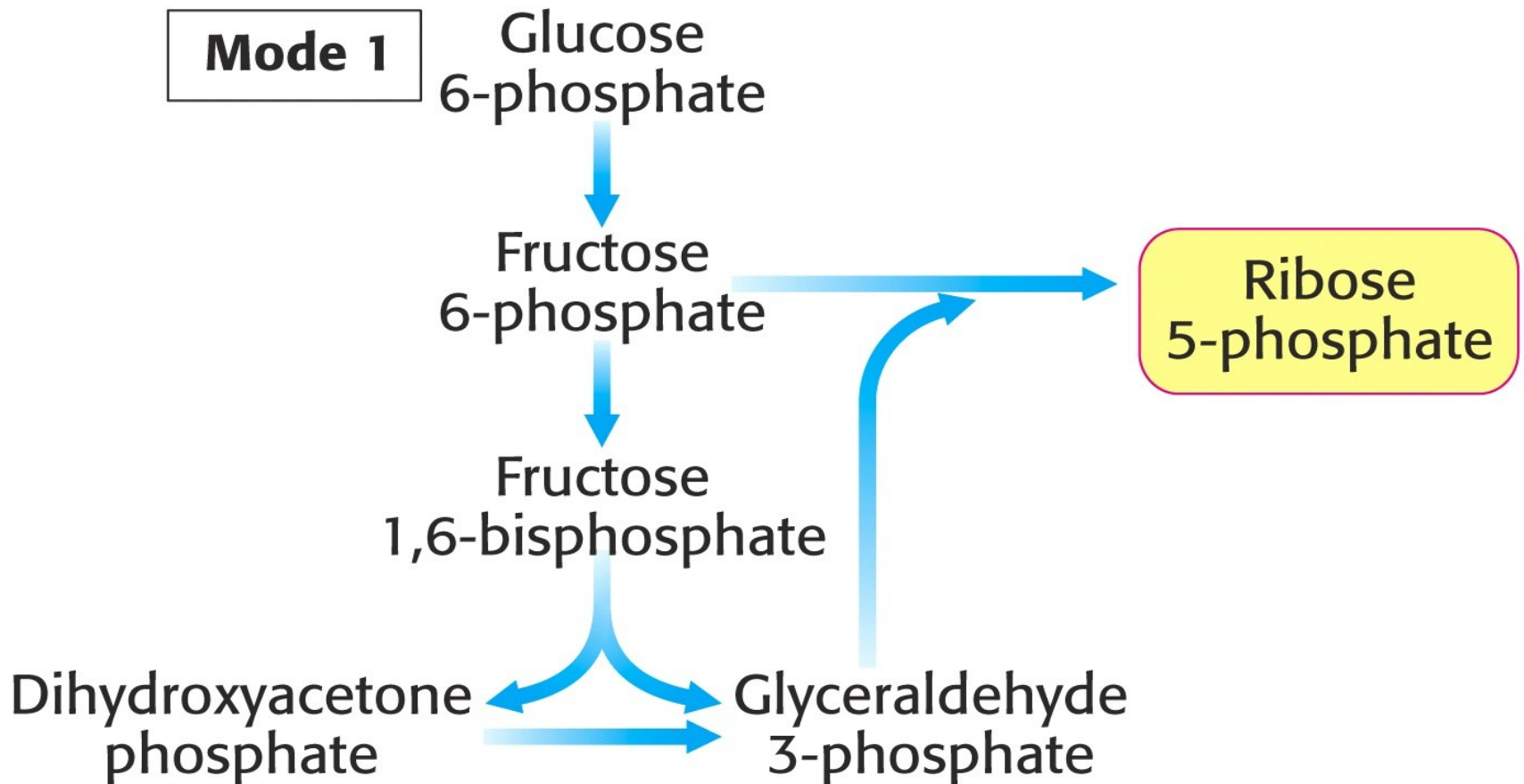
- Glucose 6-phosphate DH is the regulatory enzyme.
- NADPH is a potent competitive inhibitor of the enzyme.
- Usually when the ratio $\text{NADPH}/\text{NADP}^+$ is high so the enzyme is inhibited.
- But, with increased demand for NADPH, the ratio decreases and enzyme activity is stimulated.

Non-oxidative Phase

- The reactions of the non-oxidative portion of the pentose pathway are readily reversible.
- The concentrations of the products and reactants can shift depending on the metabolic needs of a particular cell or tissue.

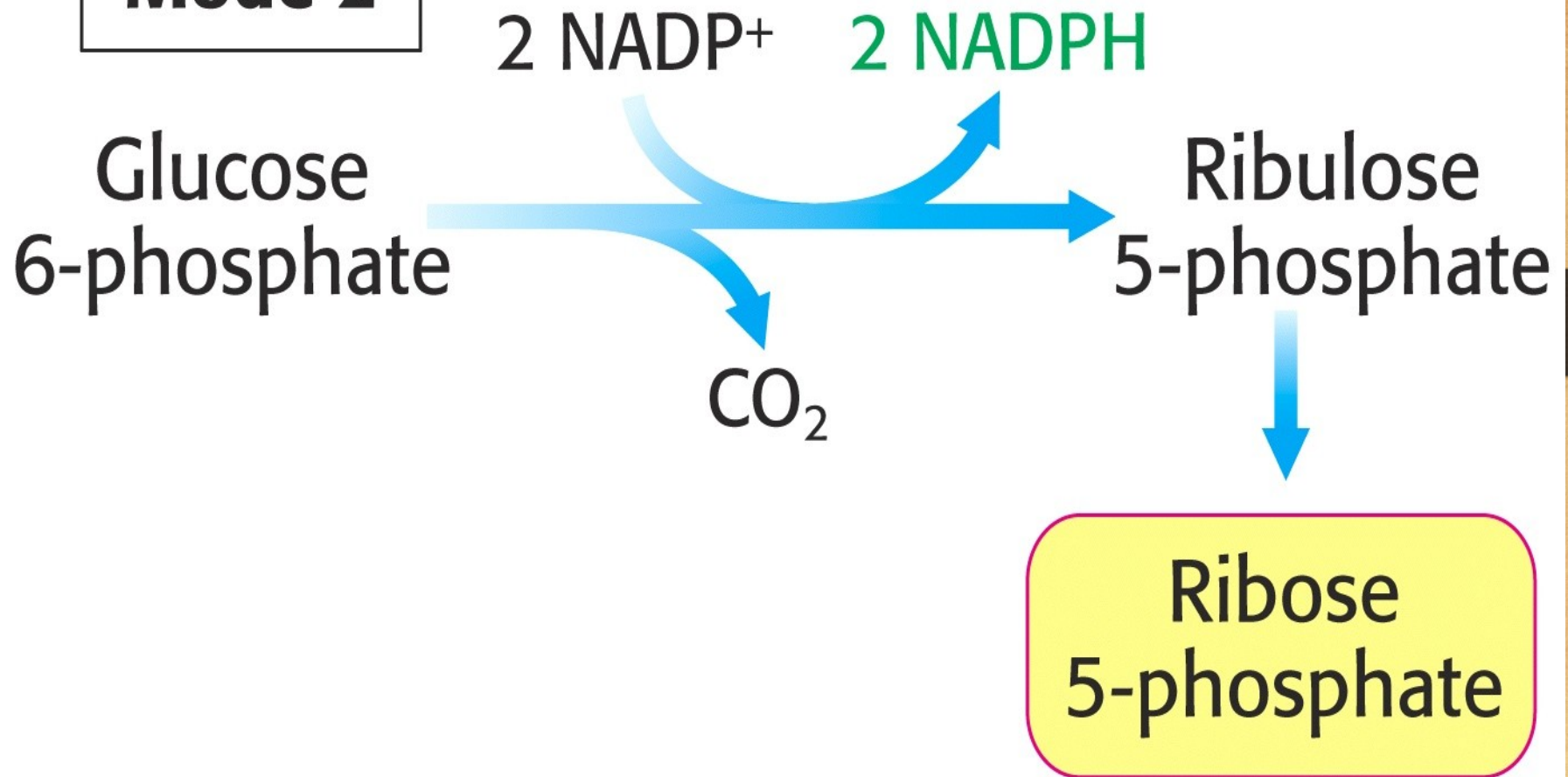
Different scenarios under which the Pentose phosphate pathway is controlled.

1. Rapidly dividing cells require more ribose 5-phosphate than NADPH.



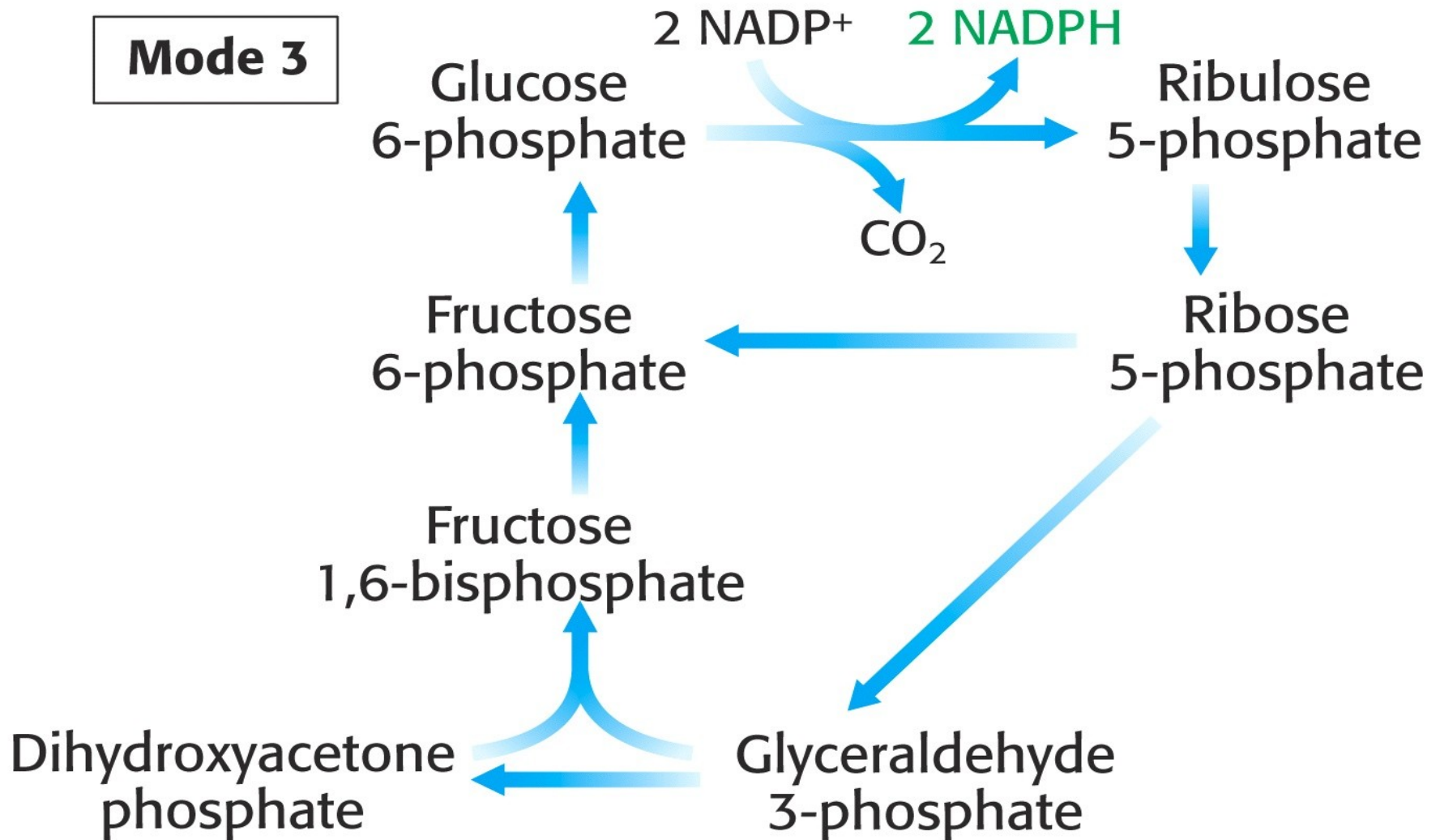
2. The need for NADPH and ribose 5-phosphate is balanced.

Mode 2



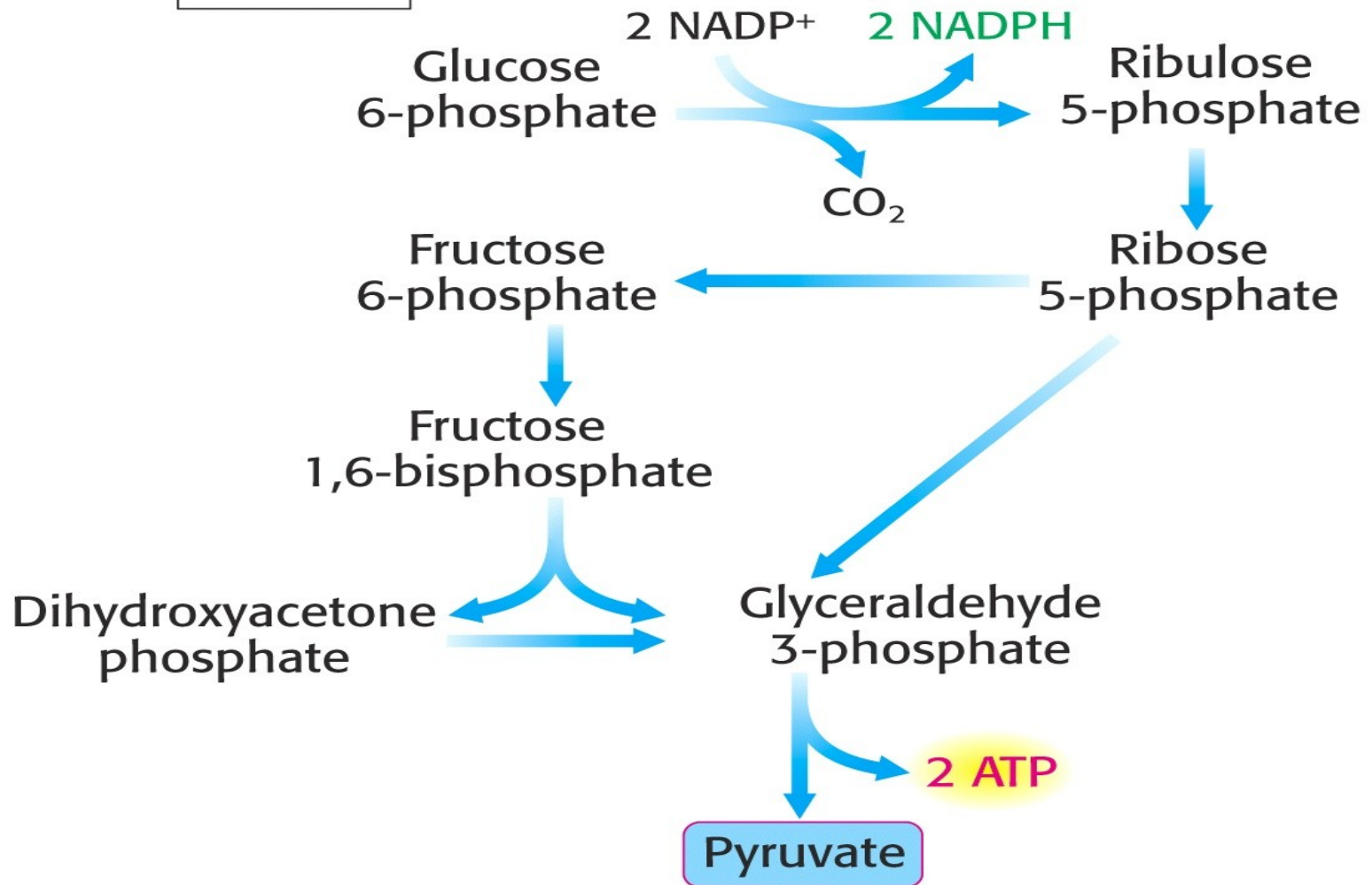
3. More NADPH is needed than ribose 5-phosphate; Fatty acid synthesis in adipose cells.

Mode 3



4. The cell needs both NADPH and ATP

Mode 4



Biological importance of HMP

1. **shunt**

NADPH produced in the shunt is used for **biosynthesis** of several important compounds in various organs.

(a) In the liver, NADPH is used for fatty acid synthesis, cholesterol synthesis, bile acid synthesis, glutamate synthesis and cytochrome P450-hydroxylase system.

(b) In the adrenal cortex and gonads, NADPH is used for cholesterol and hormone synthesis.

(c) In the adipose tissue, NADPH is used for fatty acid synthesis.

(d) NADPH is used for formation of deoxy ribonucleotides and pyrimidine nucleotides.

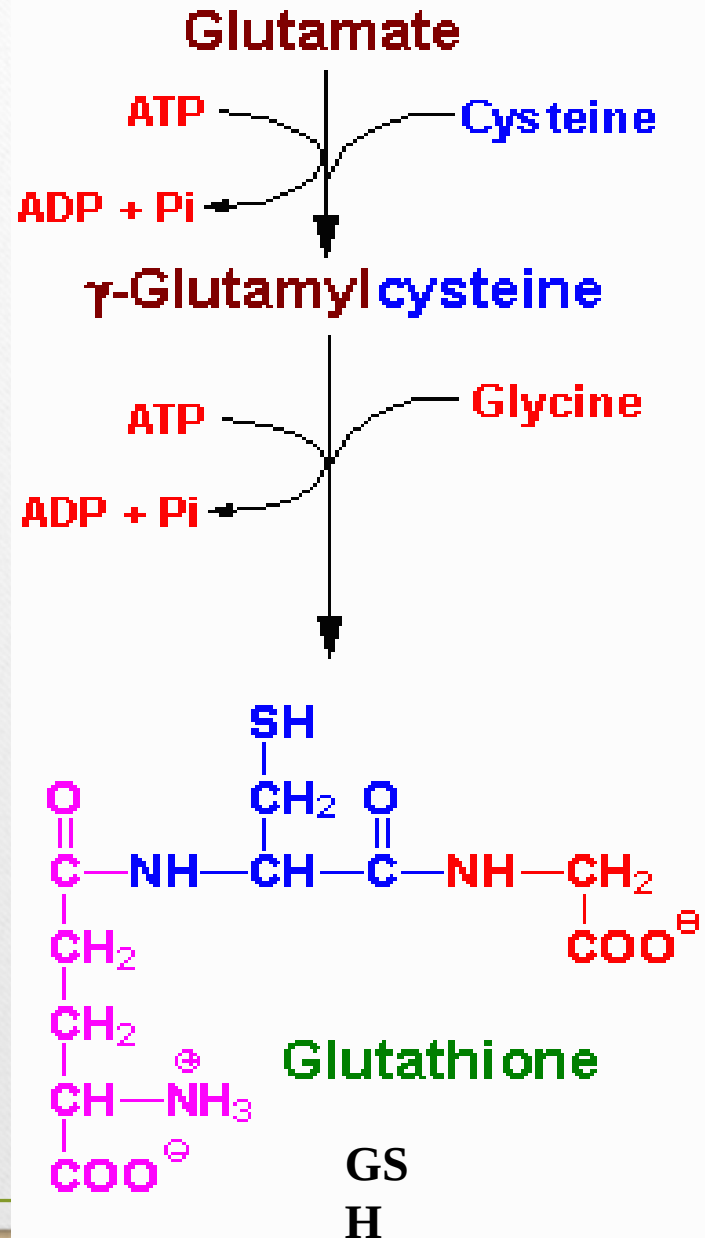
2. In RBC, NADPH produced is used for the formation of **reduced glutathione (GHS)** from **oxidized glutathione (GSSG)**, catalyzed by **Glutathione reductase**. **Reduced glutathione** is required for removal of H_2O_2 in RBCs by the enzyme glutathione peroxidase for maintenance of erythrocyte proteins.

- What is glutathione?
- Why is it important?
- How is it related to NADPH?

What is glutathione?

Glutathione is a tripeptide composed of glutamate, cysteine, glycine.

Reduced glutathione (GSH) maintains the normal reduced state of the cell.



Why is it important?

- The sulfhydryl of GSH is used to reduce peroxides (ROS) formed during oxygen transport.
 - Reactive oxygen species (ROS) damage macromolecules (DNA, RNA, and protein) and ultimately lead to cell death.
- The resulting oxidized form of GSH, two molecules of glutathione are linked by a disulfide bridge (GSSG).

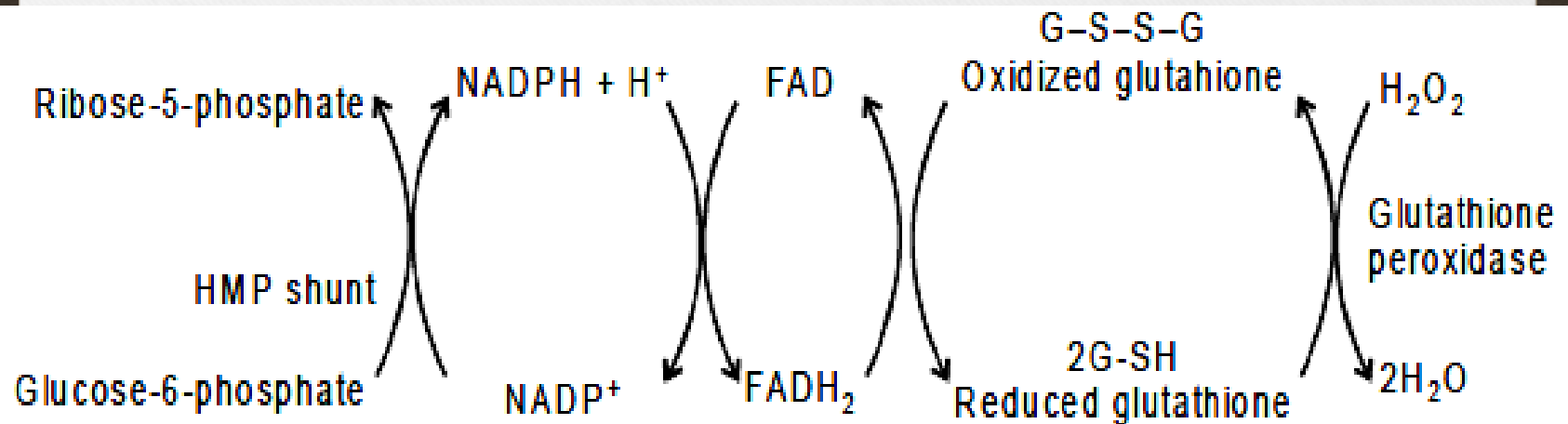
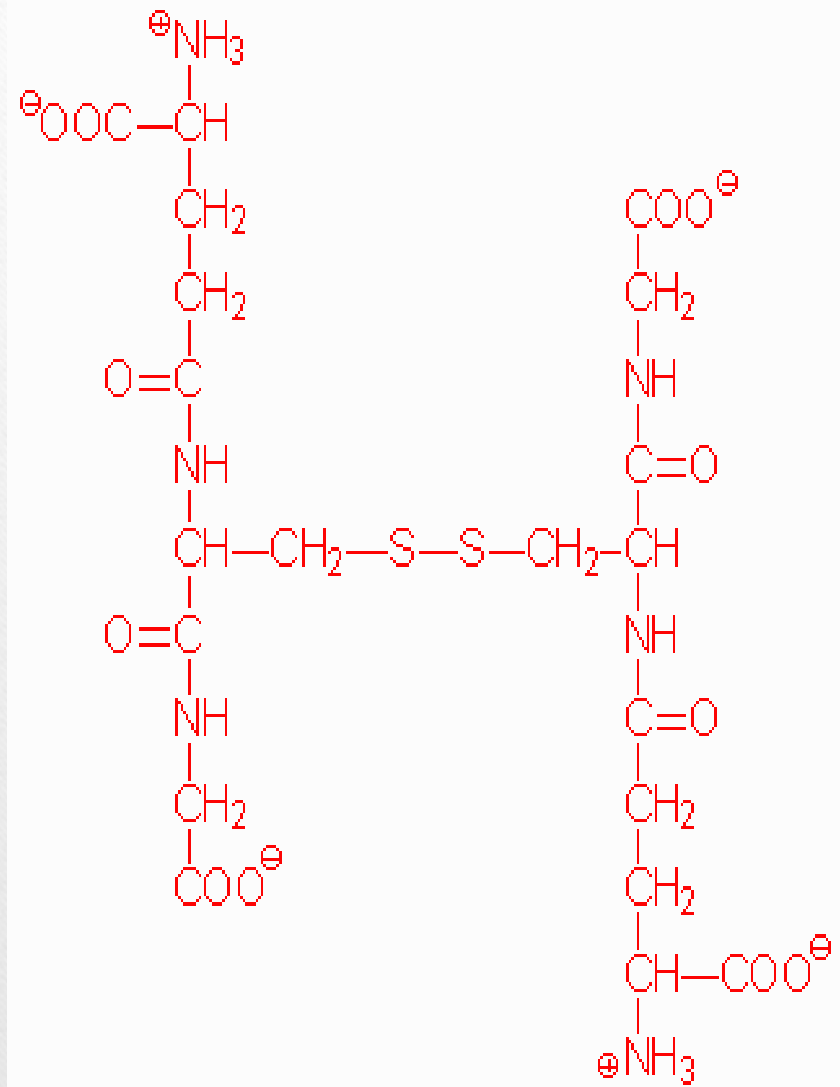


Fig. 9.11 Fate of NADPH in erythrocytes

- How is it related to NADPH?

The enzyme **glutathione reductase** uses **NADPH** as a cofactor to reduce **GSSG** back to two moles of **GSH**. Thus, the pentose pathway is linked to the supply of adequate amounts of GSH.



Glutathione disulfide (GSSG)

Biological importance of HMP shunt

cont.....

3. In neutrophils, NADPH is required for the formation of **superoxide** by **NADPH oxidase**.

❖ Respiratory burst of neutrophils during phagocytosis involves superoxide formation.

4. Pentoses produced in this pathway are used for nucleic acid synthesis and nucleotide coenzymes like NAD⁺, FAD and FMN synthesis

5. Non-oxidative phase of the pathway converts pentoses of endogenous or dietary nucleic acids into intermediates of glycolysis where they are, further oxidized to generate energy.

Medical implications of HMP

1st Unit Glucose-6-phosphate dehydrogenase

❖ deficiency

- ❖ G6PD deficiency is an inheritable X-linked recessive disorder.
- ❖ Approximately 10-14% of the male African American population is affected.
- ❖ Individuals with G6PD deficiency can not produce sufficient **GSH** to cope with the **ROS**.
- ❖ People with the disorder are not normally anemic and display no evidence of the disease until the red cells are exposed to an oxidant or stress.
i.e. The ingestion of oxidative agents that generate peroxides or reactive oxygen species (ROS).
 - Antimalarials - pamaquine
 - purine glycoside from fava beans.

The condition is often clinically seen as black urine under certain conditions.

Drugs that can precipitate this reaction:

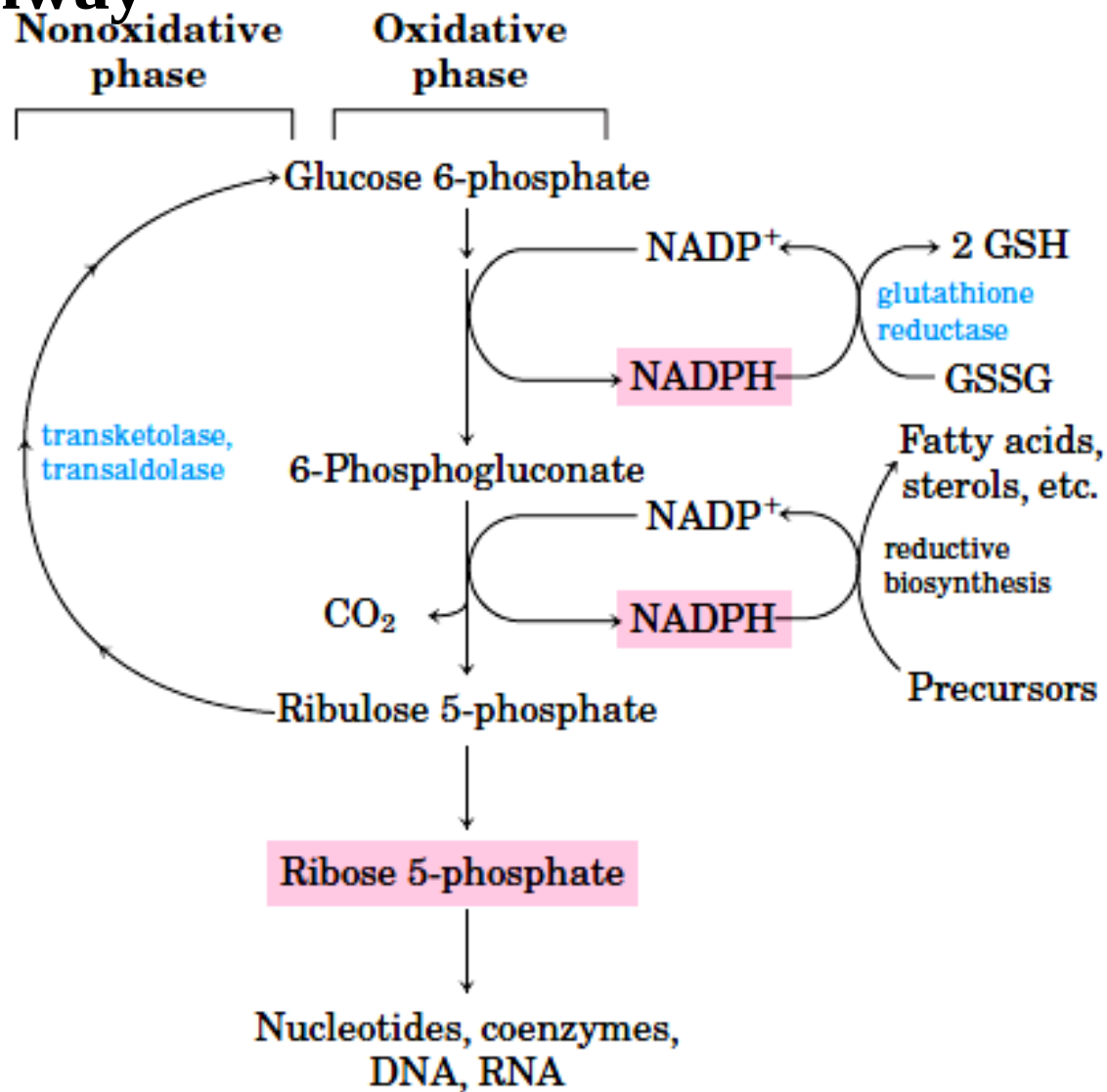
- antimalarial agents
- sulfonamides (antibiotic)
- aspirin
- nonsteroidal antiinflammatory drugs (NSAIDs)
- nitrofurantoin
- quinidine
- quinine
- exposure to certain chemicals - mothballs

Medical implications of HMP shunt cont...

2. Wernicke-korsakoff encephalopathy

- ❖ It is due to defective transketolase genes.
- ❖ Transketolase of affected individuals has lower affinity for TPP.
- ❖ The characteristic symptoms are abnormal walking and standing, memory loss and partial paralysis.
- ❖ The disease manifests only when there is thiamine deficiency.
- ❖ The syndrome is more common among alcoholics than in the general population.

Summary of the Pentose Phosphate Pathway



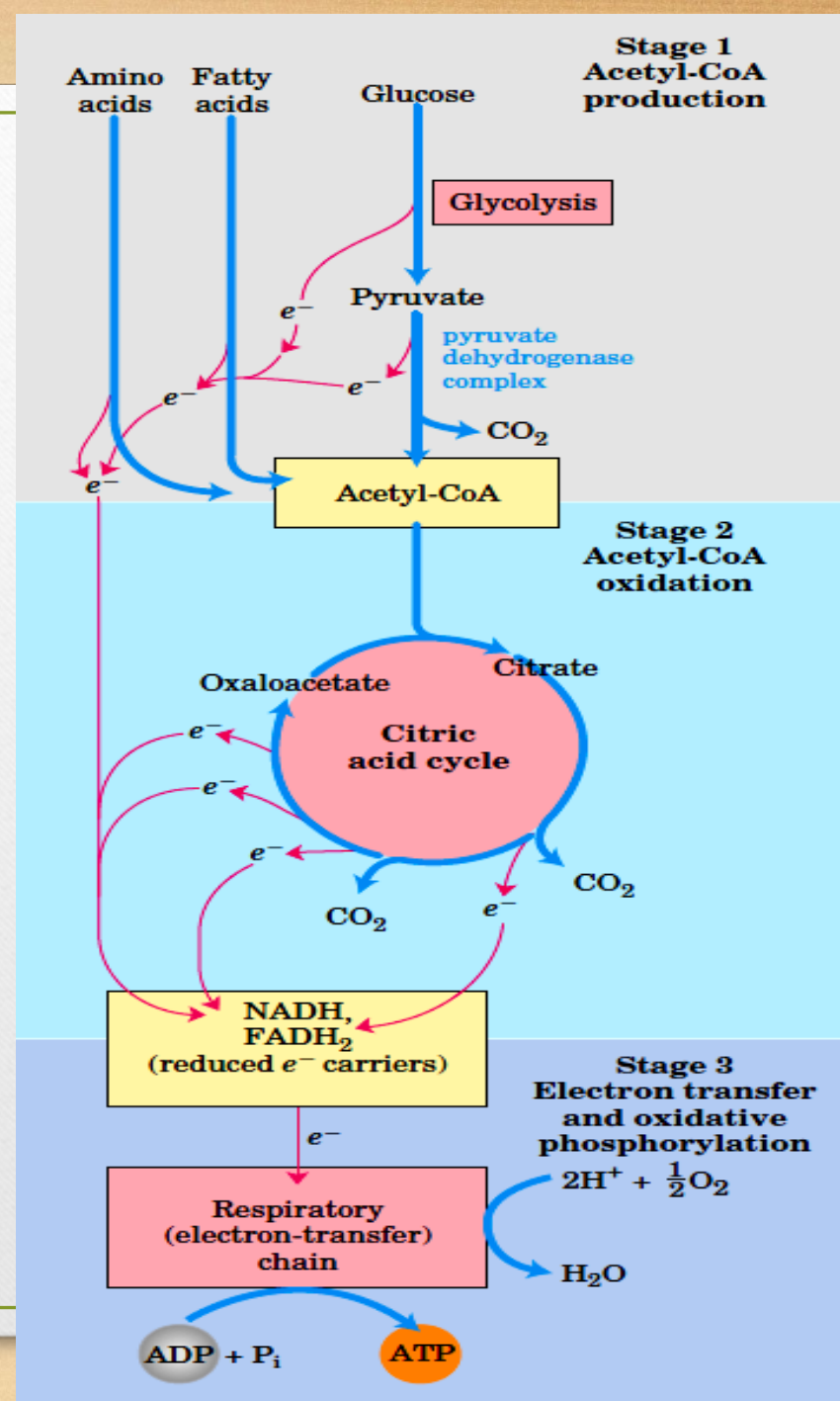
The Citric Acid Cycle

Cellular Respiration:

1. **Oxidation in three stages:** fuel molecules—glucose, fatty acids, and some amino acids—to yield two-carbon fragments in the form of the acetyl group of acetyl-coenzyme A (**acetyl-CoA**).

2. The acetyl groups are fed into the citric acid cycle, which enzymatically oxidizes them to CO_2 ; conserving energy in form of reduced electron carriers **NADH** and **FADH₂**.

3. Oxidation of **NADH** and **FADH₂** and transfer of electrons to O_2 —the final electron acceptor—via a chain of electron-carrying molecules known as the **respiratory chain**; Energy is conserved in the form of ATP (**oxidative phosphorylation**)

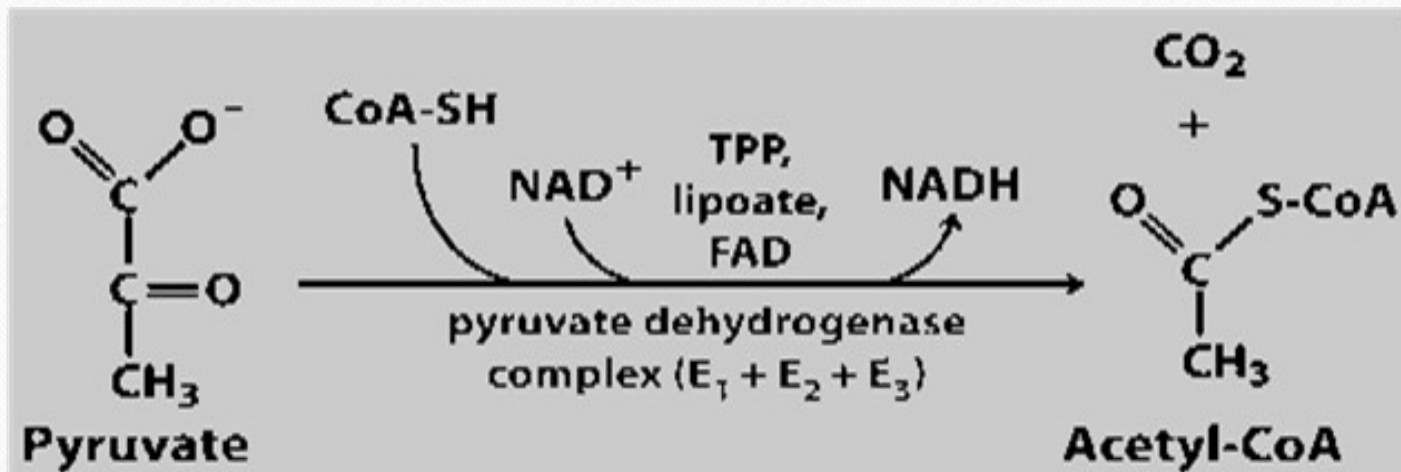


Enzymatic conversion of Pyruvate to acetyl CoA

- The fate of pyruvate produced during glycolysis and oxidation of other biofuels depends on the amount of energy in the tissues.
- If tissues have adequate energy pyruvate is oxidized through gluconeogenesis to generate glucose which can be converted glycogen for storage.
- However, when the energy is low pyruvate is preferentially oxidized to CO_2 and H_2O in the TCA cycle, with generation of 15 molecules ATP per pyruvate.
- The enzymatic activities of the TCA cycle (and of oxidative phosphorylation) are located in the mitochondrion.
- When transported into the mitochondrion, pyruvate encounters two principal metabolizing enzymes: pyruvate carboxylase (a gluconeogenic enzyme) and pyruvate dehydrogenase (PDH), the first enzyme of the PDH complex.
- The activities of these two enzymes depend on the acylation state state of Coenzyme A (CoA).

CONT'

- In the presence of CoA, NAD, FAD and TPP pyruvate is converted to acetyl CoA with release CO_2 .
- The acetyl CoA formed enters the TCA cycle as a substrate for energy generation.
- This reaction is illustrated in the reaction here below.



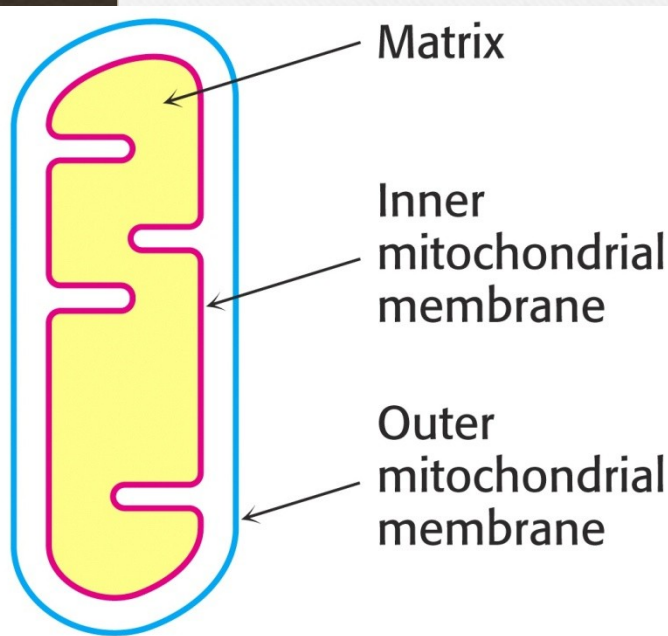
Citric Acid Cycle

Names:

- ❖ The Citric Acid Cycle
- ❖ Tricarboxylic Acid Cycle
- ❖ Krebs Cycle

Hans Adolf Krebs.

Biochemist; born in Germany. Worked in Britain. His discovery in 1937 of the 'Krebs cycle' of chemical reactions was critical to the understanding of cell metabolism and earned him the 1953 Nobel Prize for Physiology or Medicine.



In eukaryotes the reactions of the citric acid cycle take place inside mitochondria



An Overview of the Citric Acid Cycle

A **four-carbon oxaloacetate** condenses with a **two-carbon acetyl** unit to yield a **six-carbon citrate**.

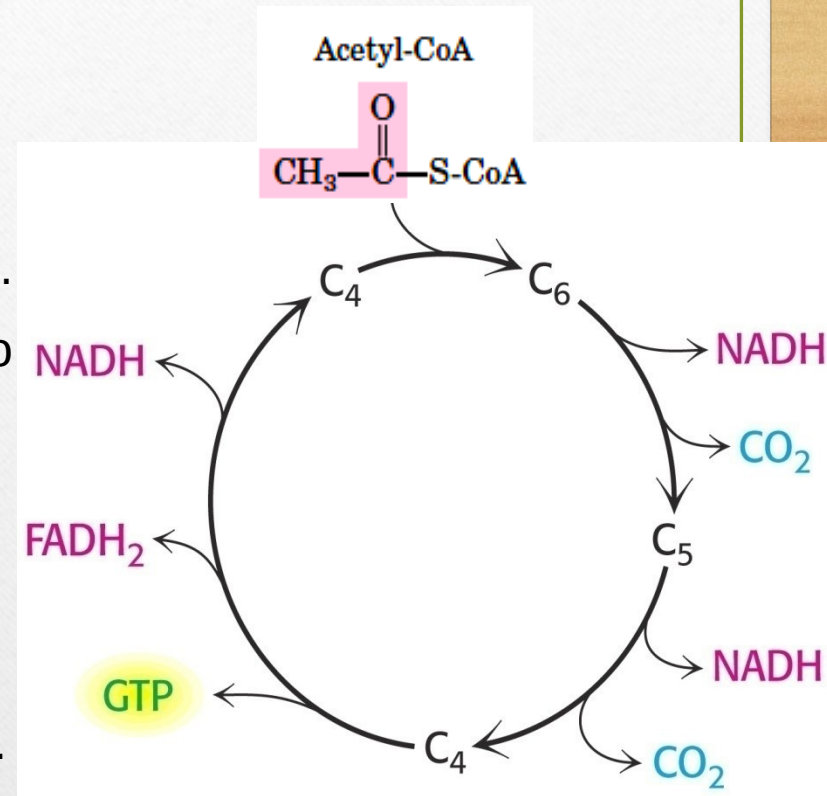
An isomer of citrate is oxidatively decarboxylated to **five-carbon α -ketoglutarate**.

α -ketoglutarate is oxidatively decarboxylated to yield a **four-carbon succinate**.

Oxaloacetate is then regenerated from **succinate**.

Two carbon atoms (**acetyl CoA**) enter the **cycle** and two carbon atoms leave the cycle in the form of two molecules of **carbon dioxide**.

Three hydride ions (**six electrons**) are transferred to three molecules of **NAD⁺**, one pair of hydrogen atoms (**two electrons**) is transferred to one molecule of **FAD**.



The function of the citric acid cycle is the harvesting of high-energy electrons from acetyl CoA.

Functions of the Citric Acid Cycle

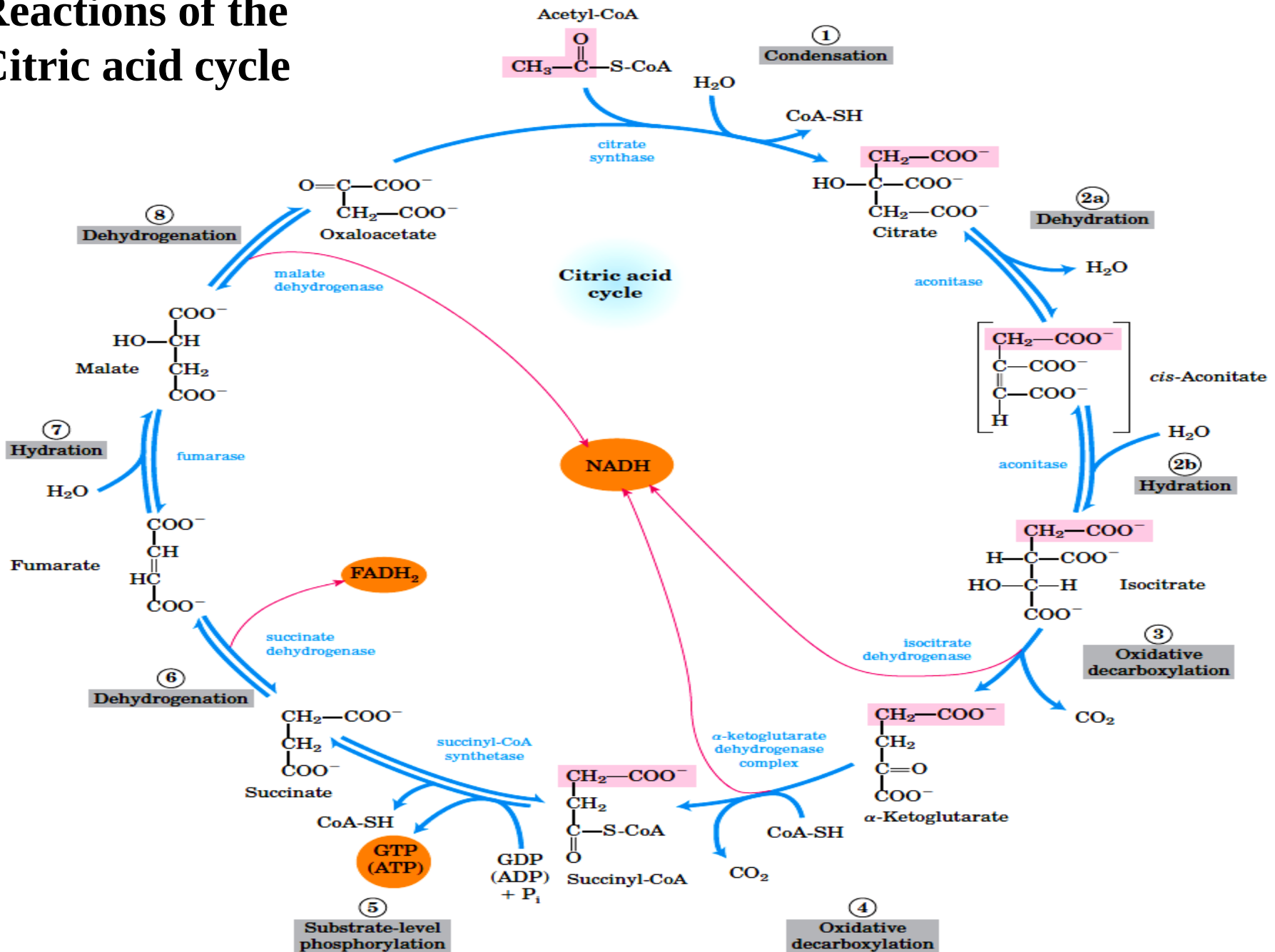
- **Integration of metabolism.** The citric acid cycle is *amphibolic* (both catabolic and anabolic).

The cycle is involved in the aerobic **catabolism** of carbohydrates, lipids and amino acids.

Intermediates of the cycle are starting points for many **anabolic** reactions.

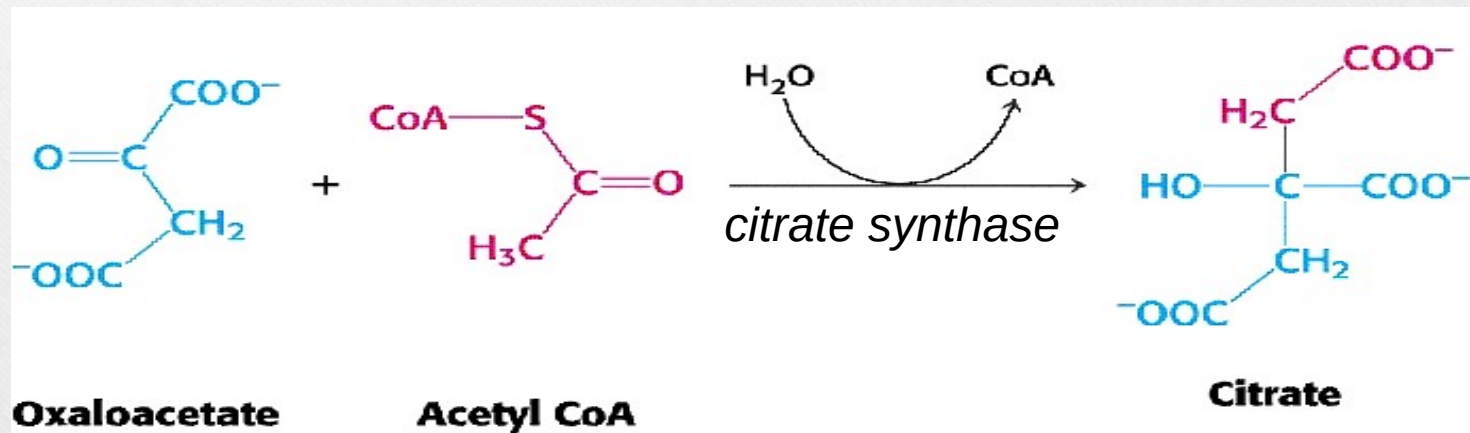
- Yields **energy** in the form of **GTP (ATP)**.
- Yields **reducing power** in the form of **NADH₂** and **FADH₂**.

Reactions of the Citric acid cycle



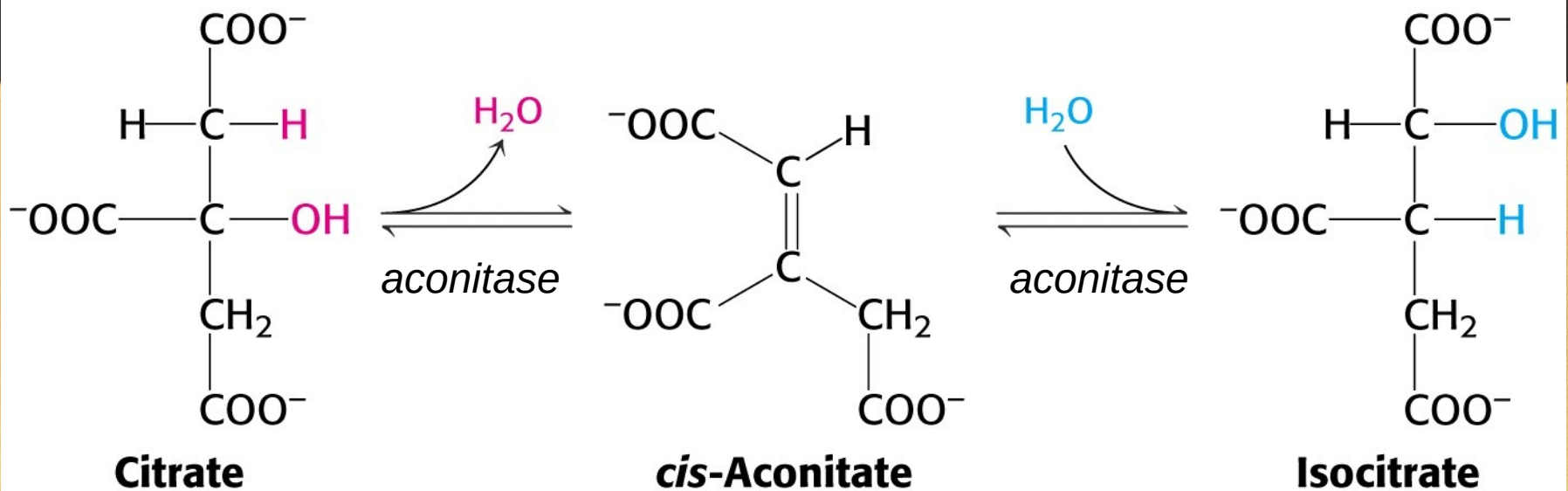
1. Citrate Synthase

- **Citrate** formed from **acetyl CoA** and **oxaloacetate**
- Only cycle reaction with C-C bond formation
- Addition of **C₂ unit** (acetyl) to the keto double bond of **C₄** acid, oxaloacetate, to produce **C₆** compound, citrate



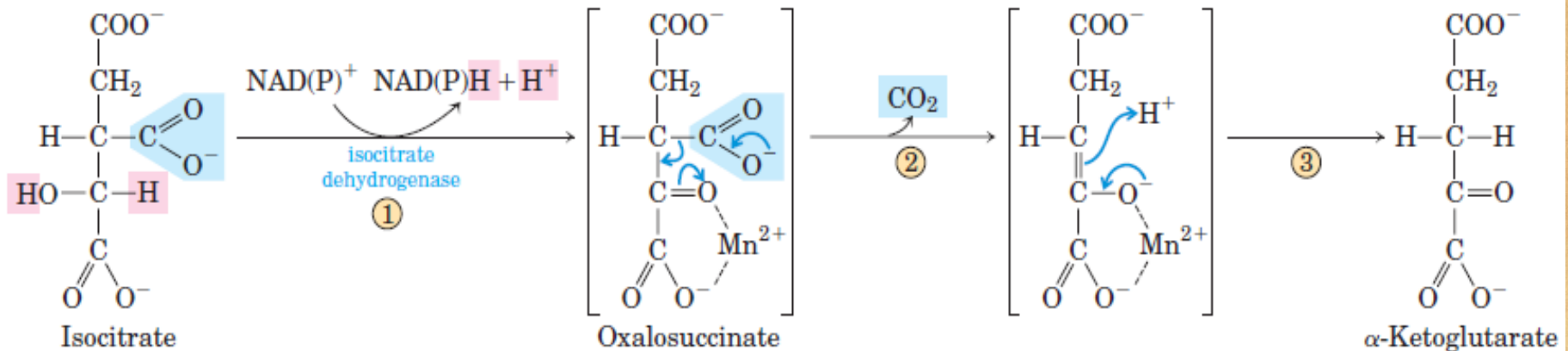
2. Aconitase

- Elimination of H_2O from citrate to form $\text{C}=\text{C}$ bond of ***cis*-aconitate**.
- Stereospecific addition of H_2O to *cis*-aconitate to form **isocitrate**.



3. Isocitrate Dehydrogenase

- Oxidative decarboxylation of **isocitrate** to **α -ketoglutarate** (a metabolically irreversible reaction)
- One of four oxidation-reduction reactions of the cycle
- Hydride ion from the C-2 of isocitrate is transferred to NAD^+ to form NADH
- **Oxalosuccinate** is decarboxylated to **α -ketoglutarate**



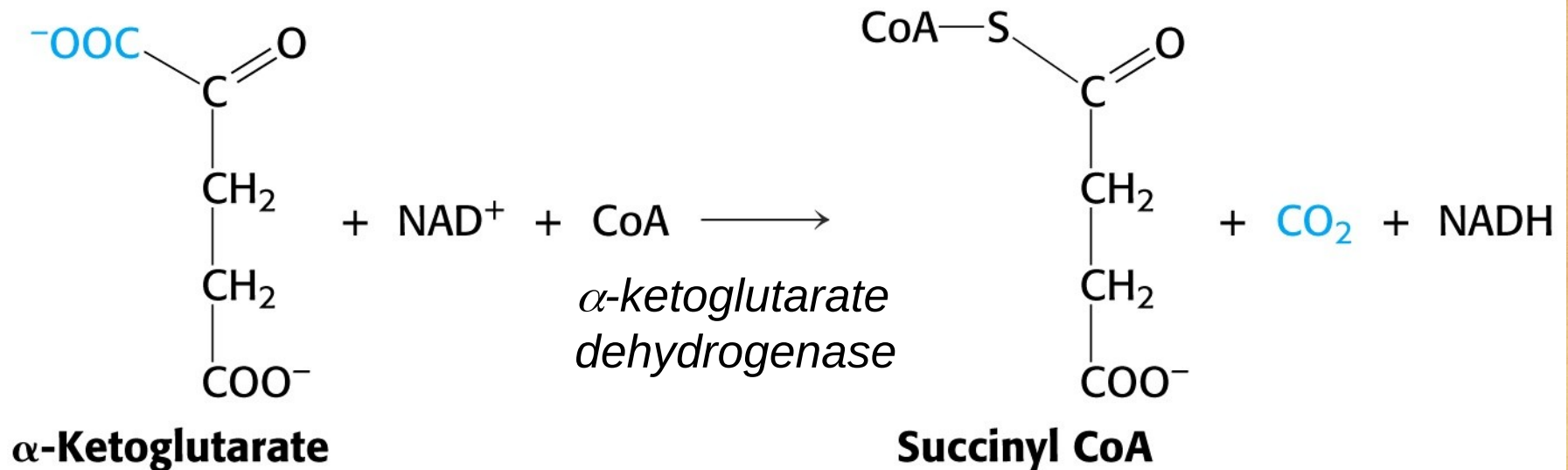
4. The α -Ketoglutarate Dehydrogenase Complex

- Similar to pyruvate dehydrogenase complex
- Same coenzymes, identical mechanisms

E₁ - *α -ketoglutarate dehydrogenase* (with **TPP**)

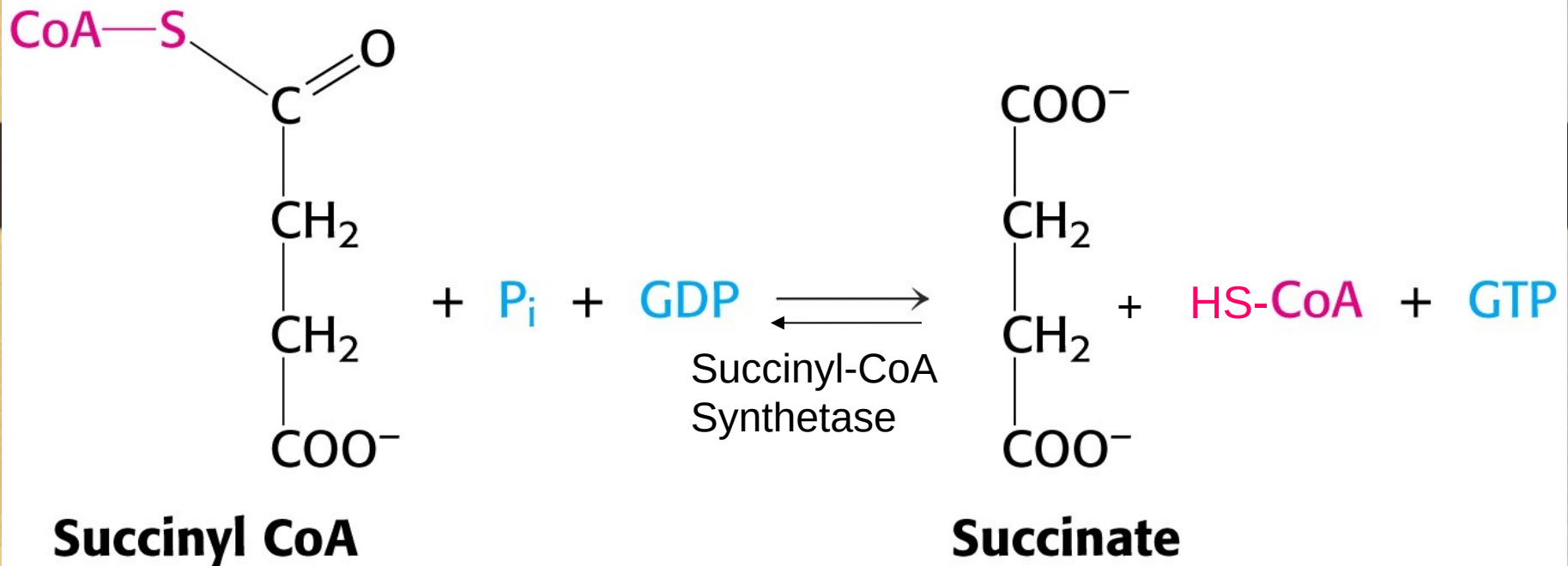
E₂ - *dihydrolipoyl succinyltransferase* (with flexible **lipoamide** prosthetic group)

E₃ - *dihydrolipoyl dehydrogenase* (with **FAD**)



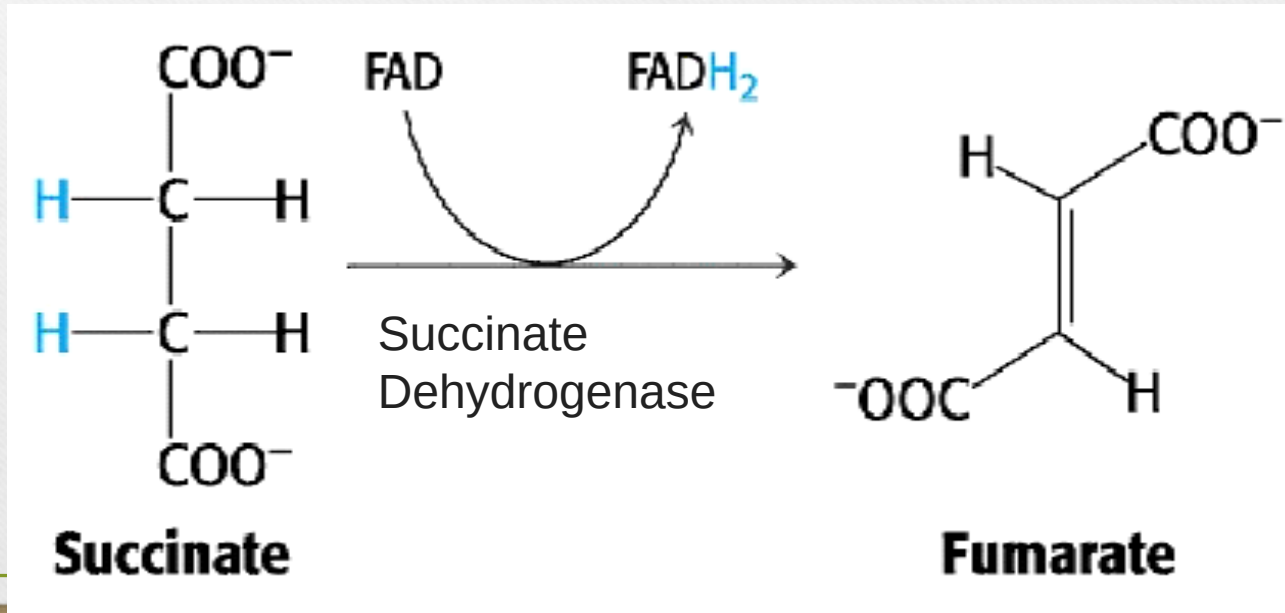
5. Succinyl-CoA Synthetase

- Free energy in thioester bond of succinyl CoA is conserved as GTP or ATP in higher animals (or ATP in plants, some bacteria)
- Substrate level phosphorylation** reaction



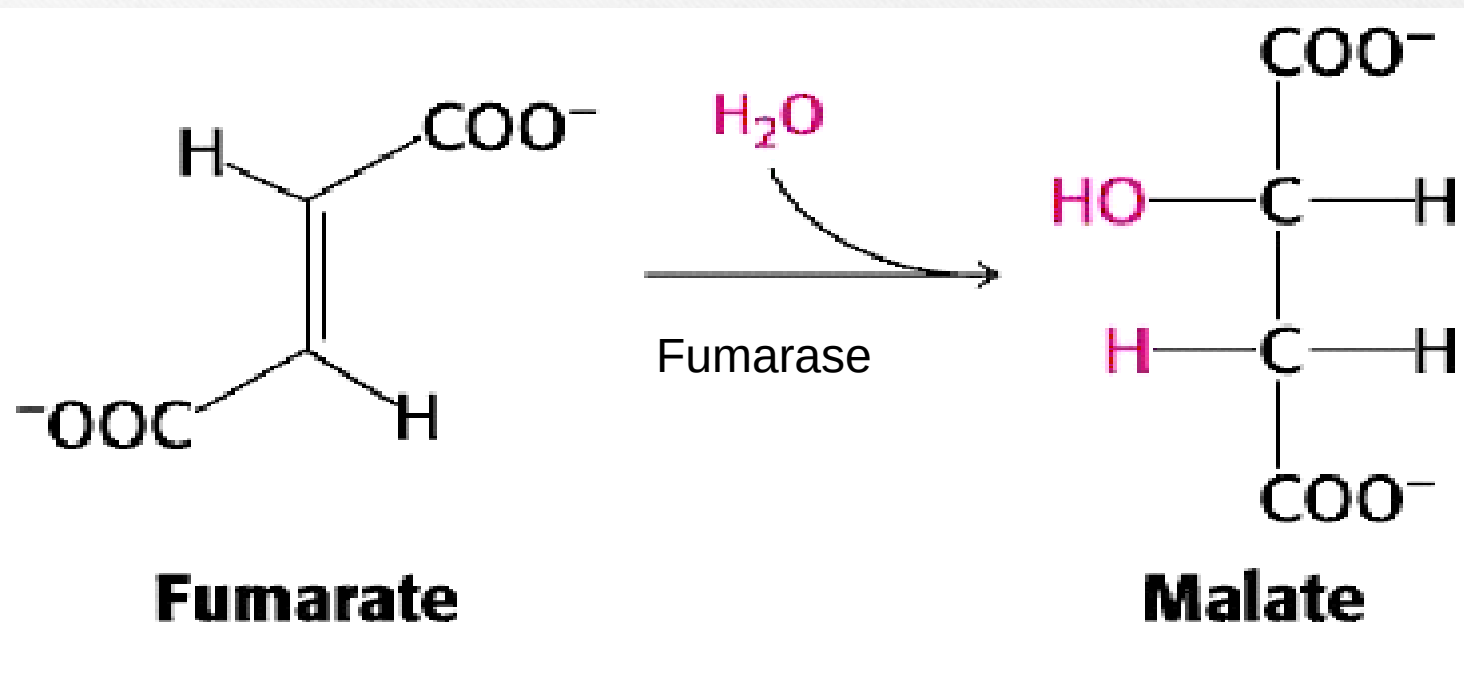
6. The Succinate Dehydrogenase Complex

- Complex of **several polypeptides**, an **FAD** prosthetic group and **iron-sulfur clusters**
- Embedded in the **inner mitochondrial membrane**
- Electrons are transferred **from succinate to FAD** and then to **ubiquinone (Q)** in electron transport chain
- Dehydrogenation is **stereospecific**; only the *trans* isomer is formed



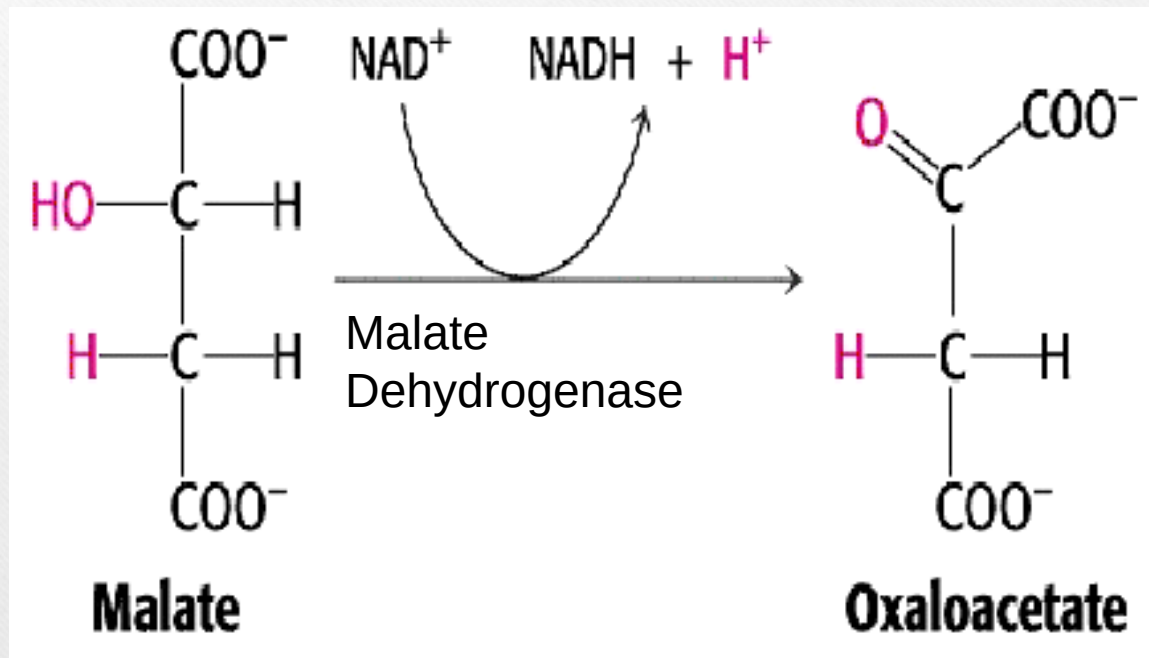
7. Fumarase

- Stereospecific *trans* addition of water to the double bond of **fumarate** to form **L-malate**
- Only the **L isomer** of malate is formed



8. Malate Dehydrogenase

Malate is oxidized to form **oxaloacetate**.



Regulation of the Citric Acid Cycle

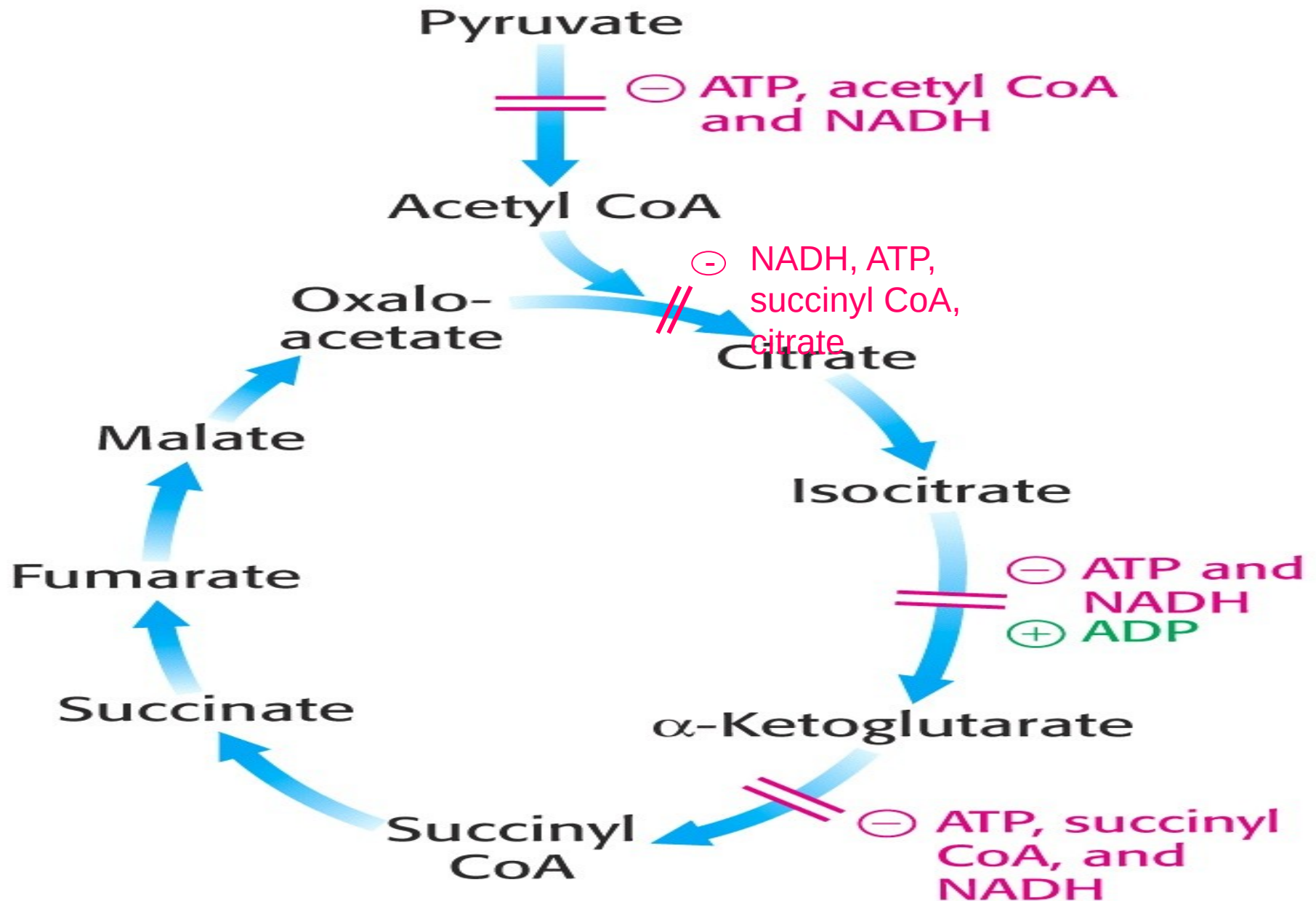
- **Pathway controlled by:**

- (1) Allosteric modulators
- (2) Covalent modification of cycle enzymes
- (3) Supply of acetyl CoA (pyruvate dehydrogenase complex)

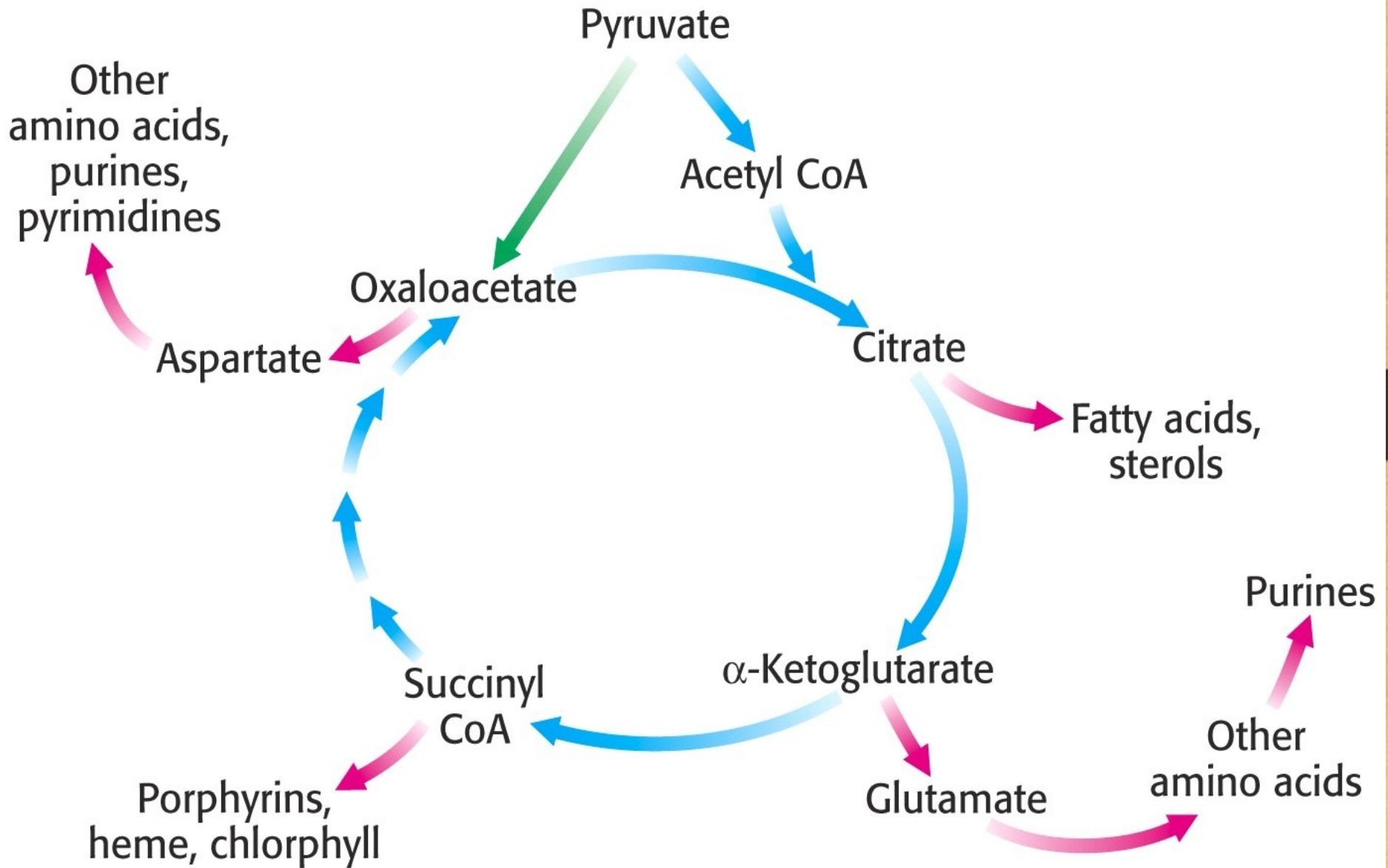
Three enzymes have regulatory properties

- ***citrate synthase*** (is allosterically inhibited by NADH, ATP, succinyl CoA, citrate – feedback inhibition)
- ***isocitrate dehydrogenase***
(allosteric effectors: (+) ADP; (-) NADH, ATP. Bacterial ICDH can be covalently modified by kinase/phosphatase)
- ***α -ketoglutarate dehydrogenase complex*** (inhibition by ATP, succinyl CoA and NADH)

Regulation of the citric acid cycle



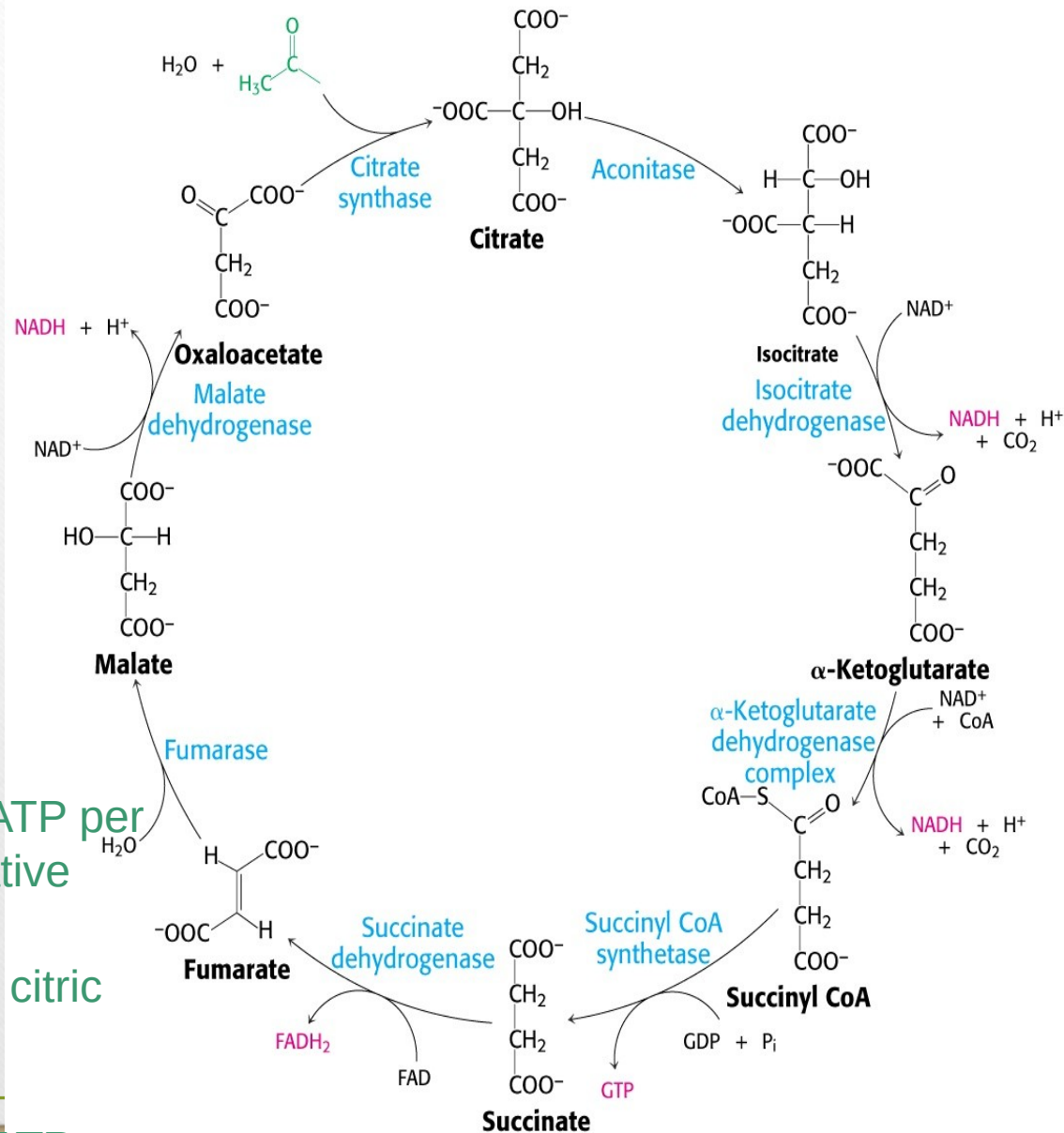
Krebs Cycle is a Source of Biosynthetic Precursors



Stoichiometry of the Citric Acid Cycle

- Two carbon atoms enter the cycle in the form of acetyl CoA.
- Two carbon atoms leave the cycle in the form of CO_2 .
- Four pairs of hydrogen atoms leave the cycle in four oxidation reactions (three molecules of NAD^+ one molecule of FAD are reduced).
- One molecule of GTP, is formed.

- 9 ATP (3 ATP per NADH, and 2 ATP per FADH_2) are produced during oxidative phosphorylation.
- 1 ATP is directly formed in the citric acid cycle.
- 1 acetyl CoA generates approximately 12 molecules of ATP.



Summary of Krebs Cycle

How many energy-producing molecules do we have per 1 glucose molecule?

Glycolysis: 2 ATP, 2 NADH

Intermediate step: Pyruvate oxidation

1 NADH x2= 2 NADH

Krebs Cycle: 3 NADHx2= 6 NADH

1 ATPx2= 2 ATP

1 FADH₂x2= 2 FADH₂

Total: 4 ATP, 10 NADH, 2 FADH₂ --> forms 38 ATP in the electron transport chain

The role of vitamins in TCA cycle

- Vitamins, mainly of the B-group play important roles in TCA cycle. Four of the B vitamins that are essential in the TCA cycle and hence energy-yielding metabolism are:
- **Riboflavin** in the form of flavin adenine dinucleotide (FAD), it is a co-factor for succinate dehydrogenase.
- **Niacin** in the form the form of adenine dinucleotide is the electron acceptor for isocitrate dehydrogenase, α -ketoglutarate dehydrogenase, and malate dehydrogenase.
- **Thiamine (Vitamin B1)** as thiamine diphosphate is the co-enzyme for decarboxylation in the α -ketoglutarate dehydrogenase reaction.
- **Pantothenic acid** is a part of coenzyme A, the co-factor attached to "active" carboxylic acid residues such as acetyl-CoA and succinyl-CoA.